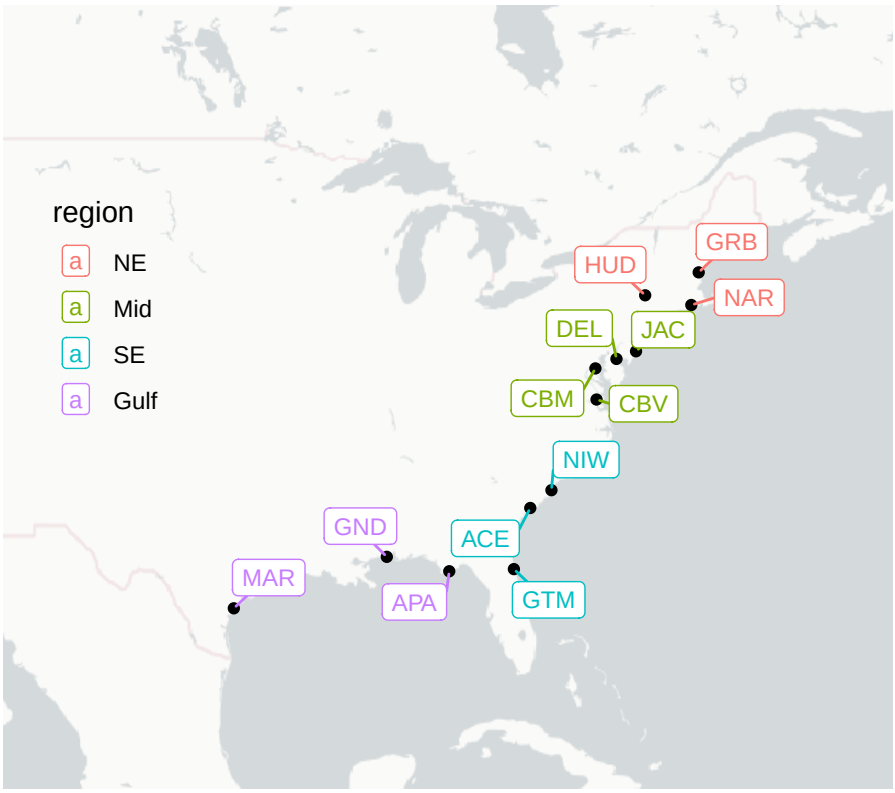
Exploring Responses in DO, pH, Conductivity, and Turbidity to Estuarine Heatwaves

Methods

I used the the heatwaveR package to estimate what (seasonally detrended) percentile daily averages of water temperature, DO, pH, conductivity, and turbidity fell under. I labeled heat wave days as days part of a period greater than 5 days long where water temperatures were greater than the 90th percentile. I then compared percentile values in the 10 days before heatwaves with percentile values during the heatwaves and for 5 days post heatwave using Kruskal-Wallis rank sum tests and post-hoc Dunn's tests. I tested each station independently to better understand spatial patterns.

Stations

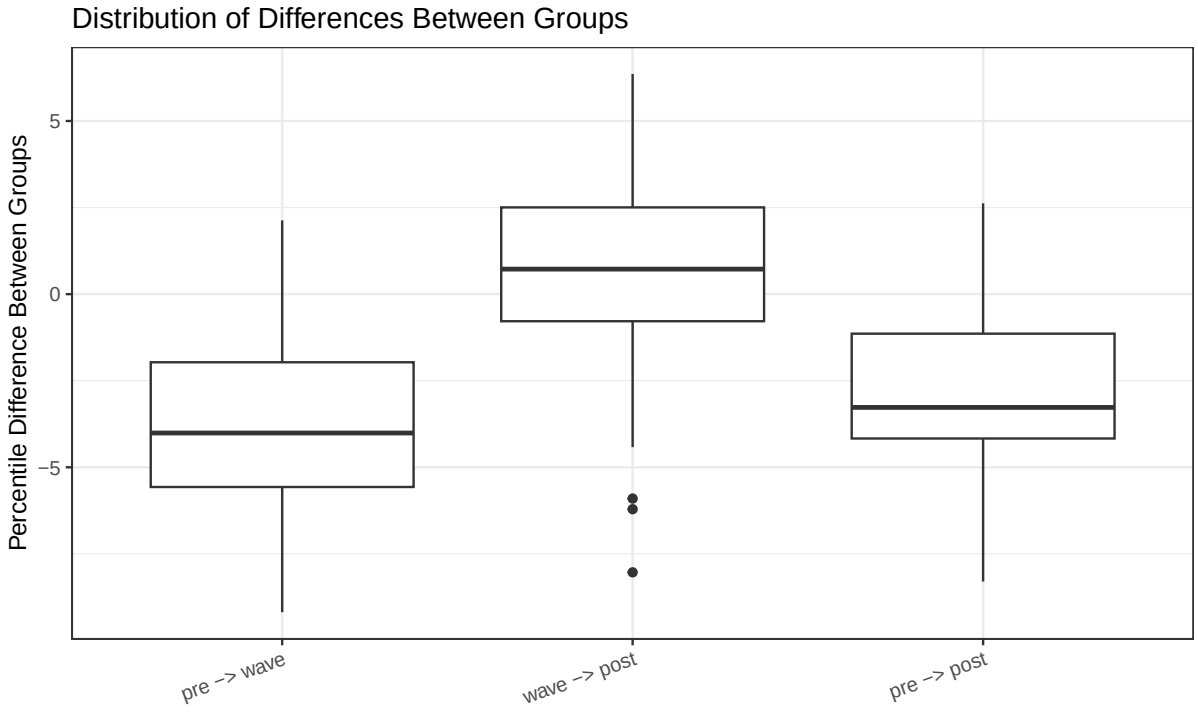
I only included SWMP stations on the Atlantic US Coast that had at least 10 years of data. This meant dropping data from a single reserve in each of PR, AK, HI, CA, and WI. Here is a map of the included reserves, colored by region.

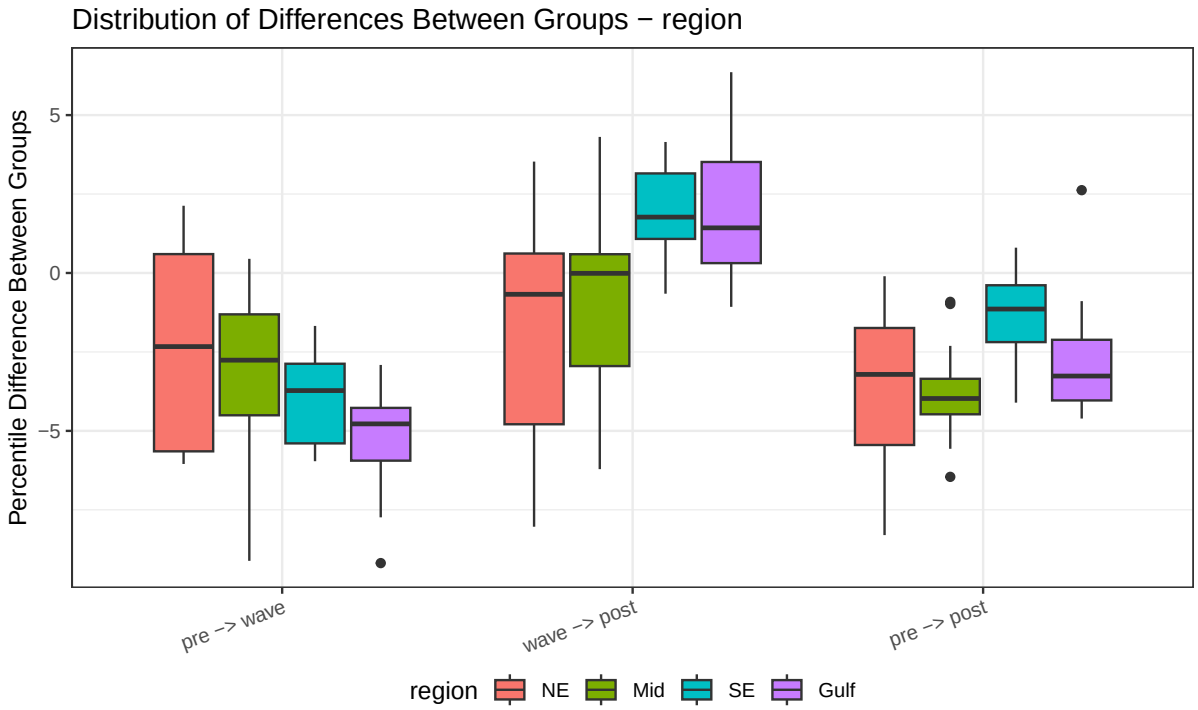

region
a NE
a Mid
a SE
a Gulf
GRB
HUD
NAR
DEL
JAC
CBM
CBV
NIW
GND
ACE
MAR
APA
GTM

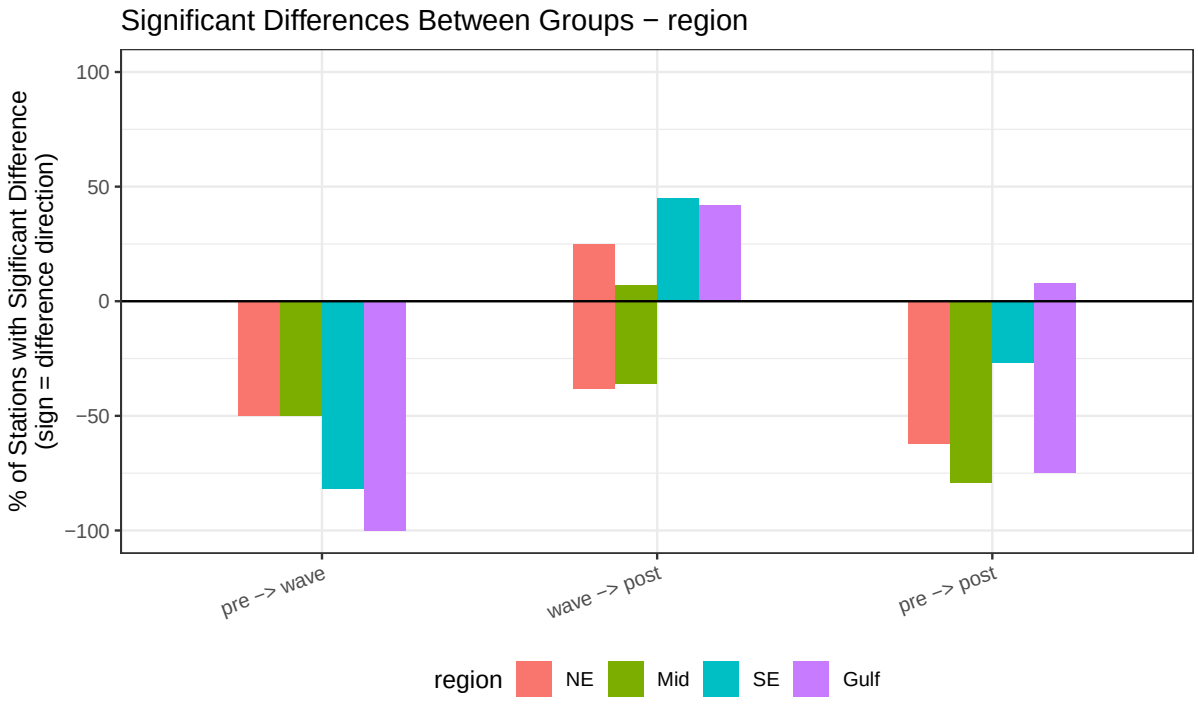
Heatwaves alone, no additional factors

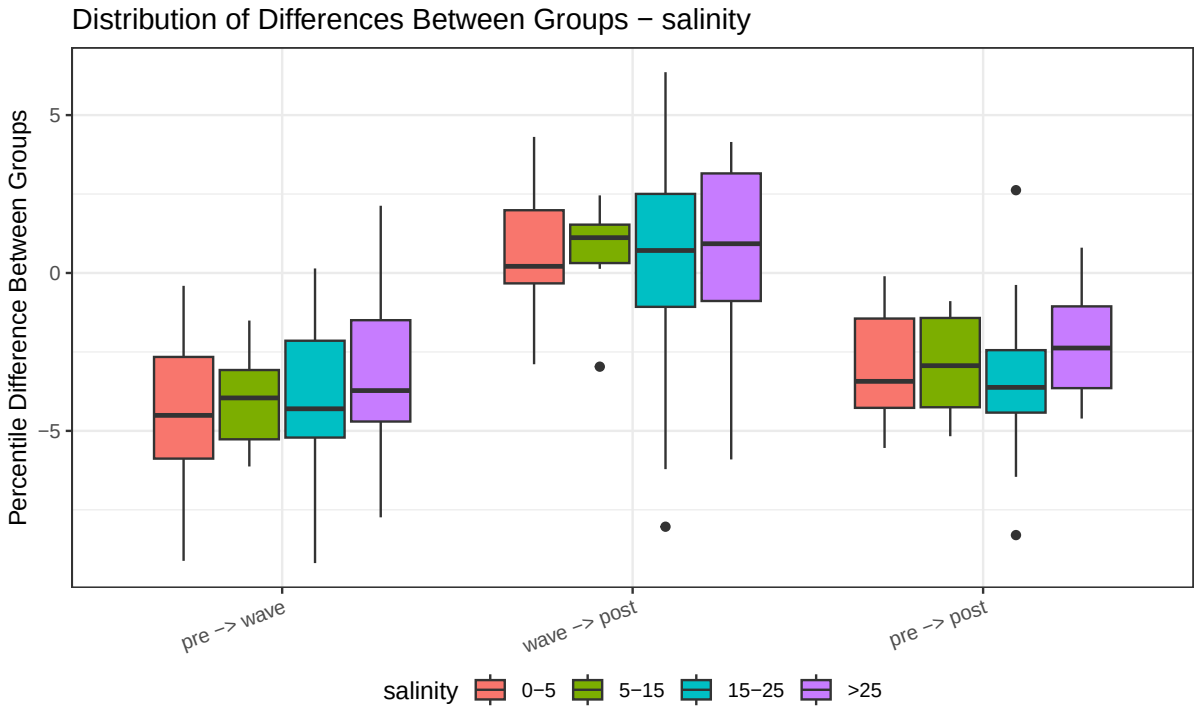
DO

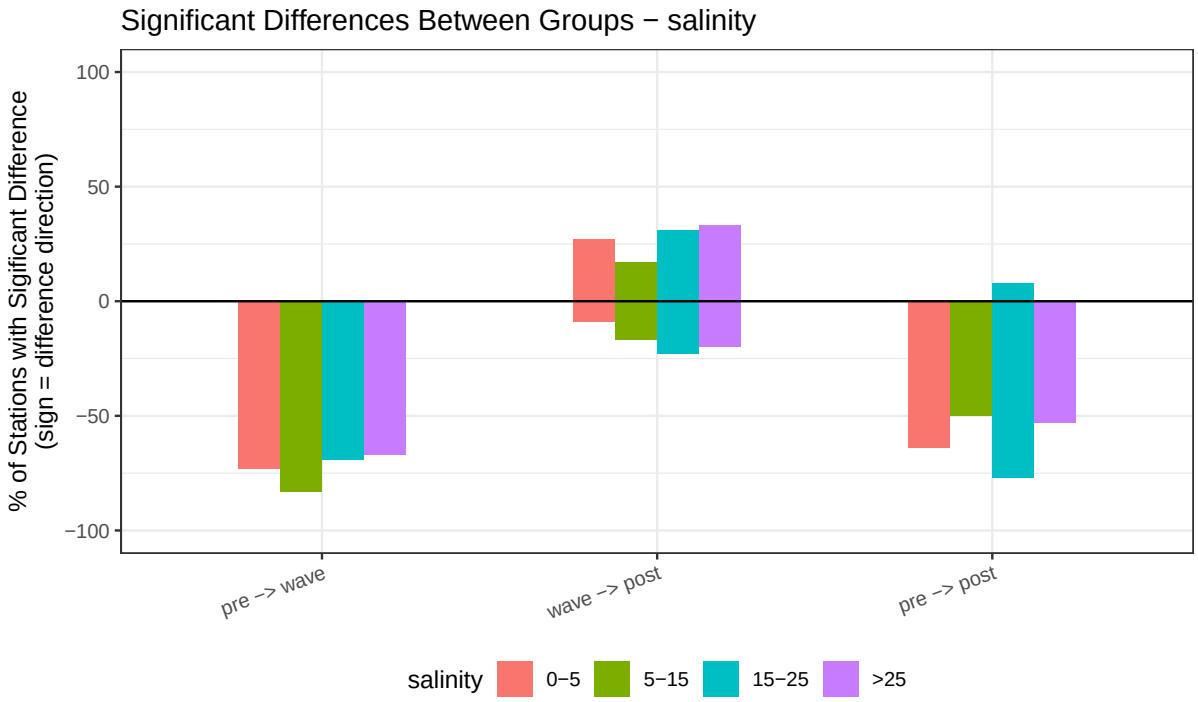
- Drops during heatwaves
- Stays lower post heatwave, though there is some rebound in SE and Gulf regions
- Pattern is present across regions
- No difference between salinity groups




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

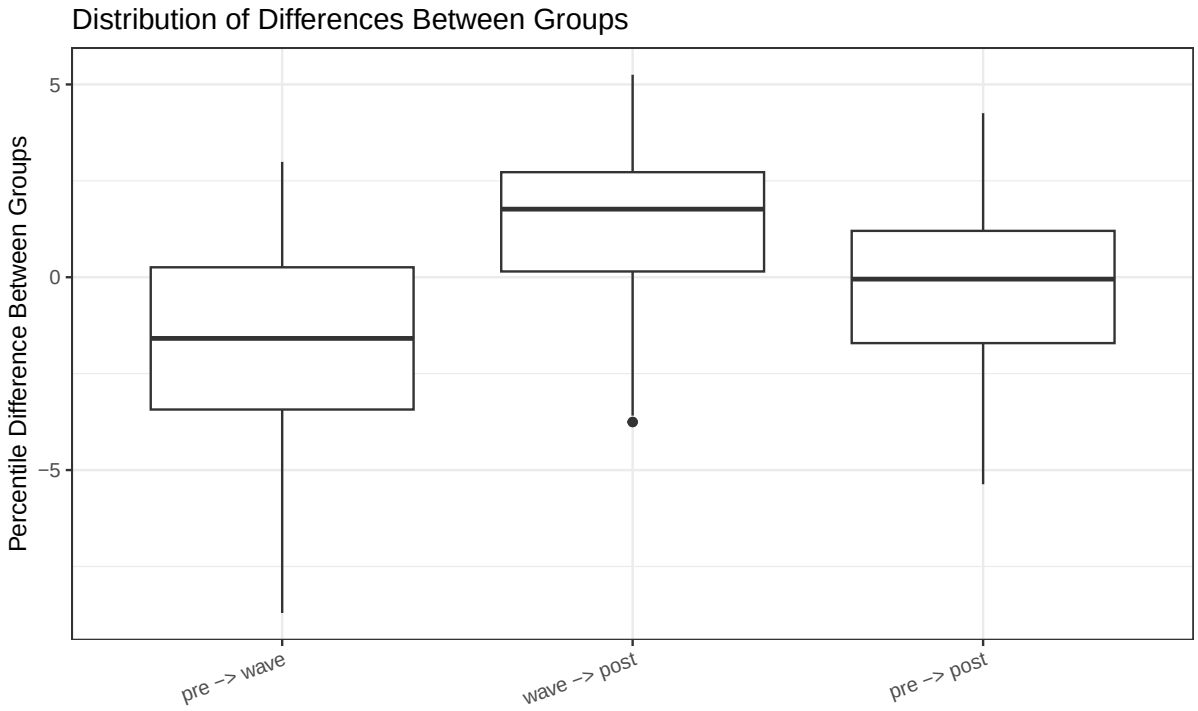

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

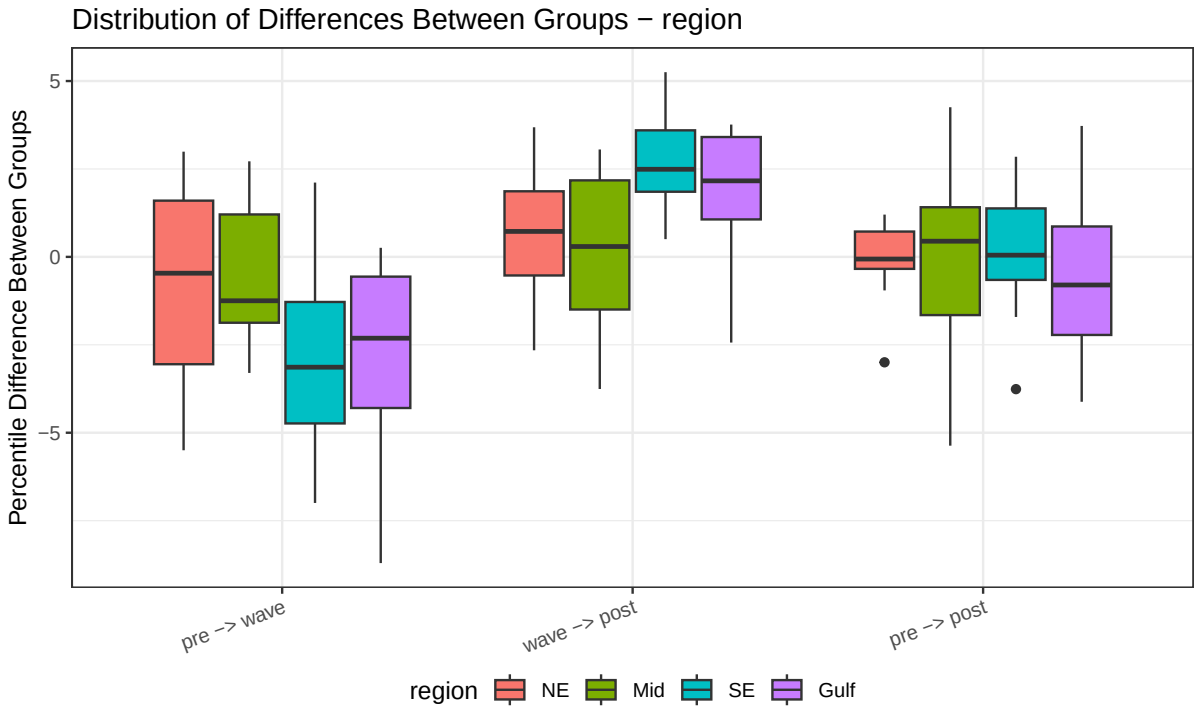

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

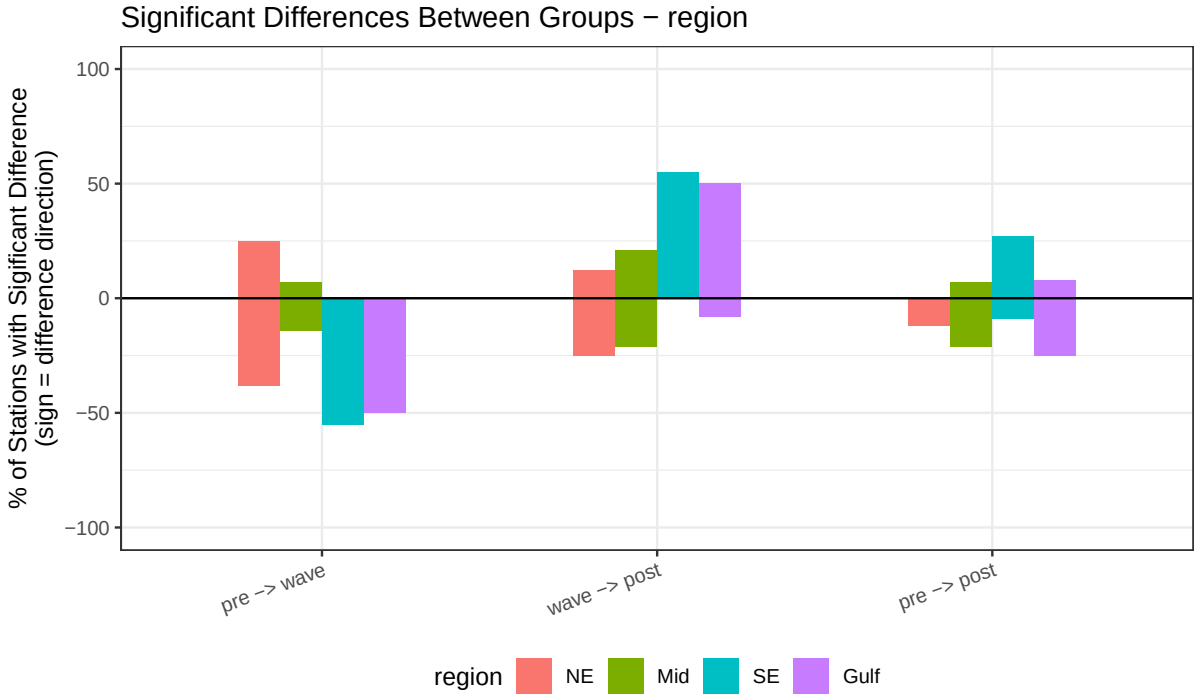

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

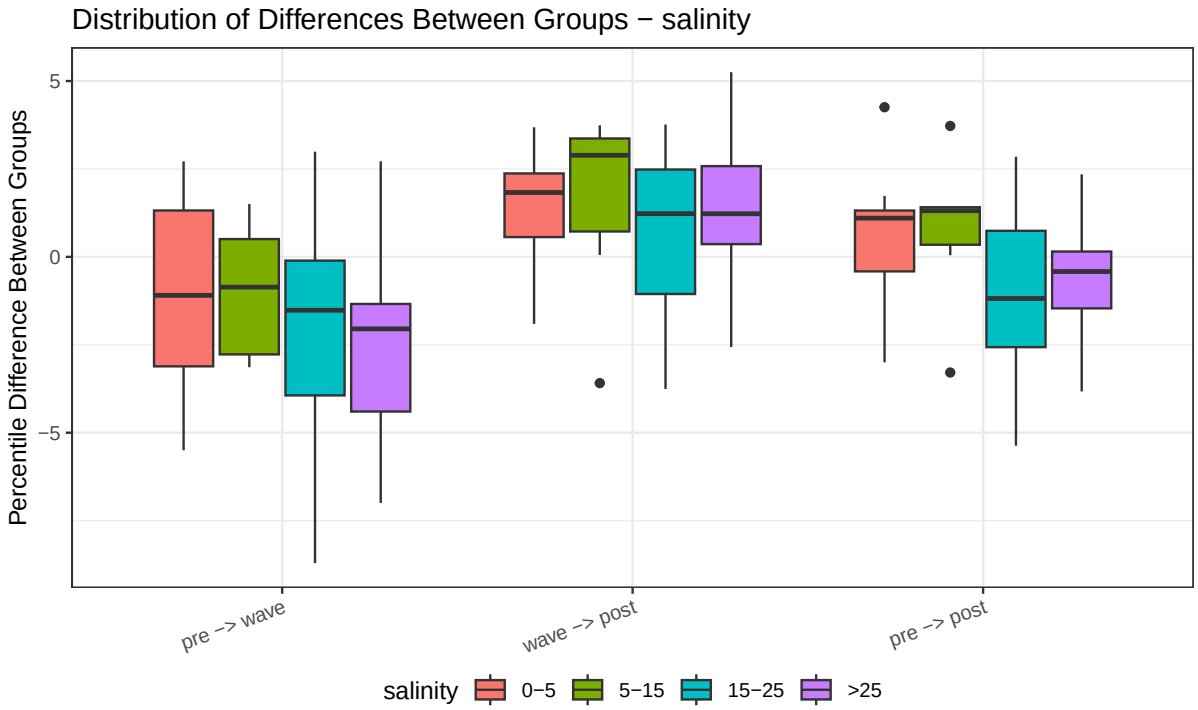
pH

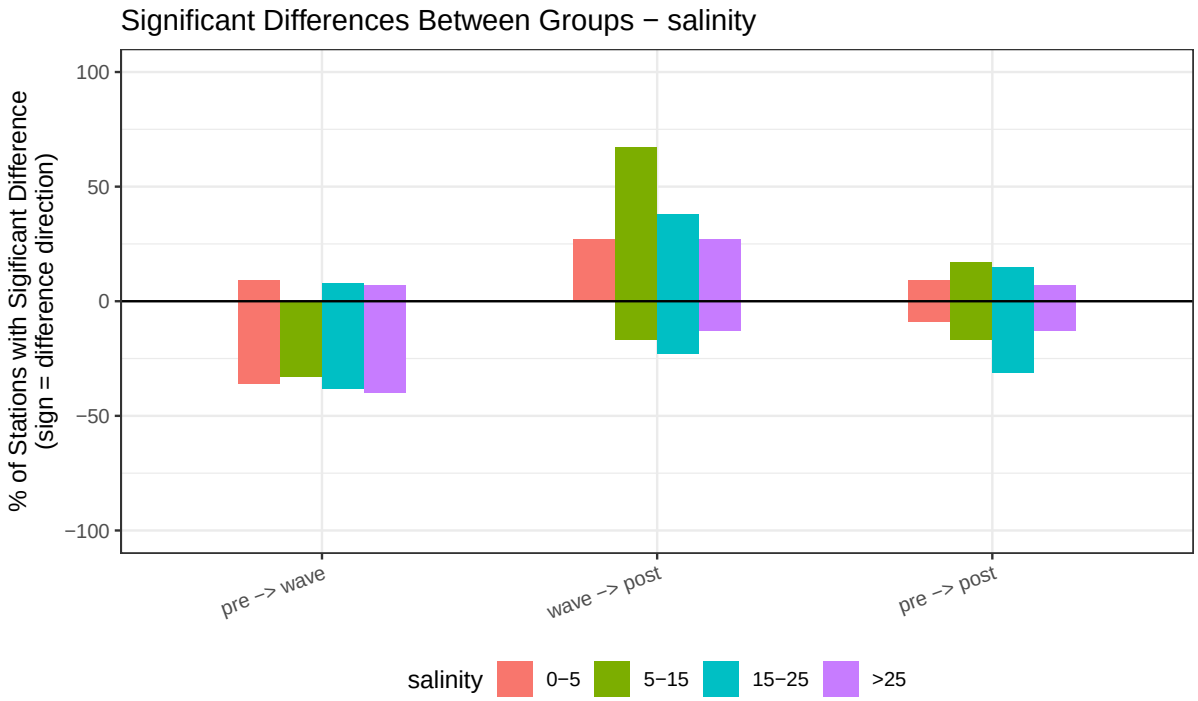
- Slight drop in pH during heatwaves, particularly at Southern sites.
- Only present during heatwaves, not post
- No difference between salinity groups




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

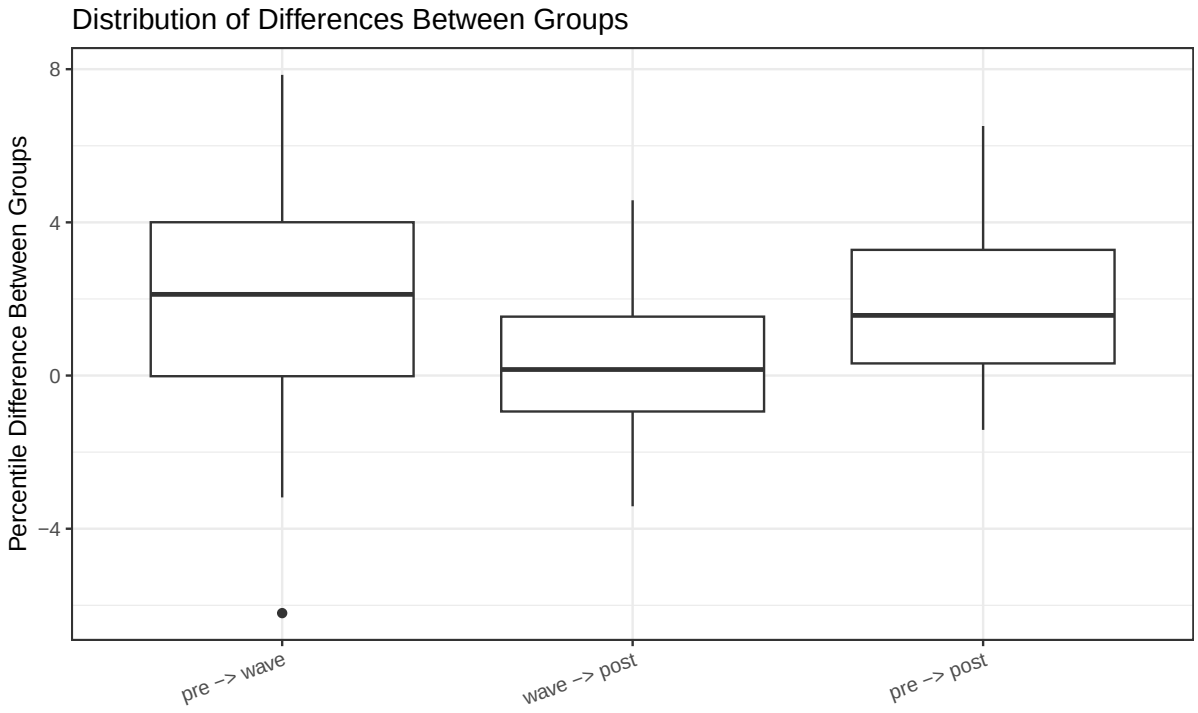

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

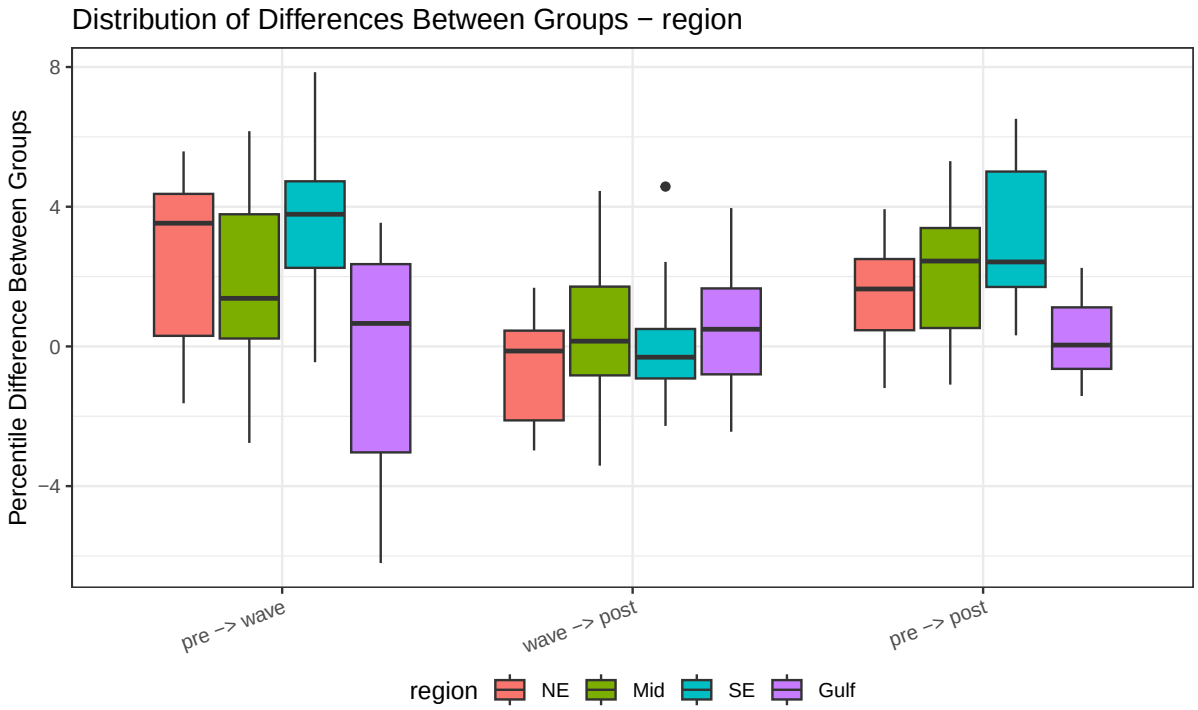

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

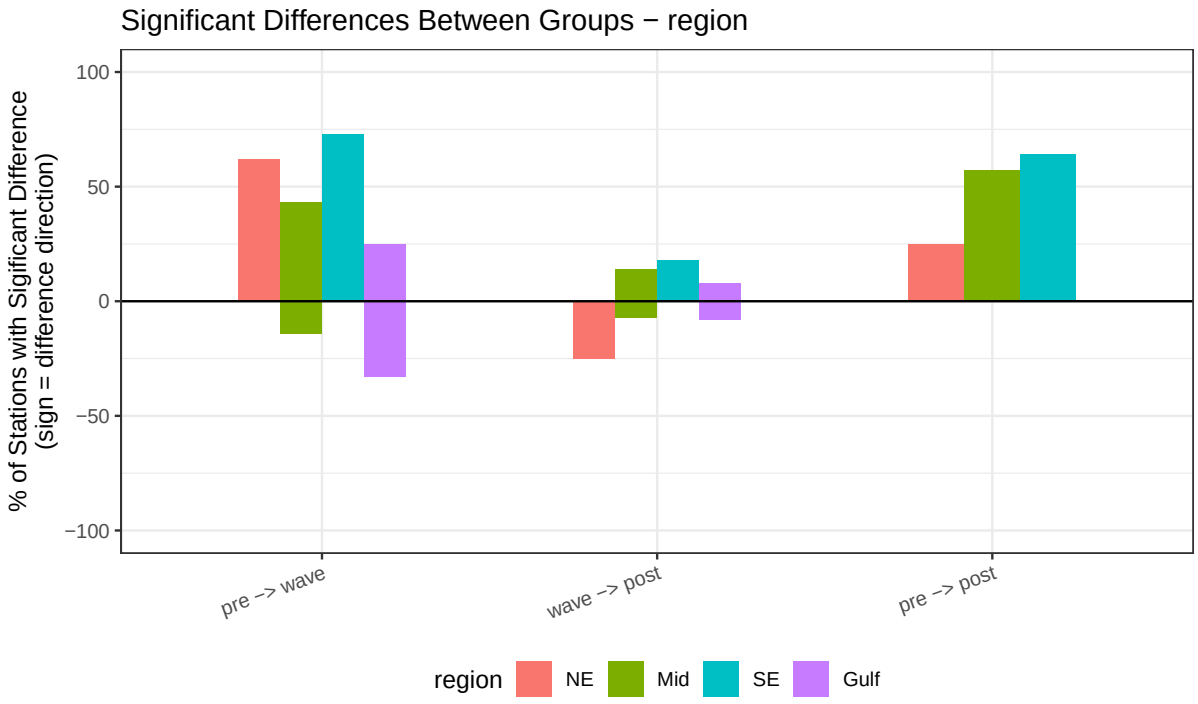

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

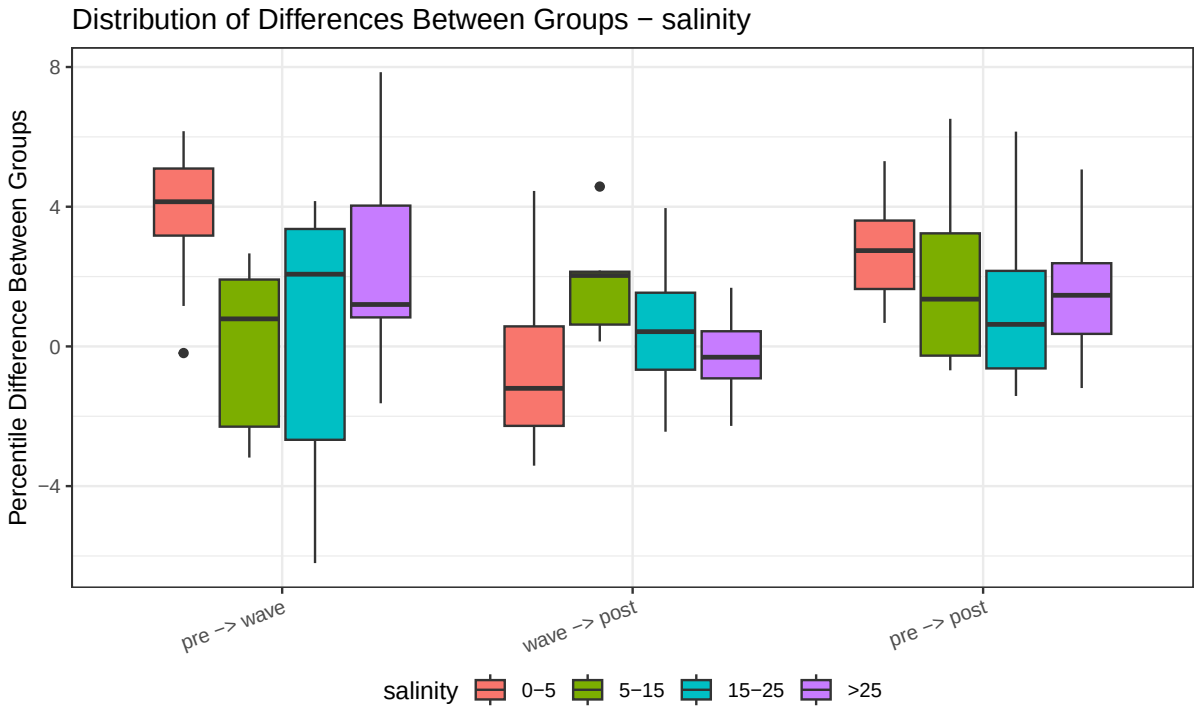
Conductivity

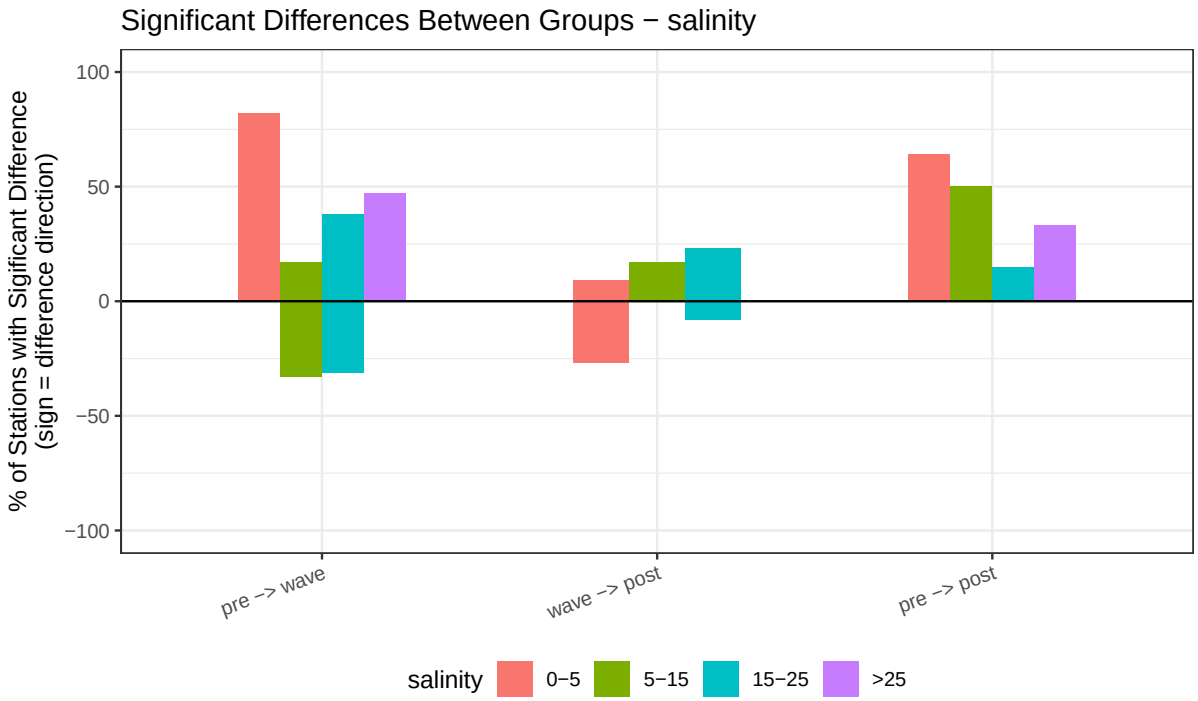
- Conductivity increases during heatwaves in all regions, but some Gulf sites show decreases
- Conductivity stays high post heatwave in all regions but Gulf
- Increases during and post wave in low and high salinity groups, but not moderate




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
8
4
0
−4
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

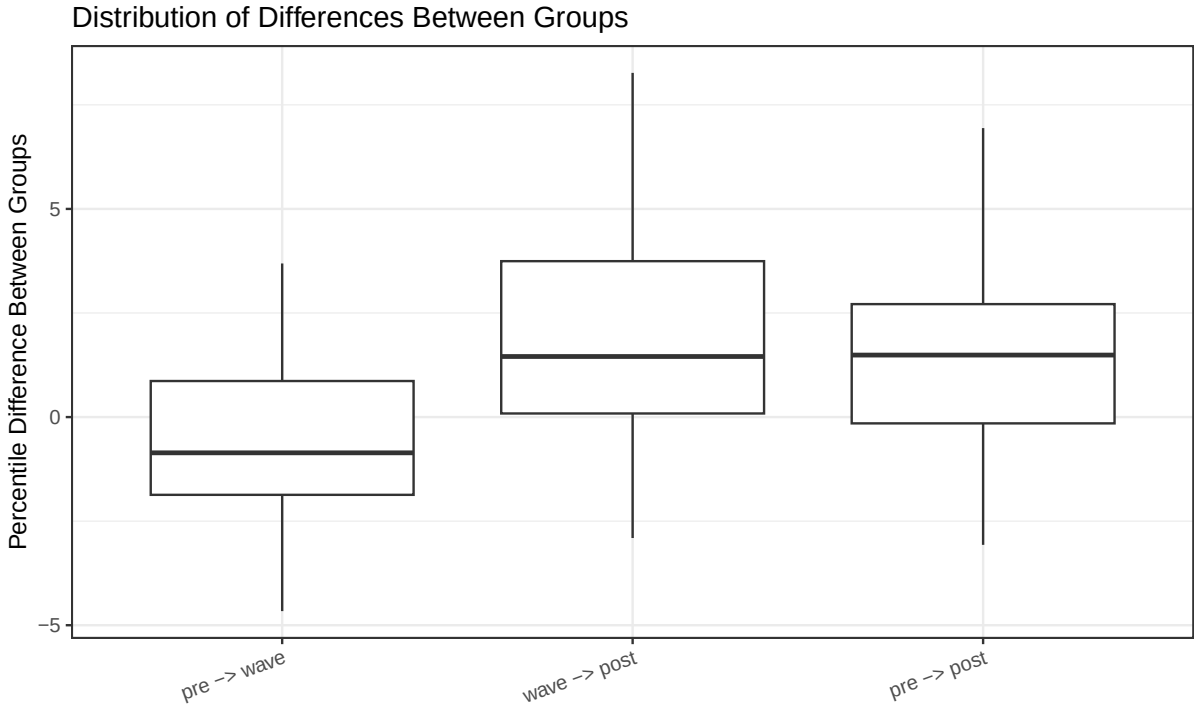

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

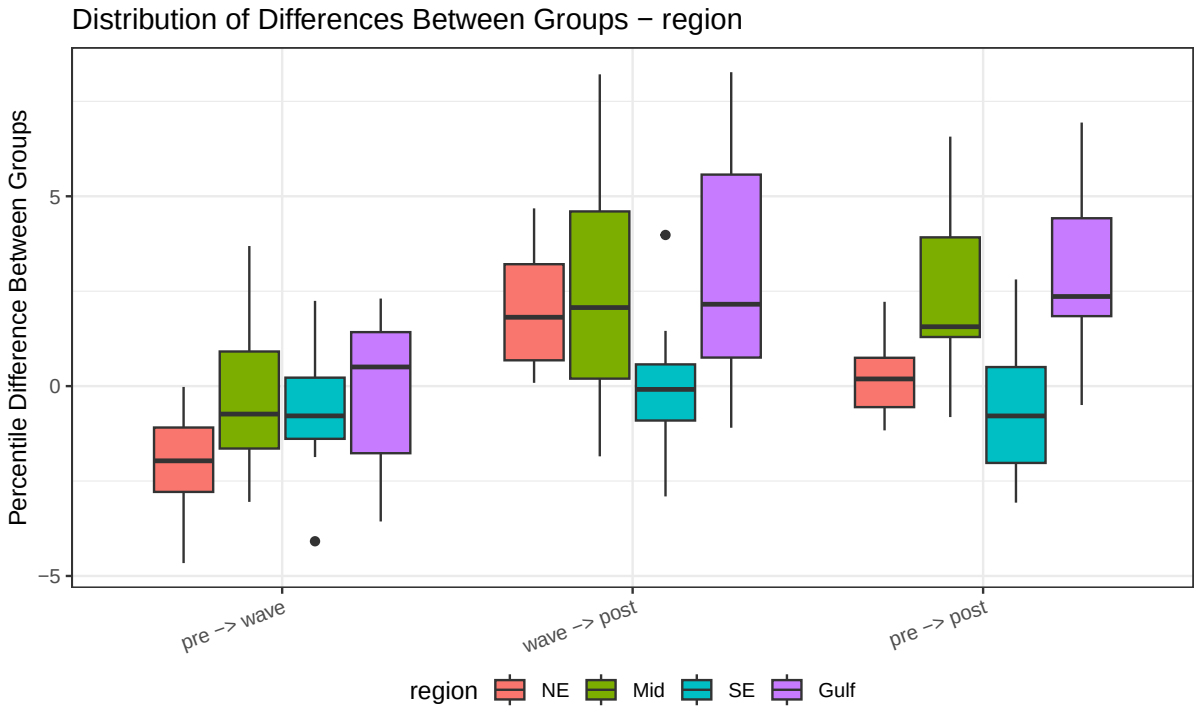

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
8
4
0
−4
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

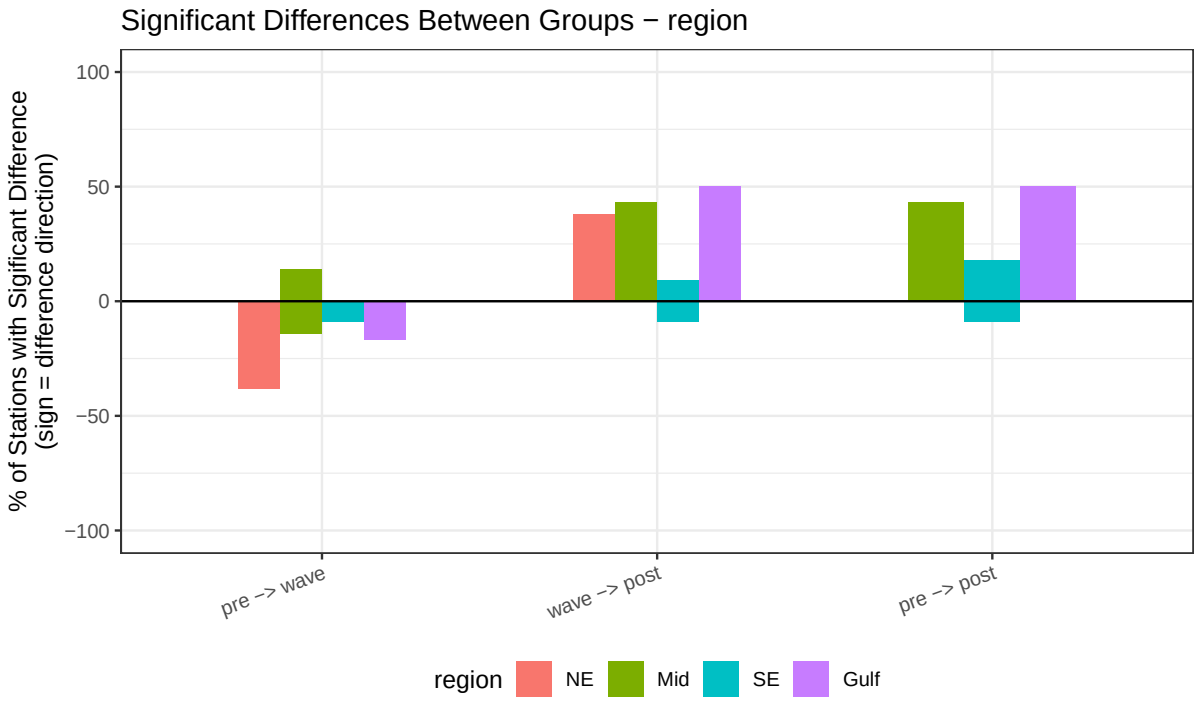

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

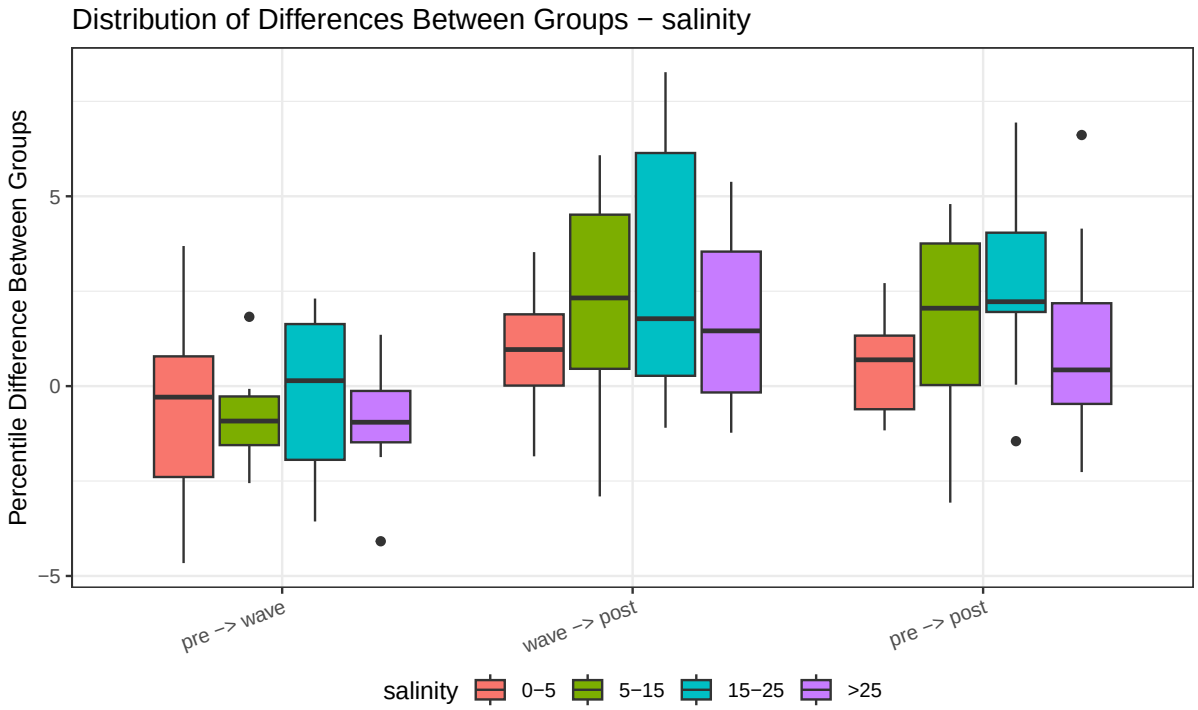
Turbidity

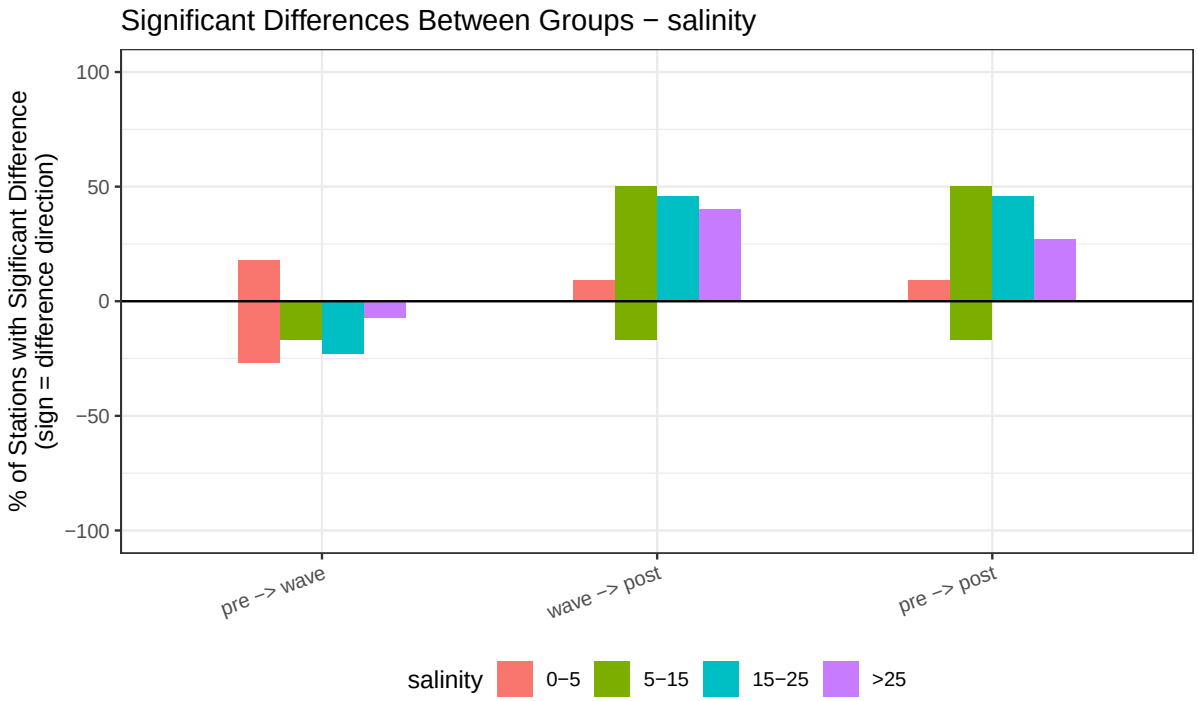
- Turbidity increases post heatwave in all regions but SE
- Turbidity decreases during heatwaves in NE, so the increase in turbidity post-heatwave just returns it to baseline
- Increases post wave in all salinity groups but low




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf


Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf


Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

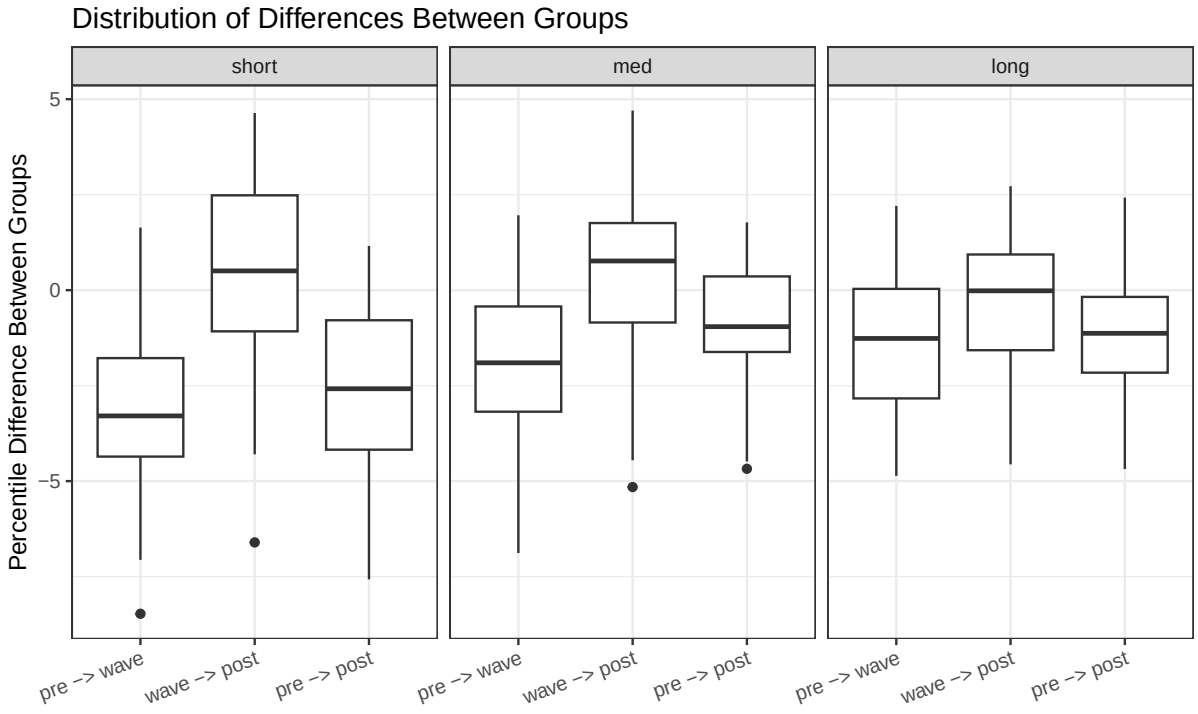

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

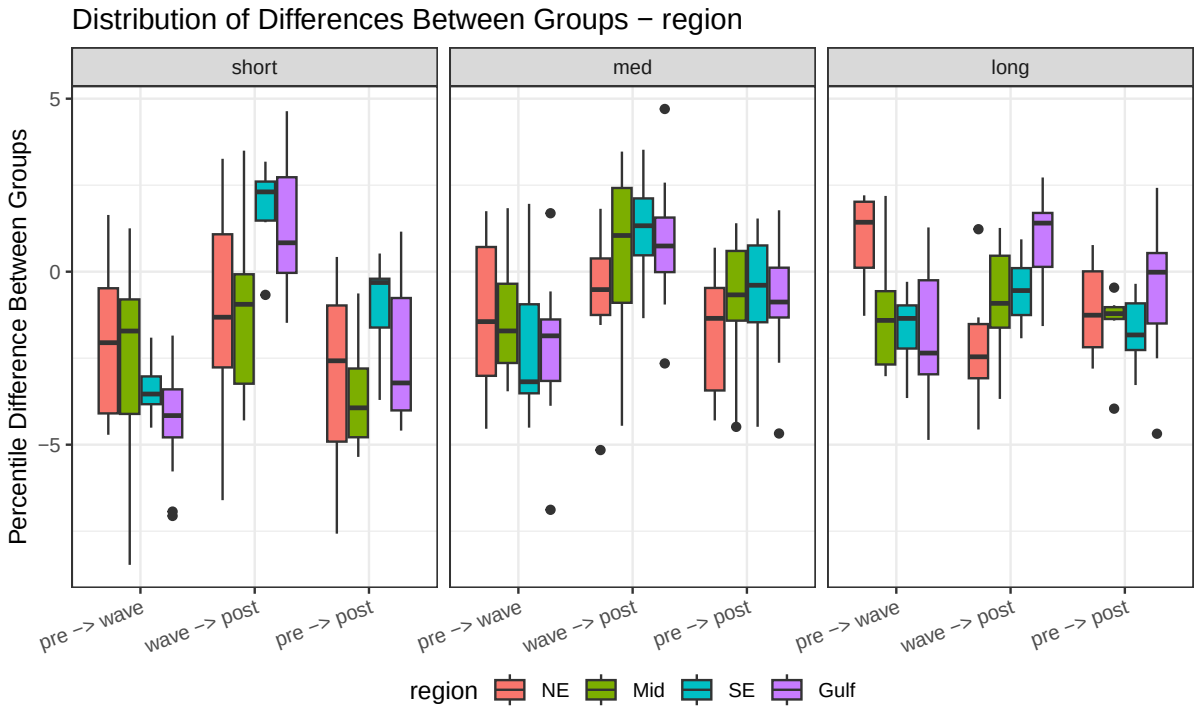
Heatwave Length

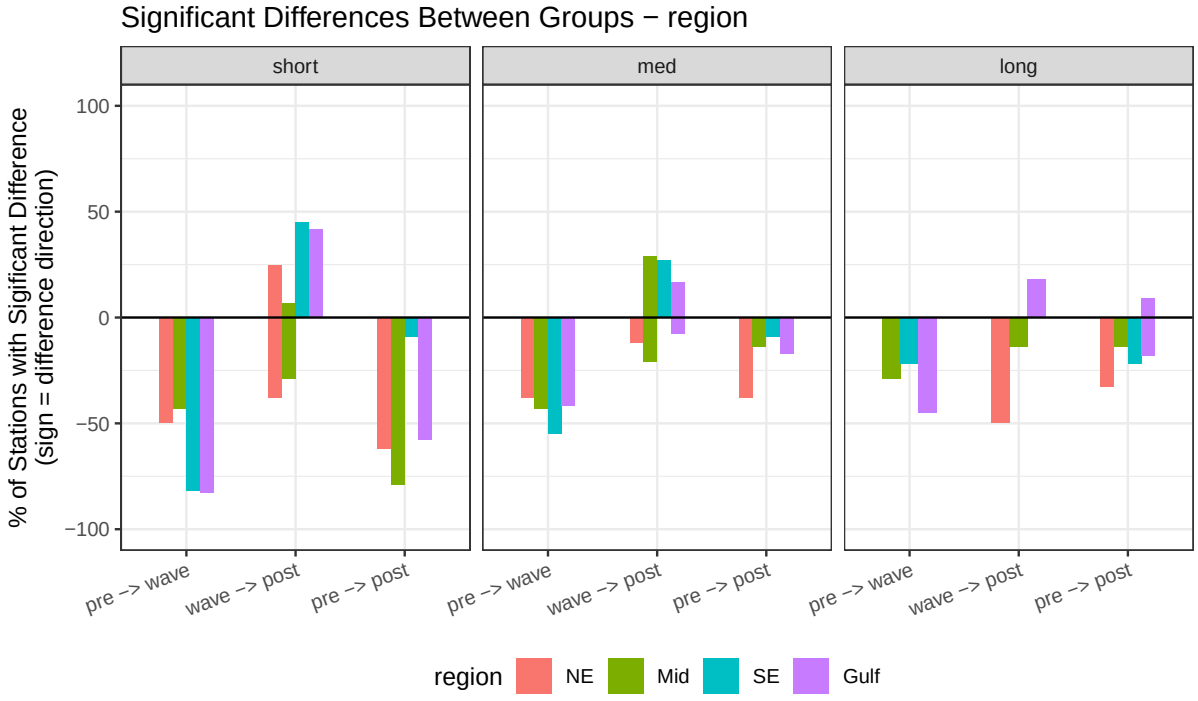
I split heatwaves into 3 groups based on size: 5-10 days (short), 11-15 days (medium), 16+ days (long). I also split post-heatwave groups along these lines.

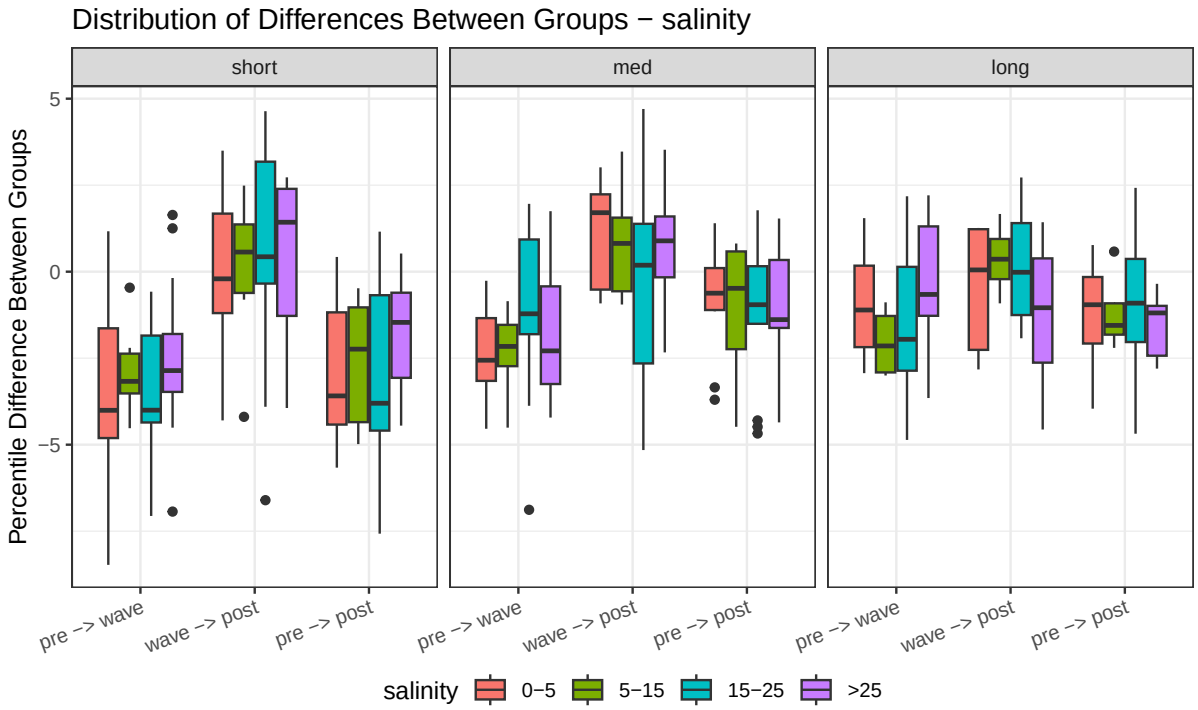
DO

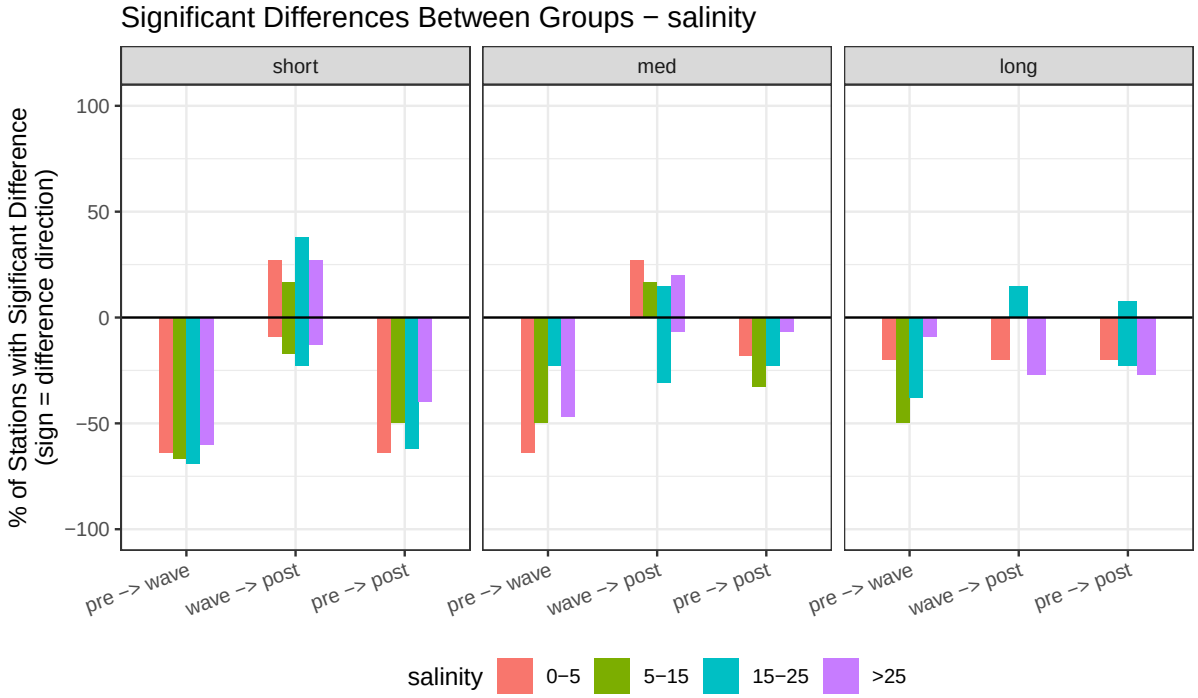
- DO drop during and post-heatwave is most prominent during short heatwaves
- NE DO increases during long heatwaves




Distribution of Differences Between Groups − region
short
med
long
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

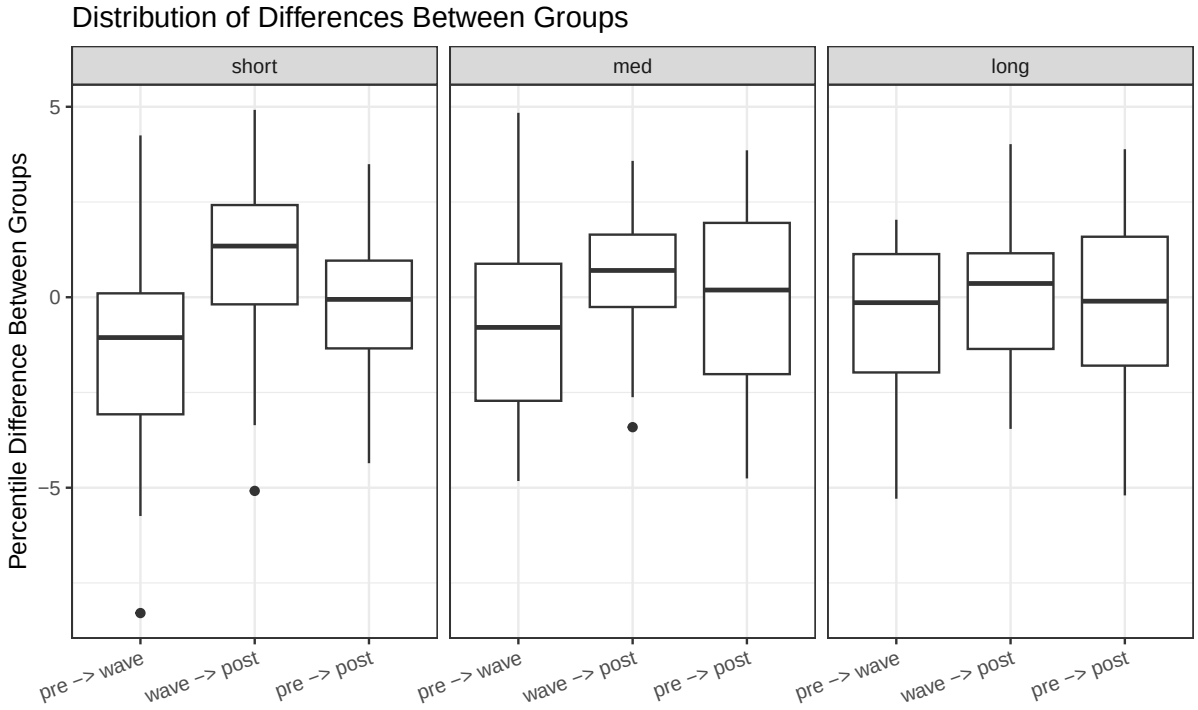

Significant Differences Between Groups − region
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

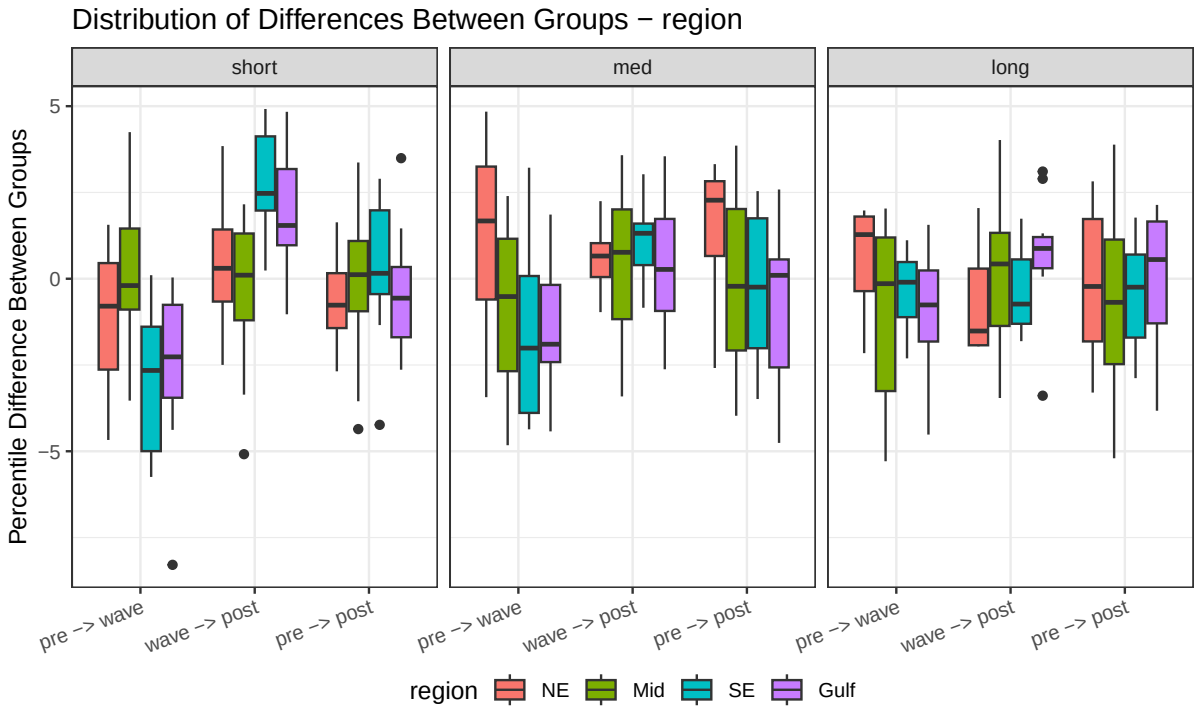

Distribution of Differences Between Groups − salinity
short
med
long
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

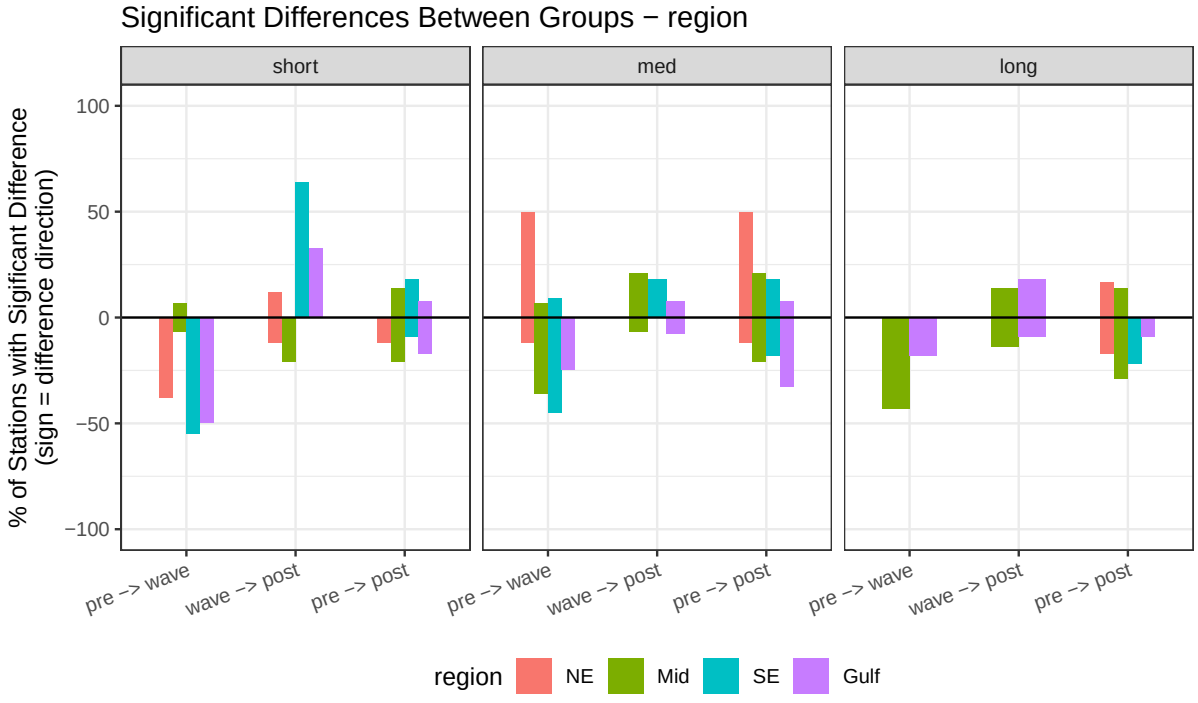

Significant Differences Between Groups − salinity
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

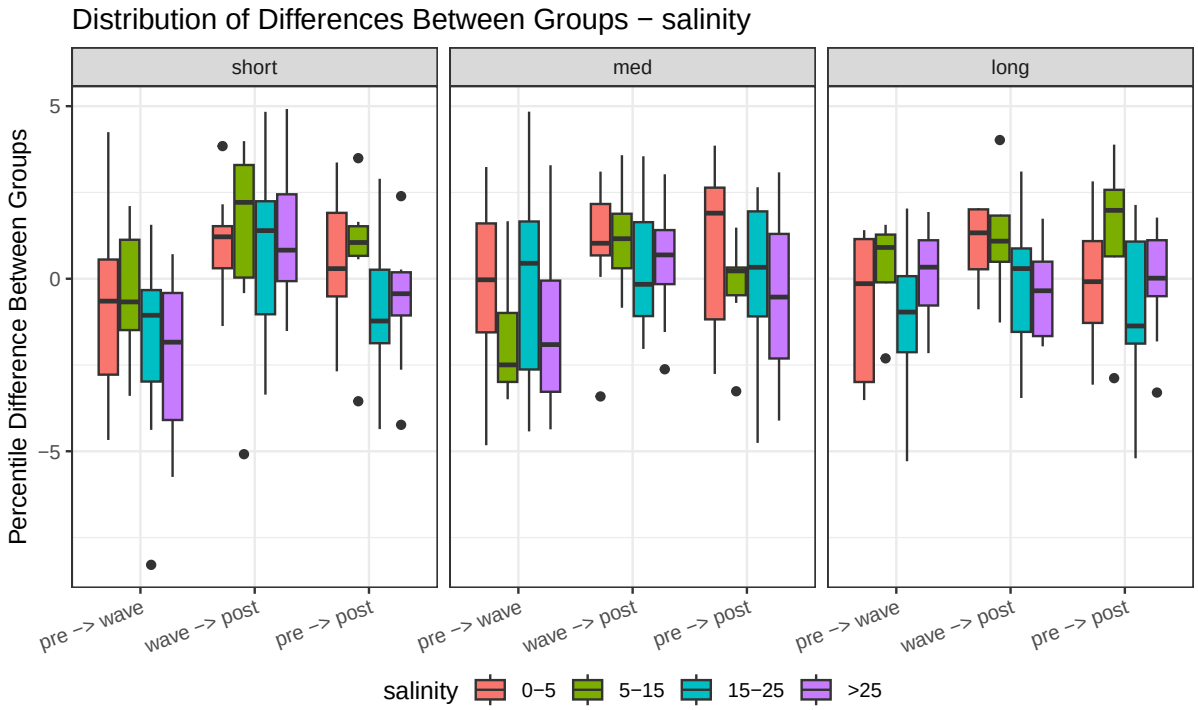
pH

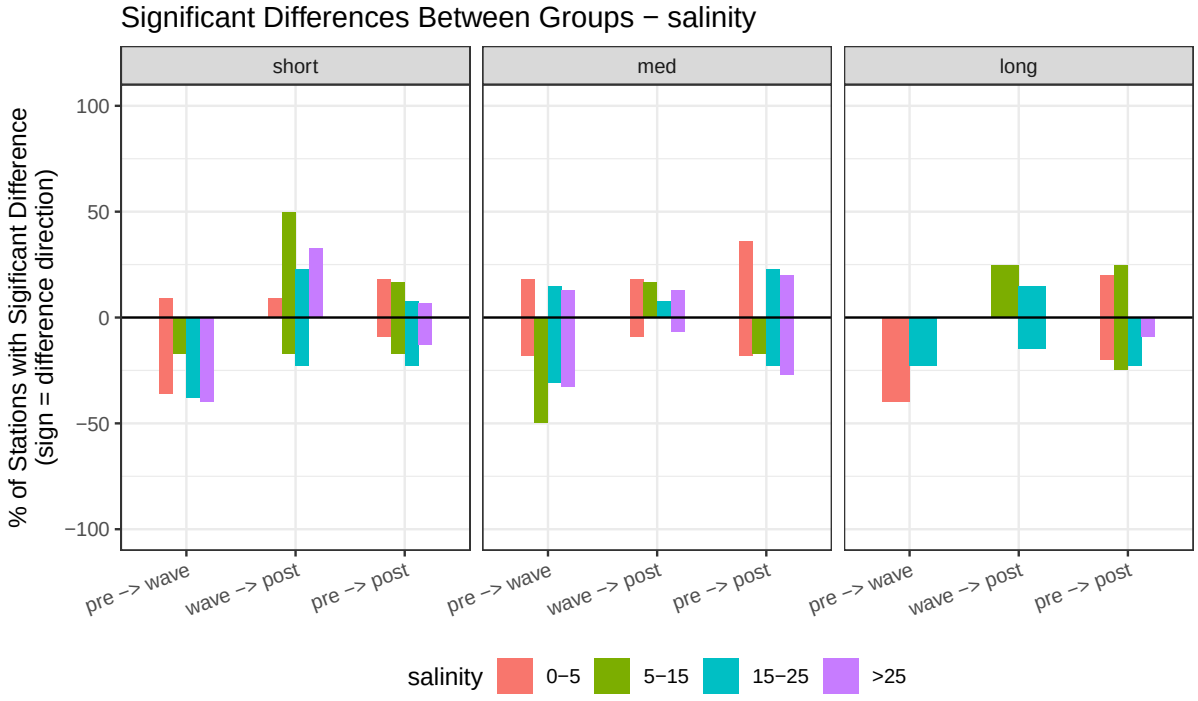
- pH drop during heatwaves is mostly present during short waves
- Inconsistent patterns with pH within most regions, particularly during medium and long waves




Distribution of Differences Between Groups − region
short
med
long
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

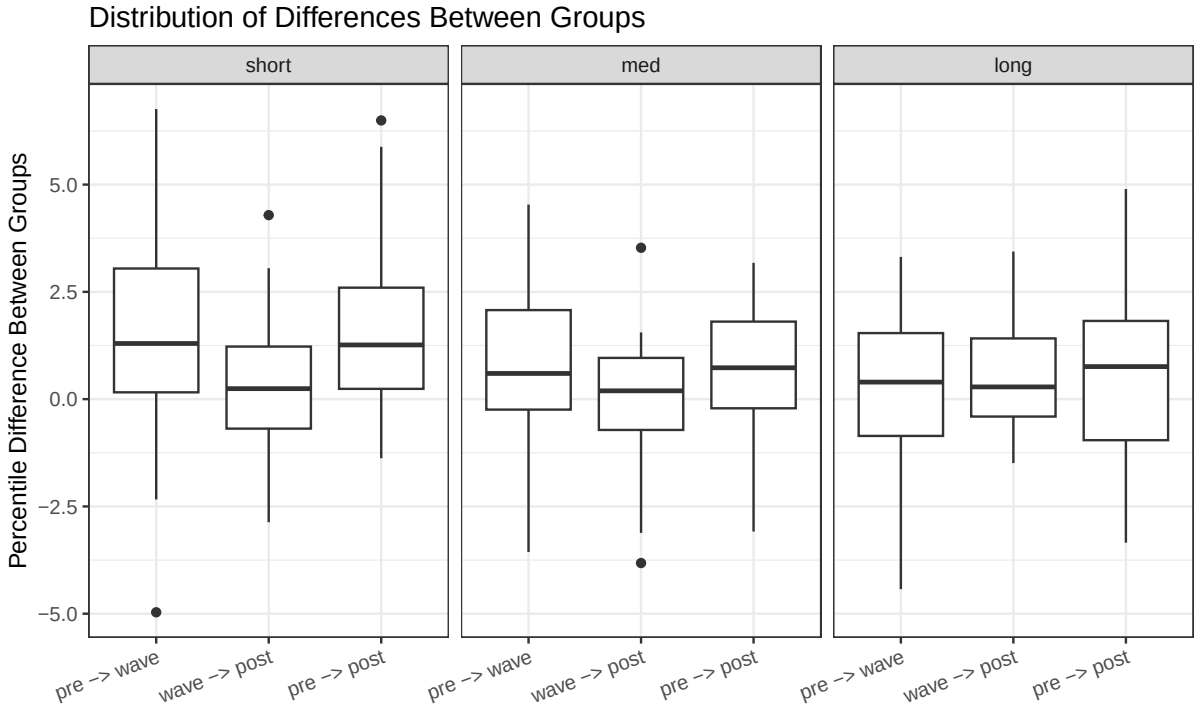

Significant Differences Between Groups − region
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

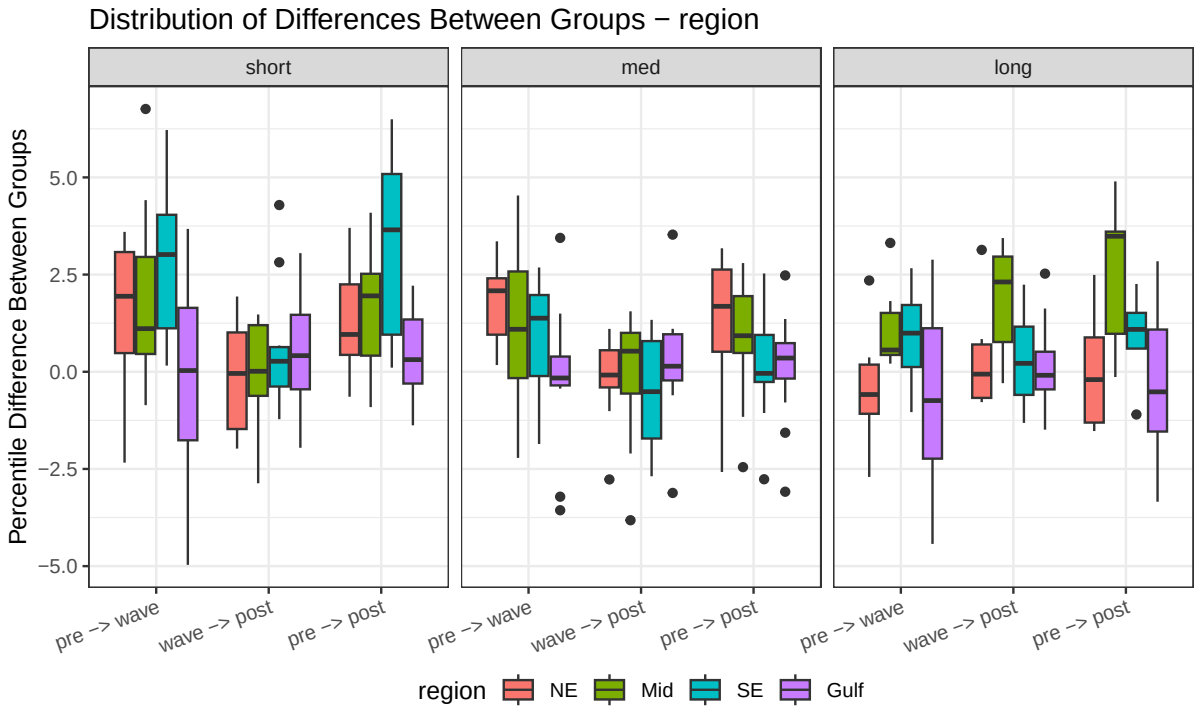

Distribution of Differences Between Groups − salinity
short
med
long
Percentile Difference Between Groups
5
0
−5
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

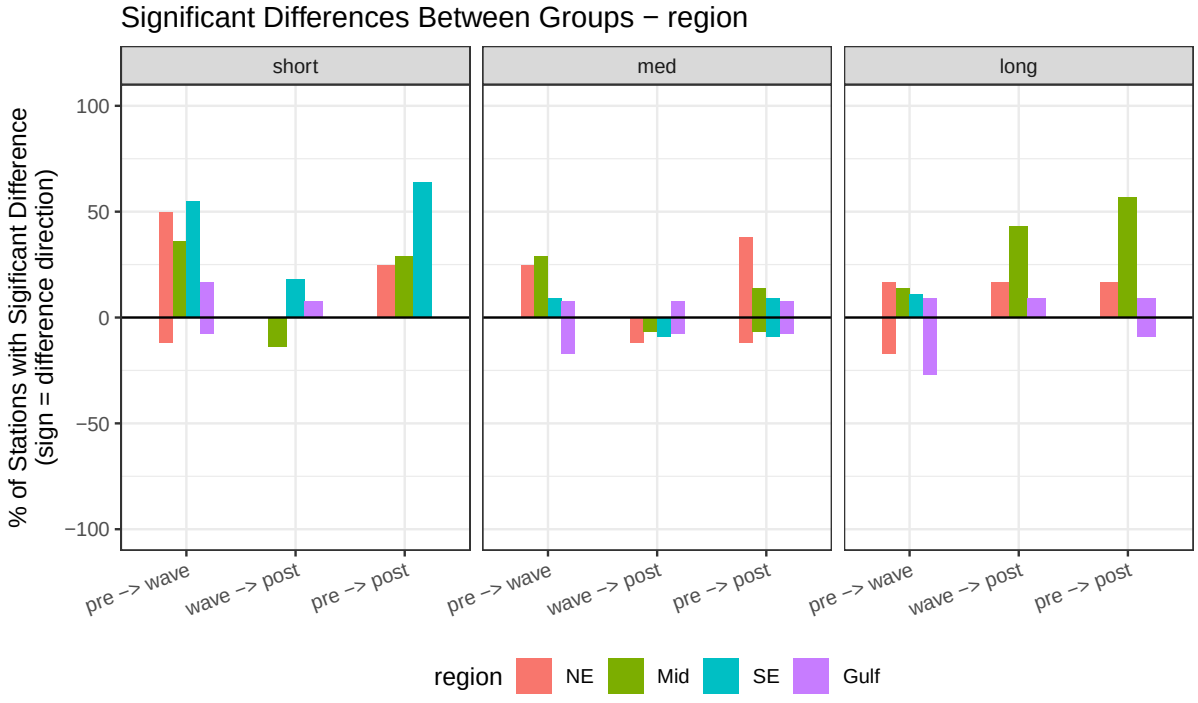

Significant Differences Between Groups − salinity
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

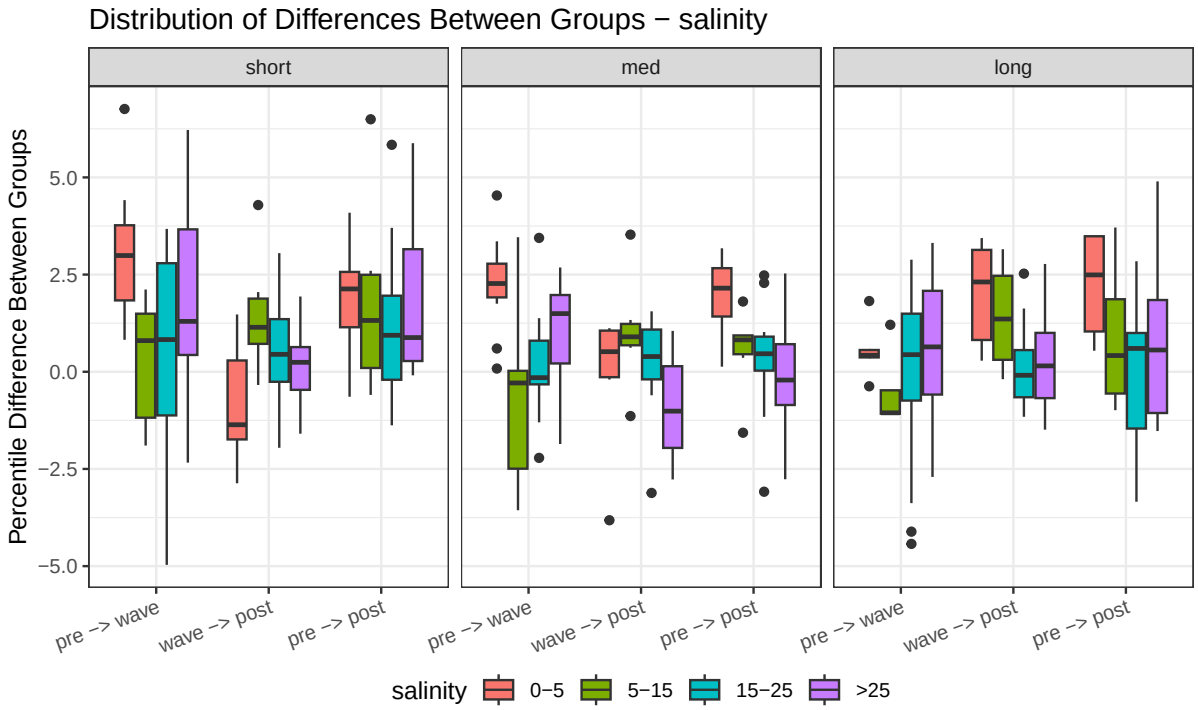
Conductivity

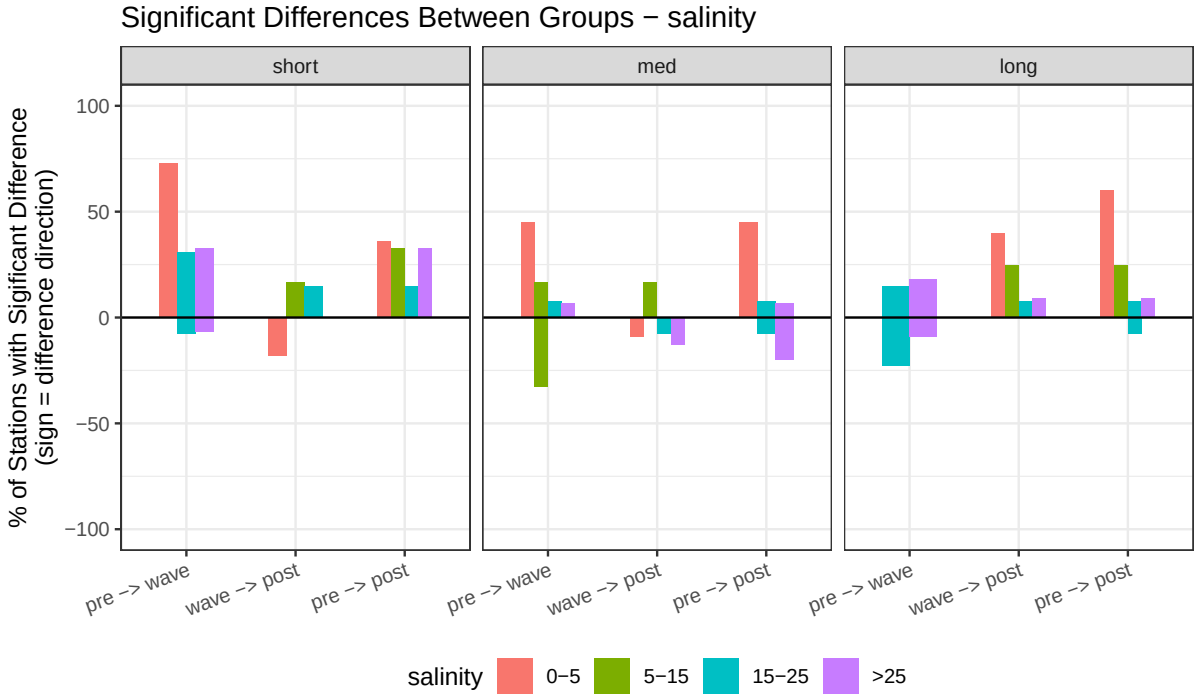
- Conductivity is higher during and post short waves, except in Mid region where it is high during and post wave regardless of wave length.
- In NE, higher during short and medium waves, but not high.
- In low salinity sites, post heatwave increase persists across all heatwave lengths (fades with heatwave length for high salinities)




Distribution of Differences Between Groups − region
short
med
long
Percentile Difference Between Groups
5.0
2.5
0.0
−2.5
−5.0
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

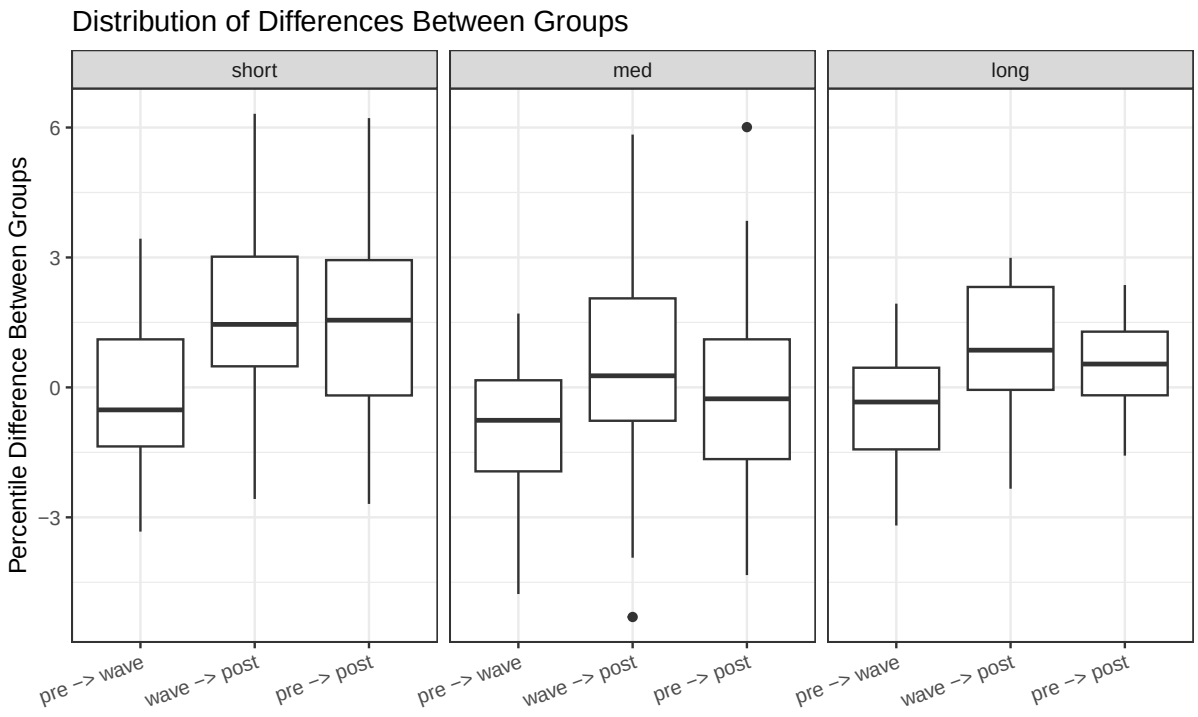

Significant Differences Between Groups − region
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf

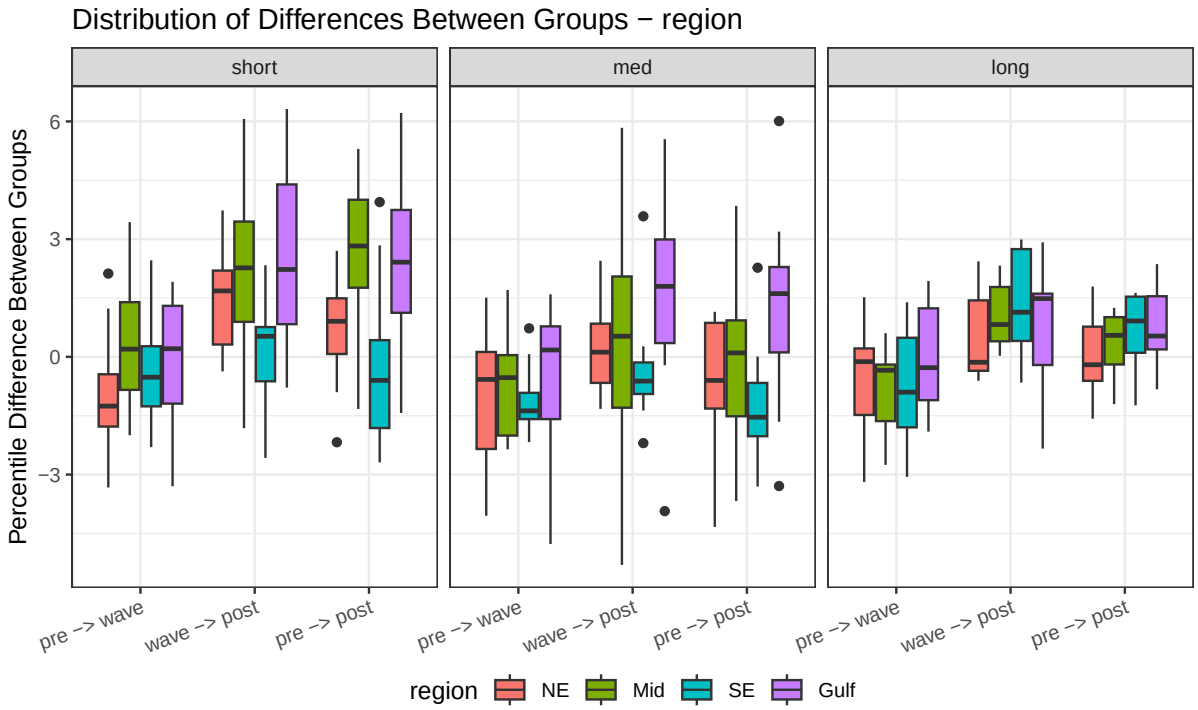

Distribution of Differences Between Groups − salinity
short
med
long
Percentile Difference Between Groups
5.0
2.5
0.0
−2.5
−5.0
pre –> wave
wave –> post
pre –> post
salinity
0−5
5−15
15−25
>25

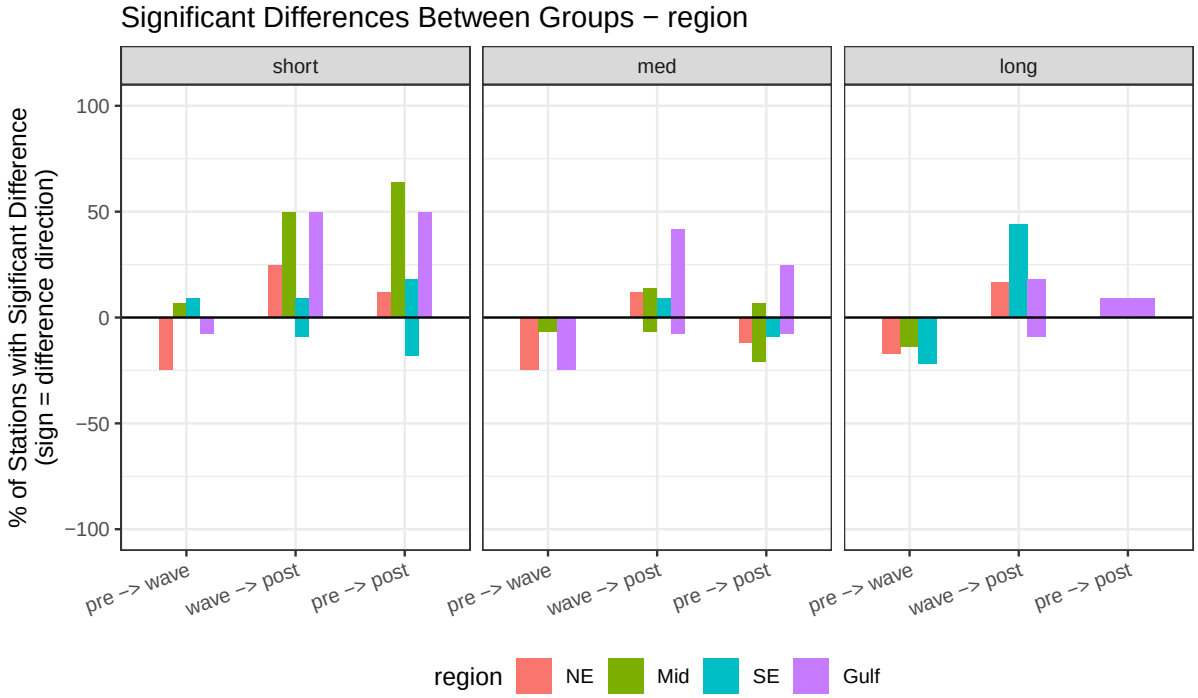

Significant Differences Between Groups − salinity
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0−5
5−15
15−25
>25

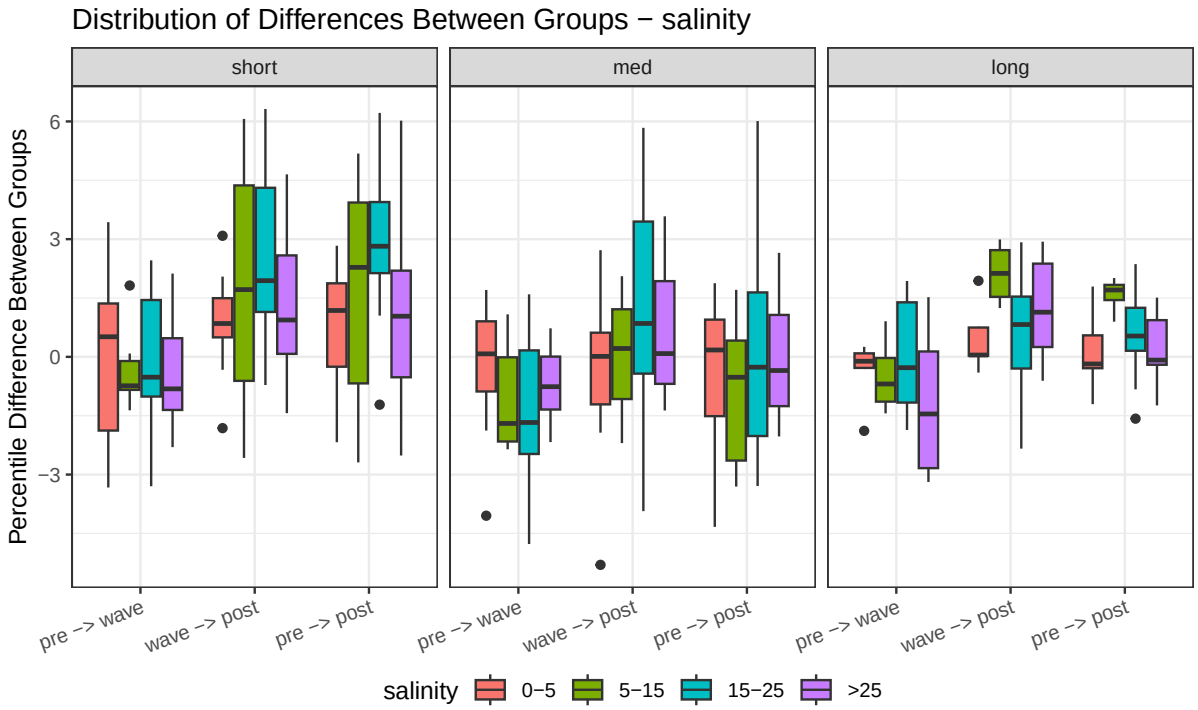
Turbidity

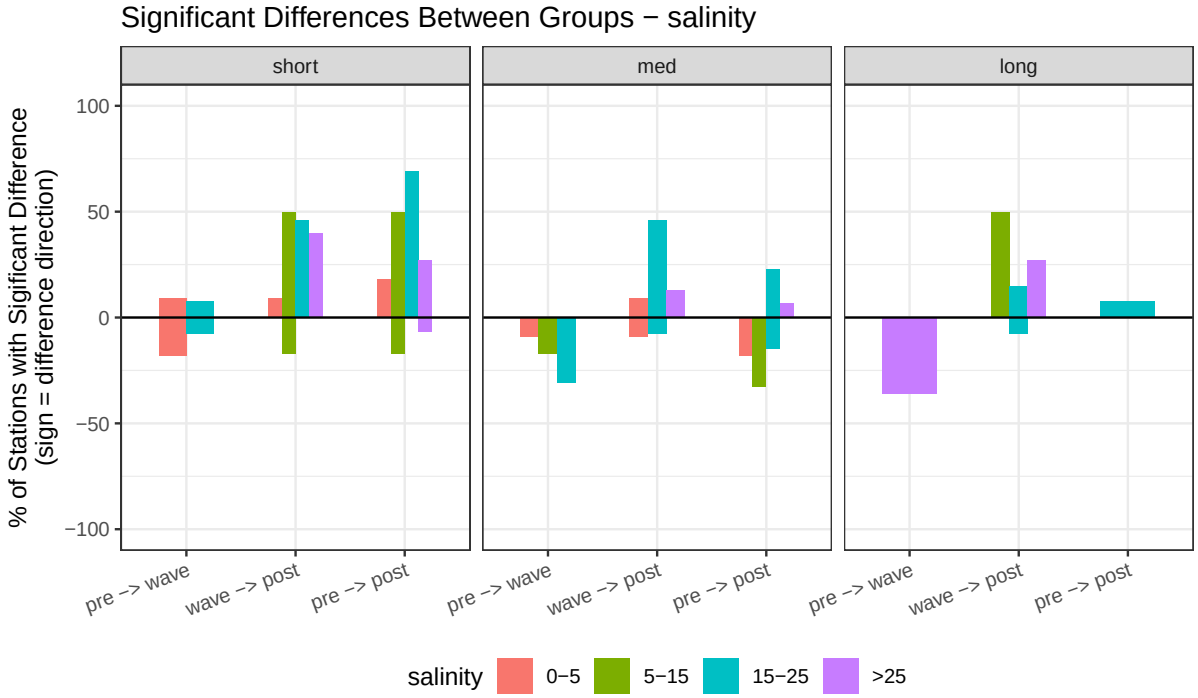
- Turbidity increase during and following heatwaves in most prominent in short waves
- Gulf sites also show turbidity increase during and post medium waves
- Some signs of turbidity decreasing during and post medium and long waves in most regions
- Turbidity increase is strongest in medium salinity groups




Distribution of Differences Between Groups − region
short
med
long
Percentile Difference Between Groups
6
3
0
−3
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf


Significant Differences Between Groups − region
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
region
NE
Mid
SE
Gulf


Distribution of Differences Between Groups − salinity
short
med
long
Percentile Difference Between Groups
6
3
0
−3
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

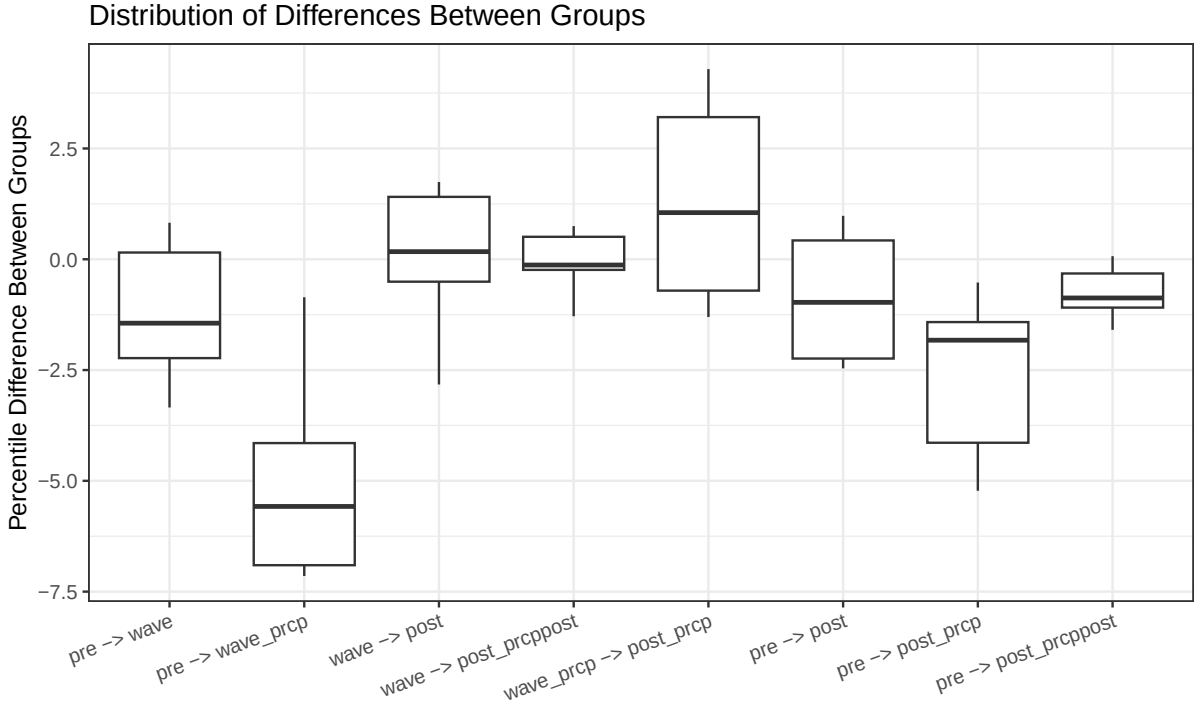

Significant Differences Between Groups − salinity
short
med
long
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
wave –> post
pre –> post
salinity
0–5
5–15
15–25
>25

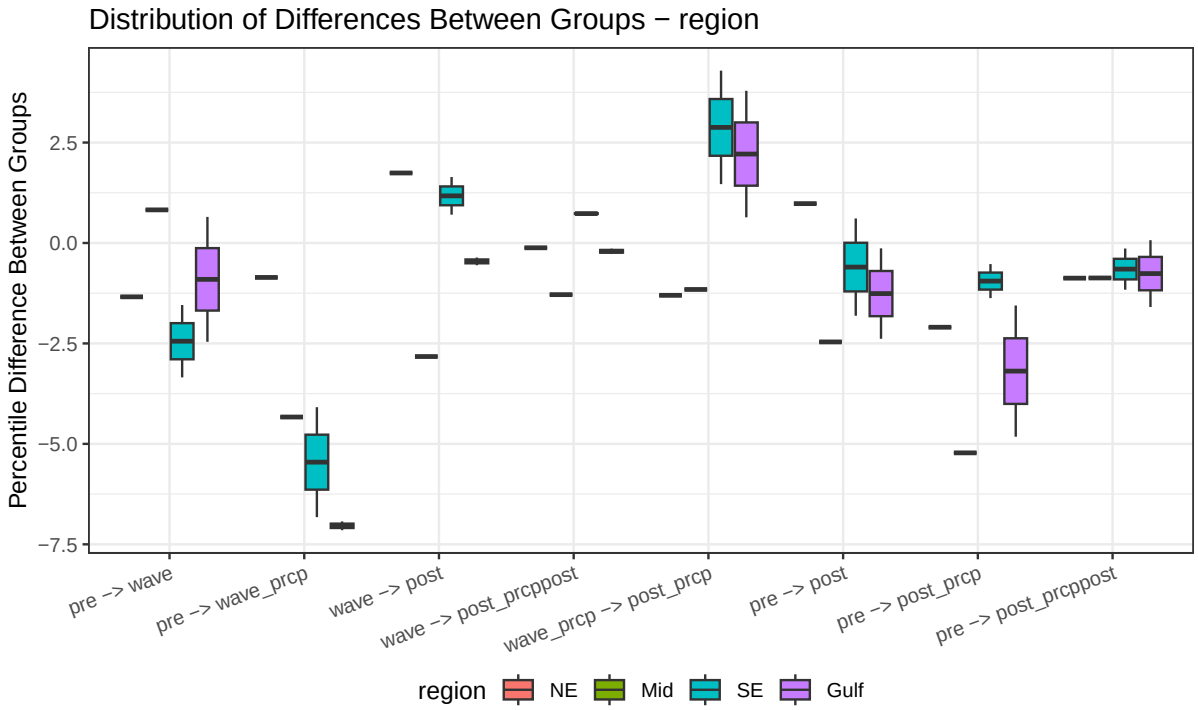
Heatwaves and Precipitation

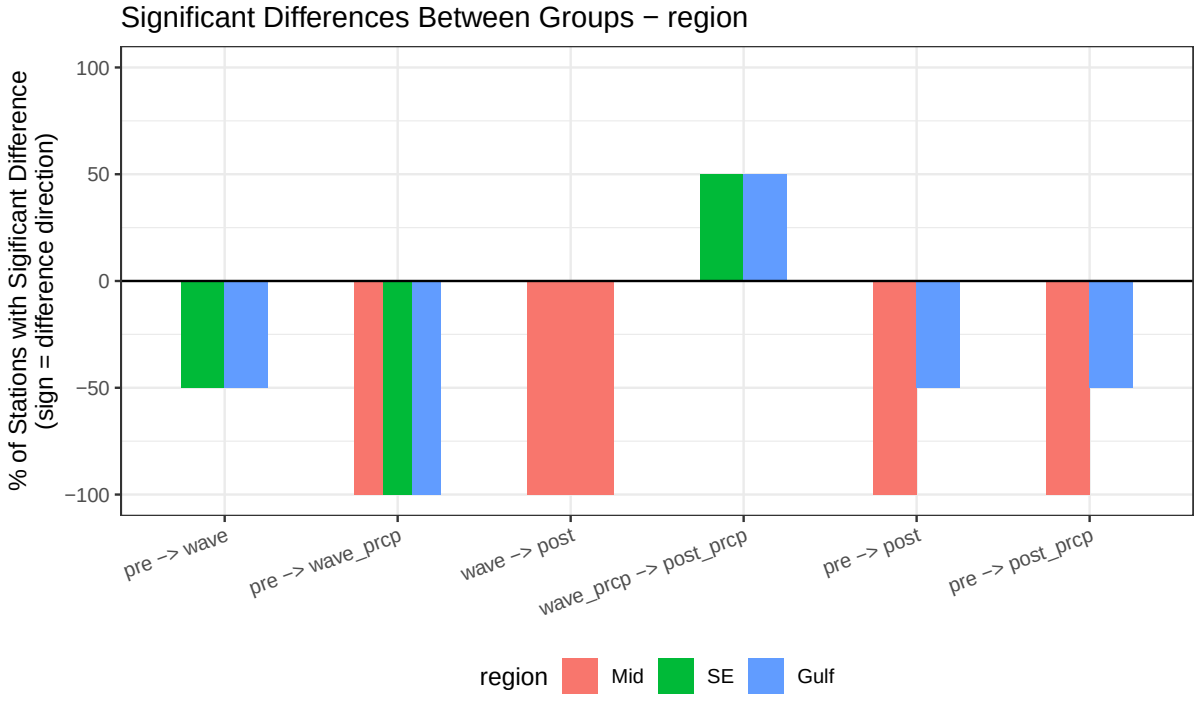
I split heatwave days into two groups: waves where a large precipitation event (90+ percentile) occurred and those without. I also split post-heatwave groups along this line, and added an additional group, for when a large precipitation event occurred in the post heatwave period but not during the heatwave. I only included stations with a meteorological station at the same site (6 stations: "apaeb", "cbvtc", "gtmpc", "marce", "narpc", "niwol"). Because there is only 1 station in NE and Mid, the regional plots may be capturing site differences more than regional differences.

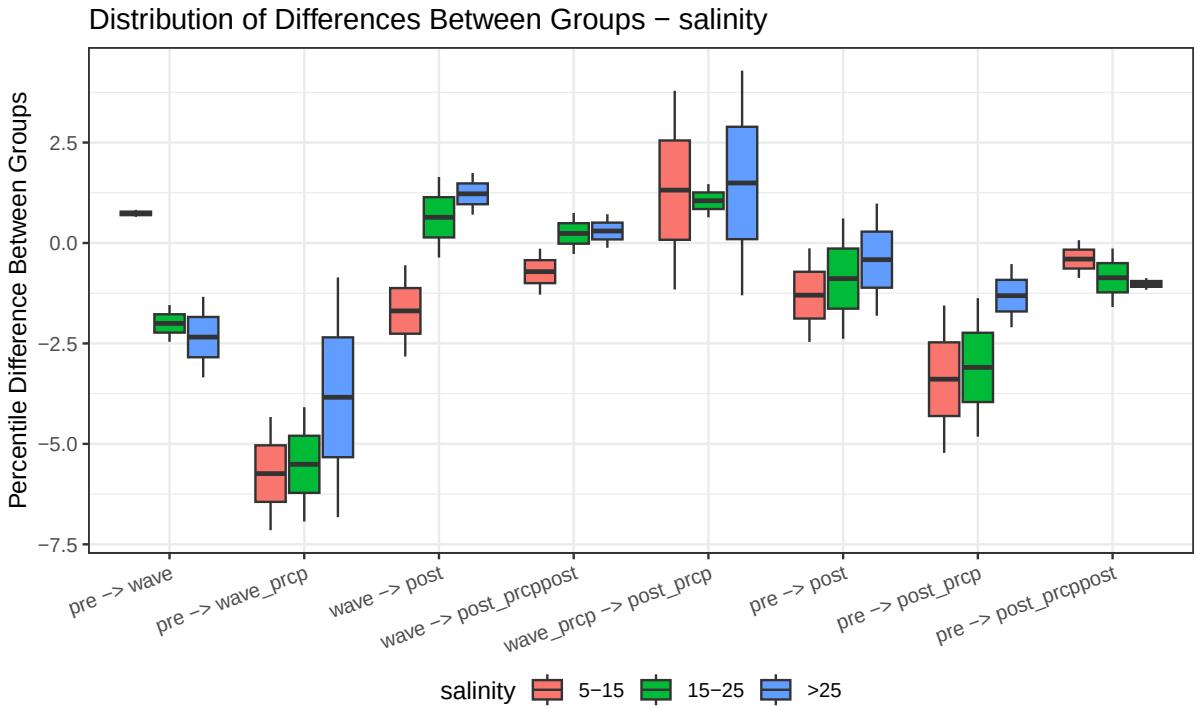
DO

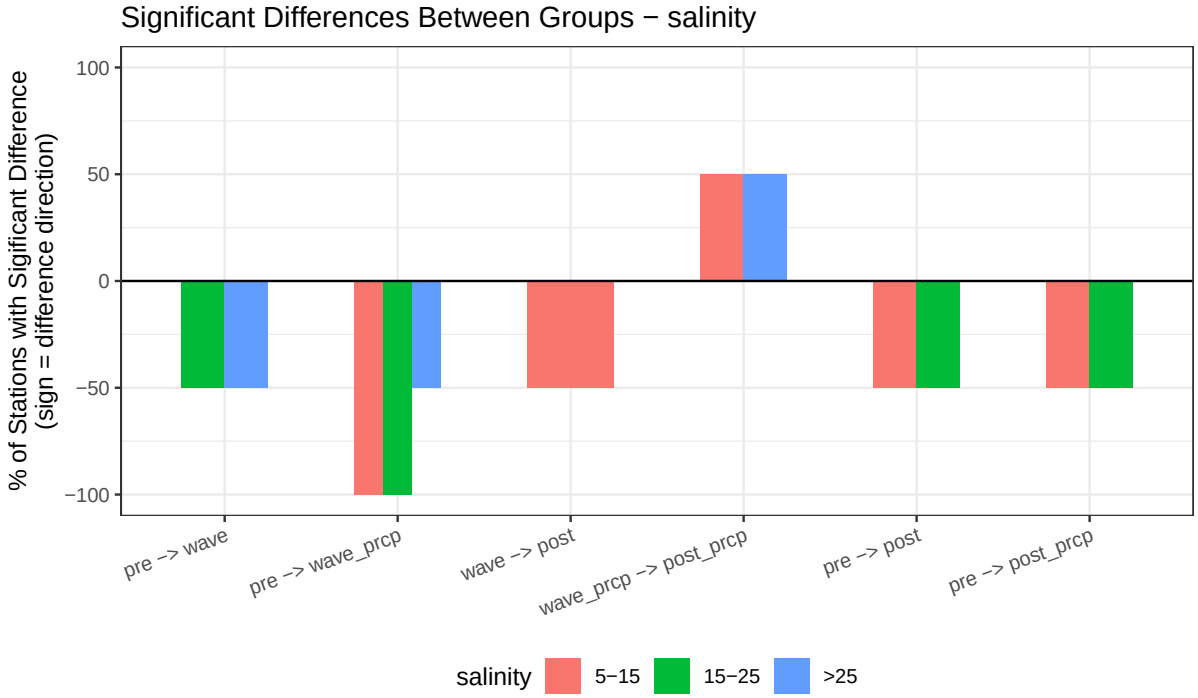
- Heatwaves with an extreme precipitation event had much larger DO drops (not in NE)
- In some cases, DO rebounds following the heatwave with precipitation (SE, Gulf), but in others (mid) it stays low. The rebound in Gulf sites also typically doesn't make up for the initial drop, leading to lower DO post heatwave with precipitation




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
2.5
0.0
−2.5
−5.0
−7.5
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
region
NE
Mid
SE
Gulf

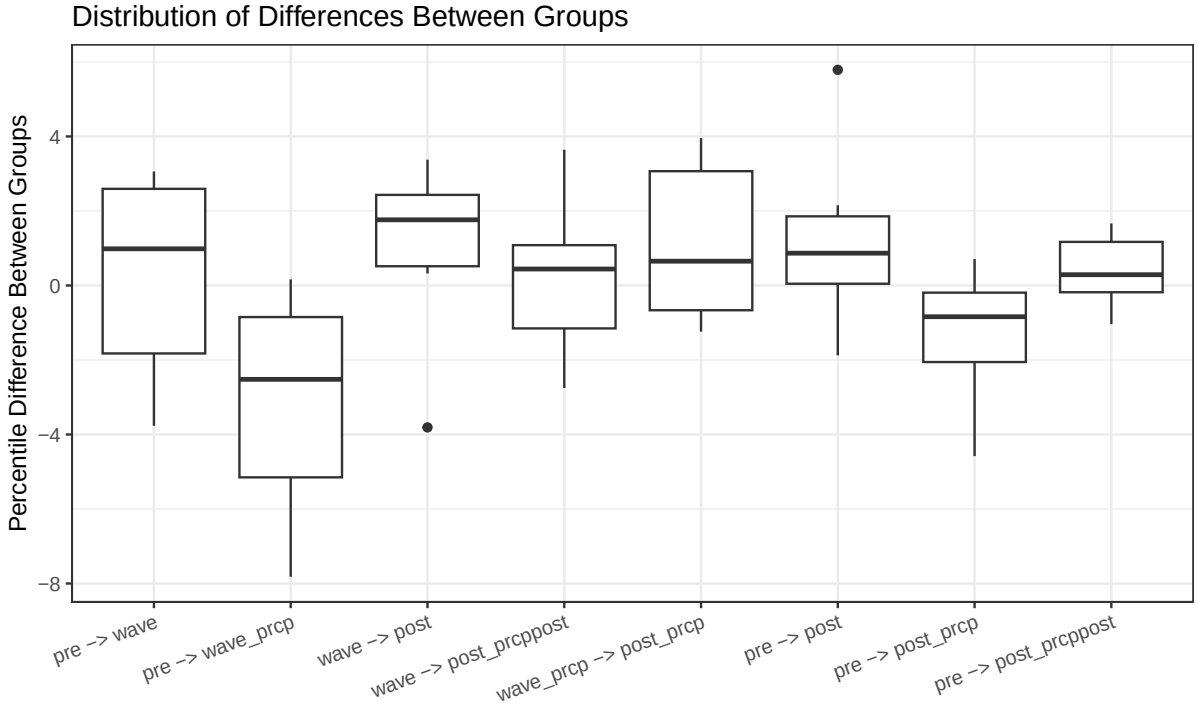

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
pre –> wave_prcp
wave –> post
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
region
Mid
SE
Gulf

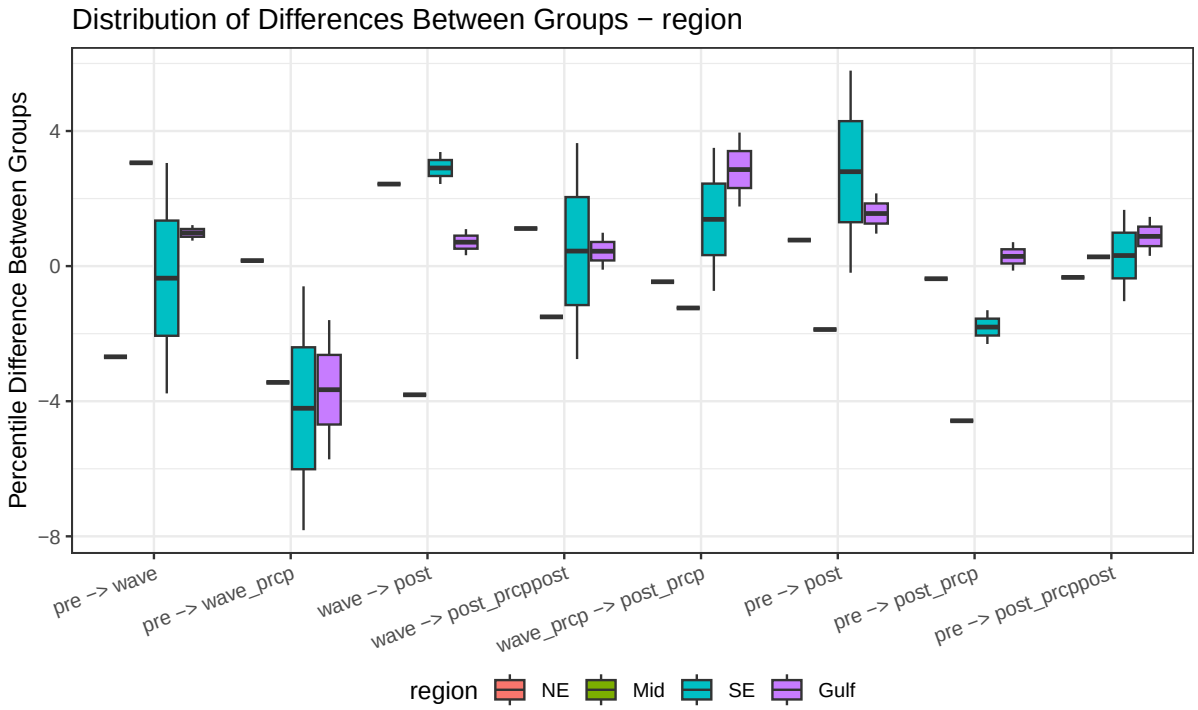

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
2.5
0.0
−2.5
−5.0
−7.5
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
salinity
5−15
15−25
>25

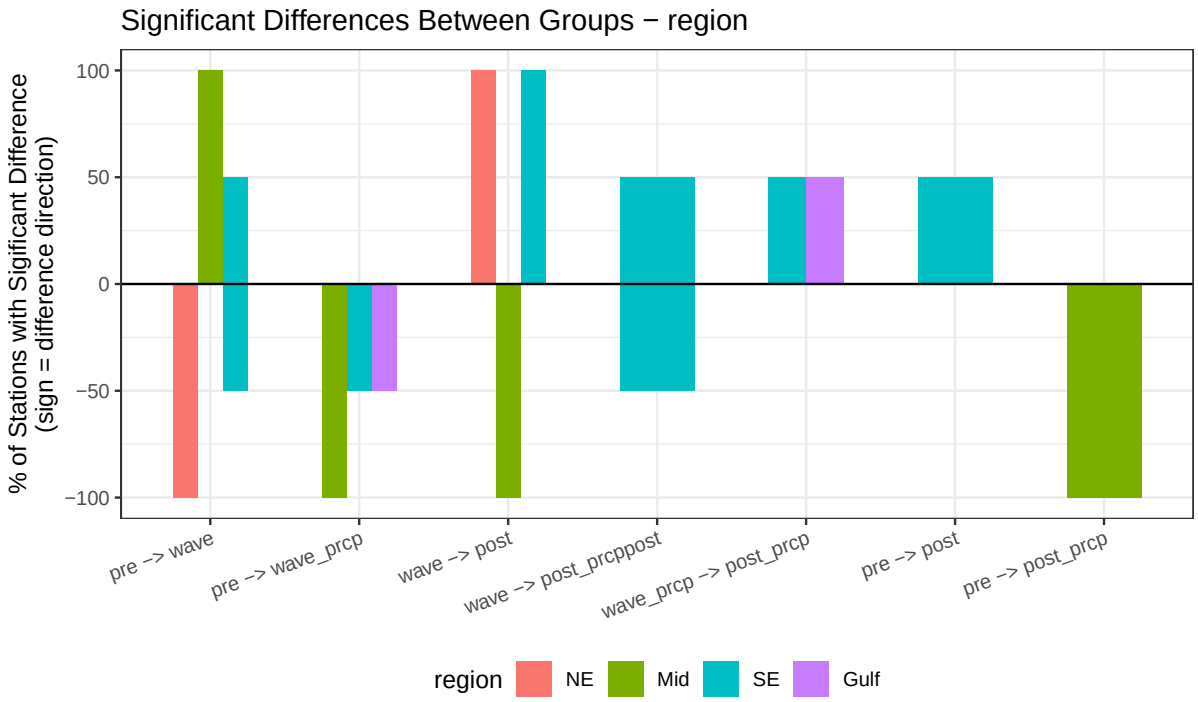

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
pre –> wave_prcp
wave –> post
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
salinity
5−15
15−25
>25

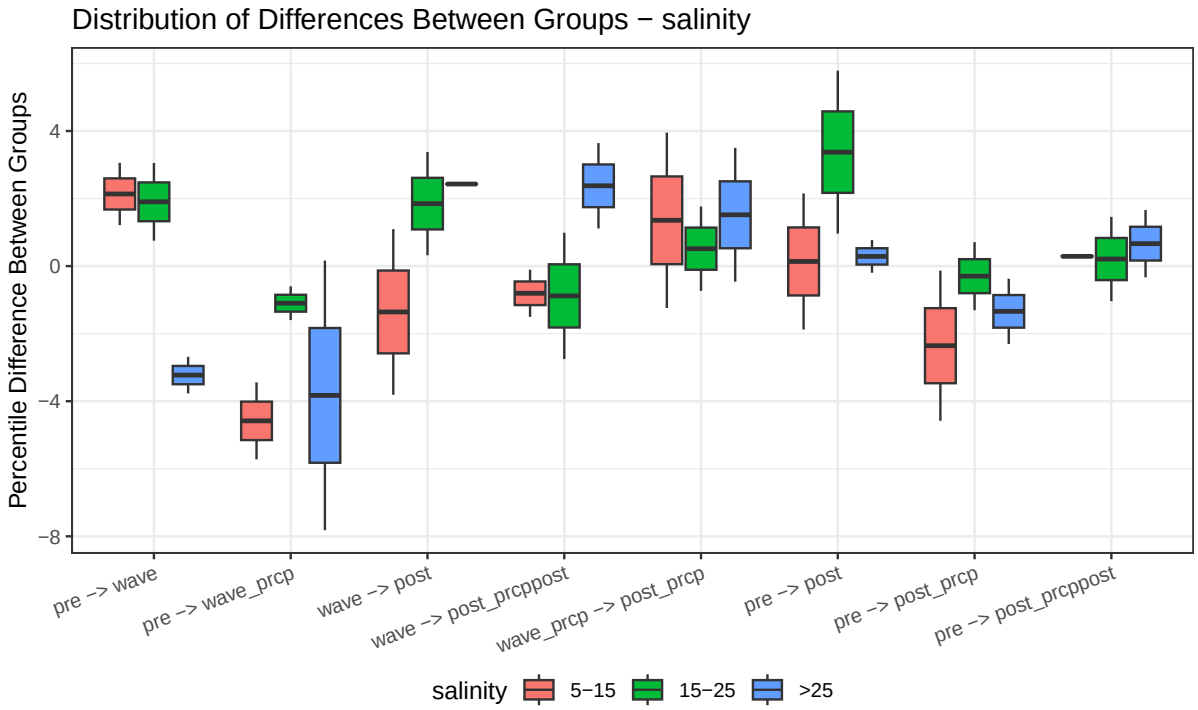
pH

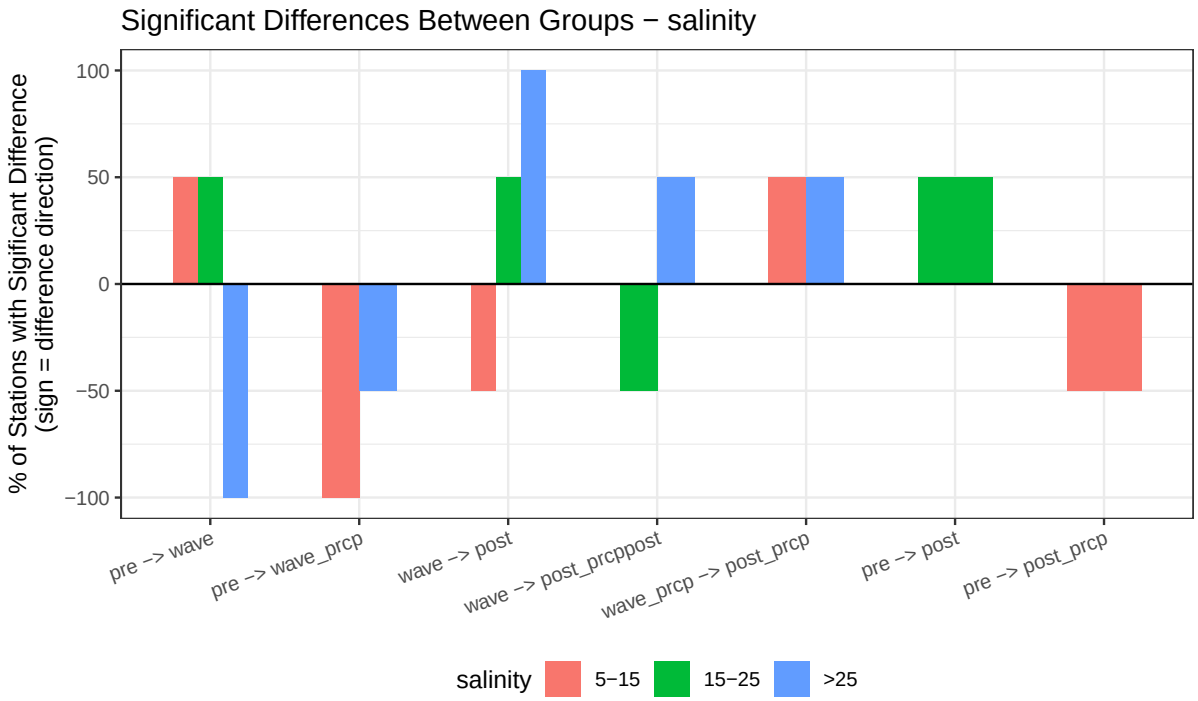
- pH is lower during heatwaves with an extreme precipitation event (outside of NE)
- Following heatwaves with extreme precipitation, pH rebounds in SE and Gulf
- In Mid, pH increases during heatwaves without precip, but decreases during waves with precip
- No clear sign that pH is different post-heatwave compared to pre-heatwave
- pH increases during heatwaves without precip in moderate salinity sites, decreases during precip heatwaves
- Decreases during all heatwaves at high salinity sites




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
4
0
−4
−8
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
region
NE
Mid
SE
Gulf

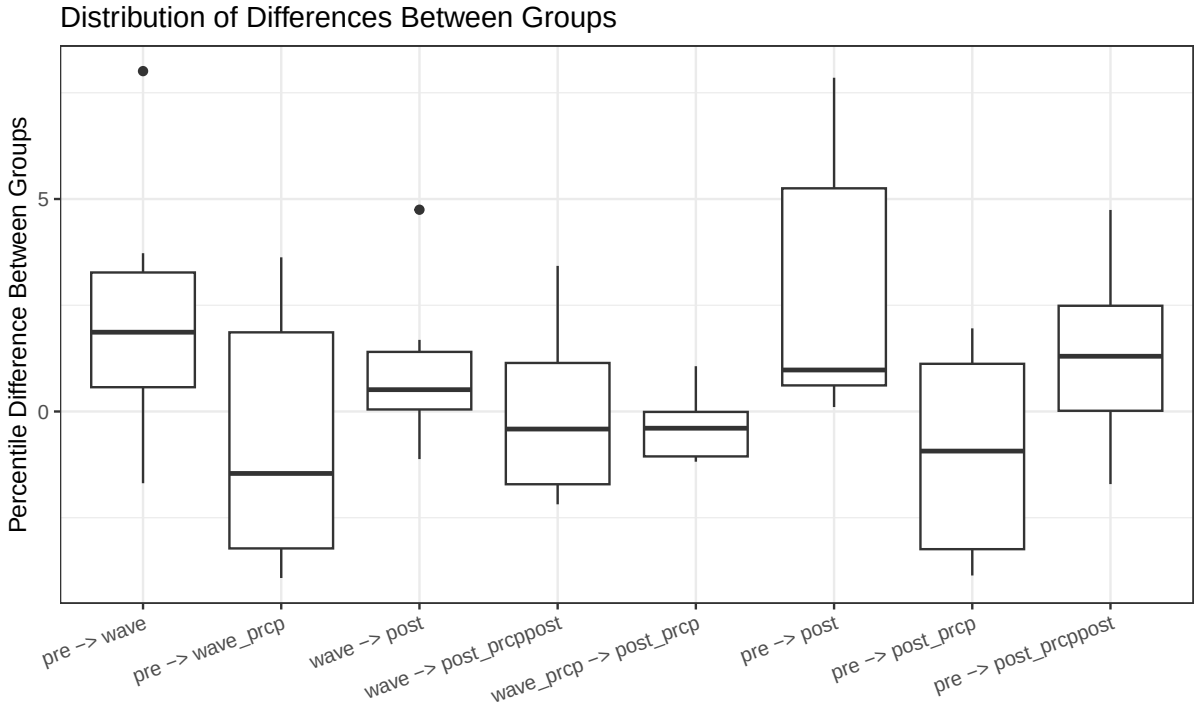

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
region
NE
Mid
SE
Gulf

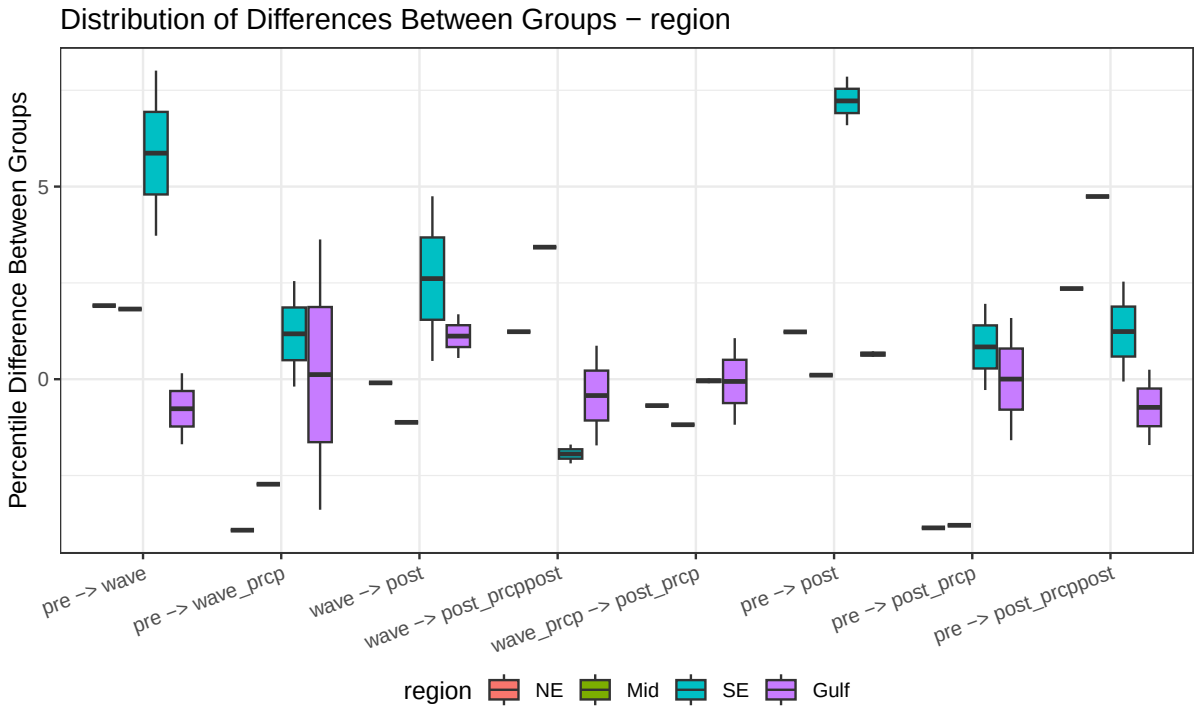

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
4
0
−4
−8
pre −> wave
pre −> wave_prcp
wave −> post
wave −> post_prcppost
wave_prcp −> post_prcp
pre −> post
pre −> post_prcp
pre −> post_prcppost
salinity
5−15
15−25
>25

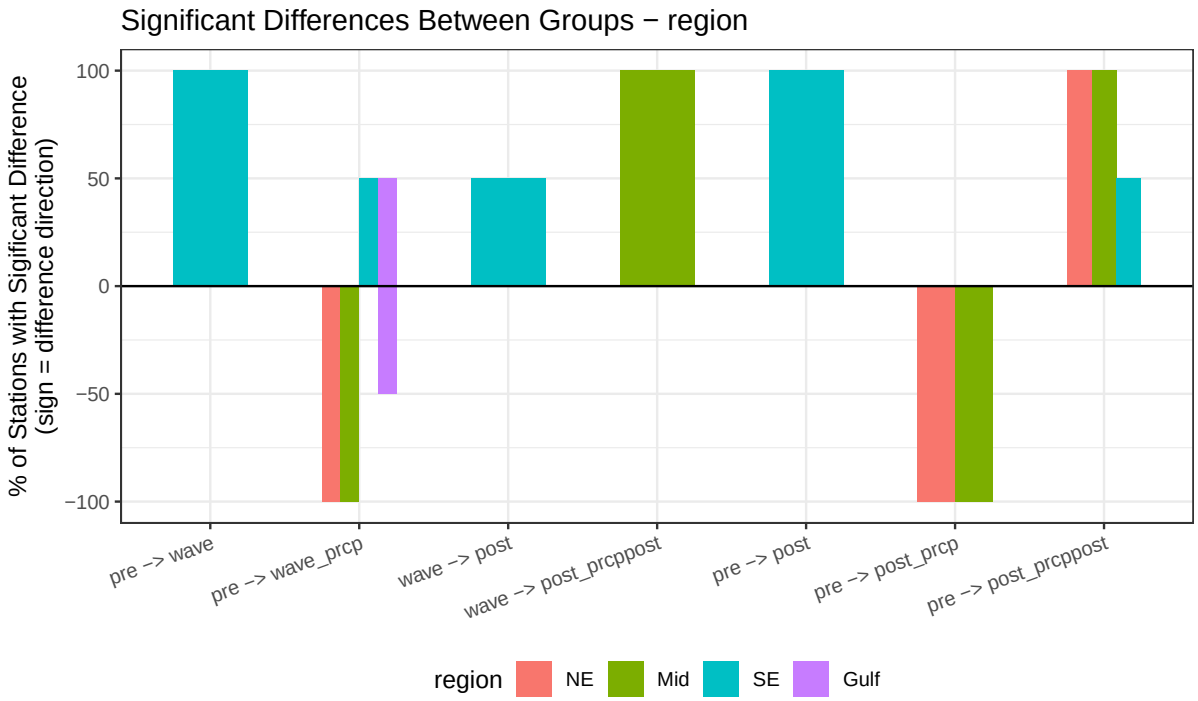

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre −> wave
pre −> wave_prcp
wave −> post
wave −> post_prcppost
wave_prcp −> post_prcp
pre −> post
pre −> post_prcp
salinity
5−15
15−25
>25

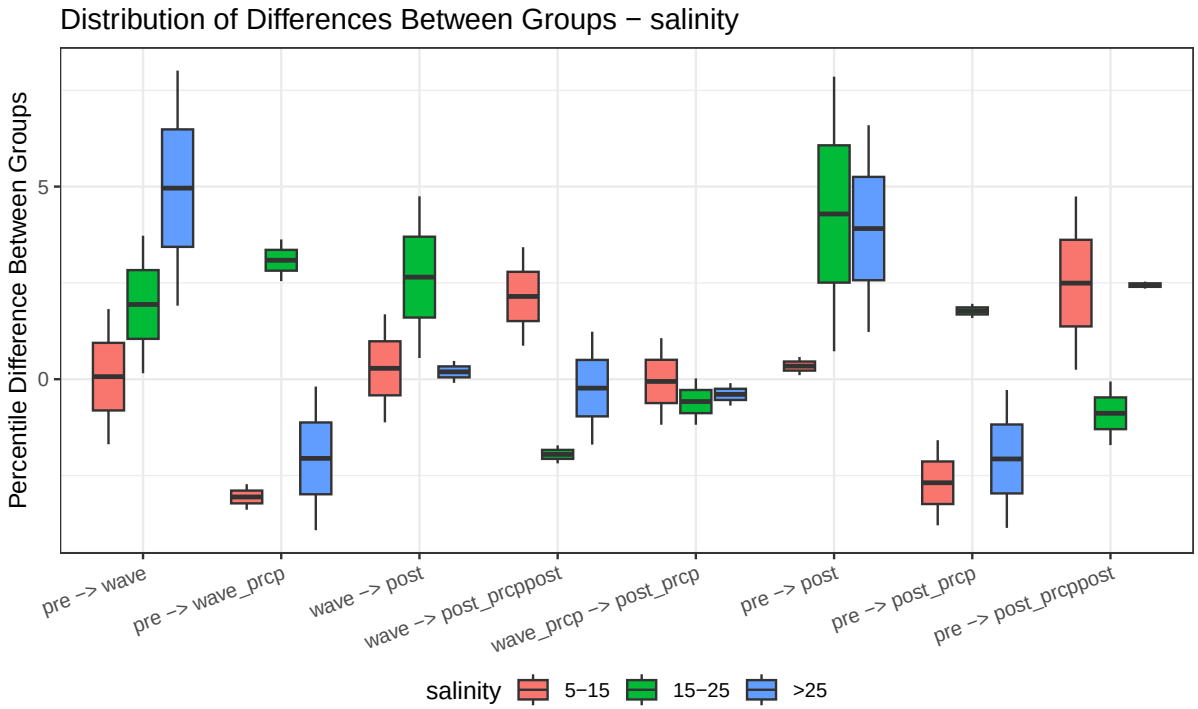
Conductivity

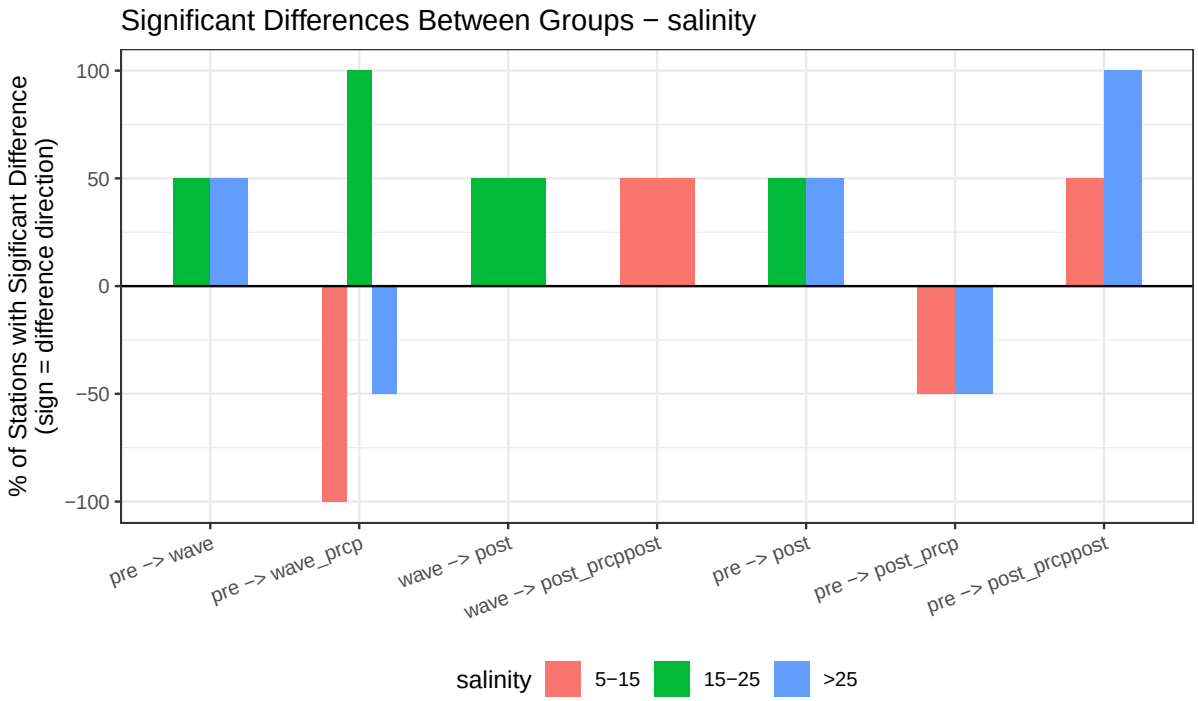
- Conductivity is higher during and post heatwaves without large precipitation events in SE
- Conductivity is lower during and post heatwaves with large precipitation events in NE and Mid
- Non-precip heatwaves had increases in conductivity that were higher at high salinity sites
- While conductivty dropped at lower and higher salinity sites during precip heatwaves, it still increased at moderate salinity sites (weird)




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
5
0
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
region
NE
Mid
SE
Gulf

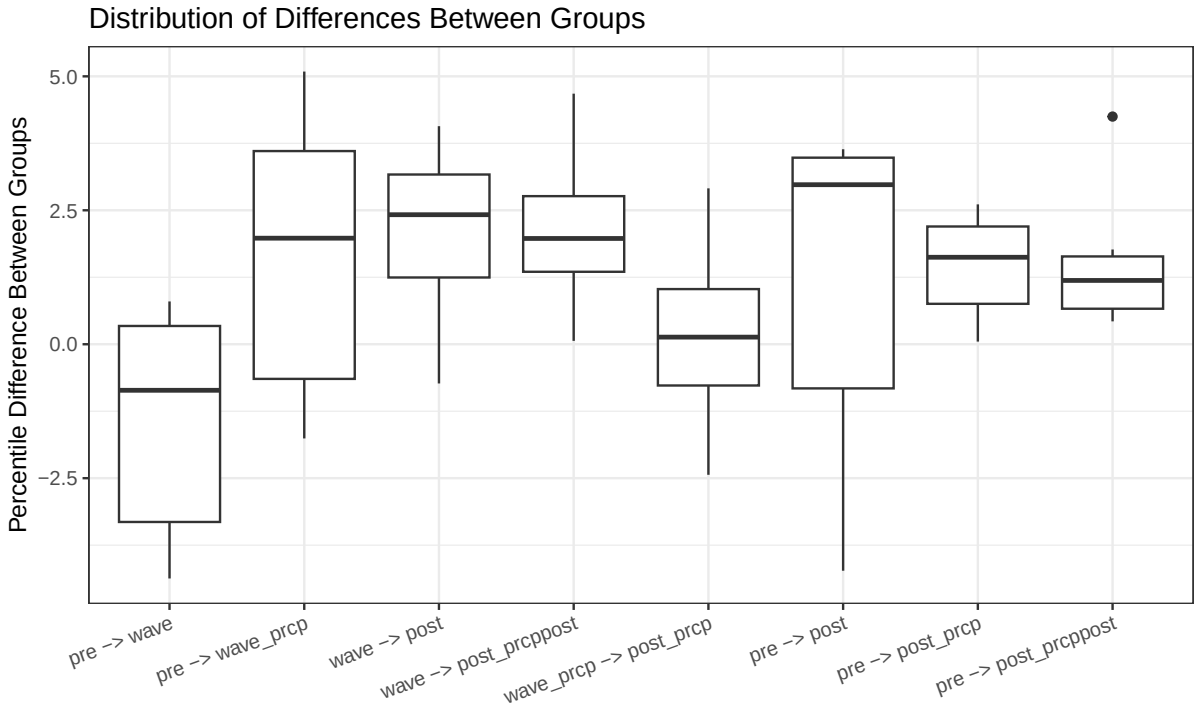

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
pre –> post
pre –> post_prcp
pre –> post_prcppost
region
NE
Mid
SE
Gulf

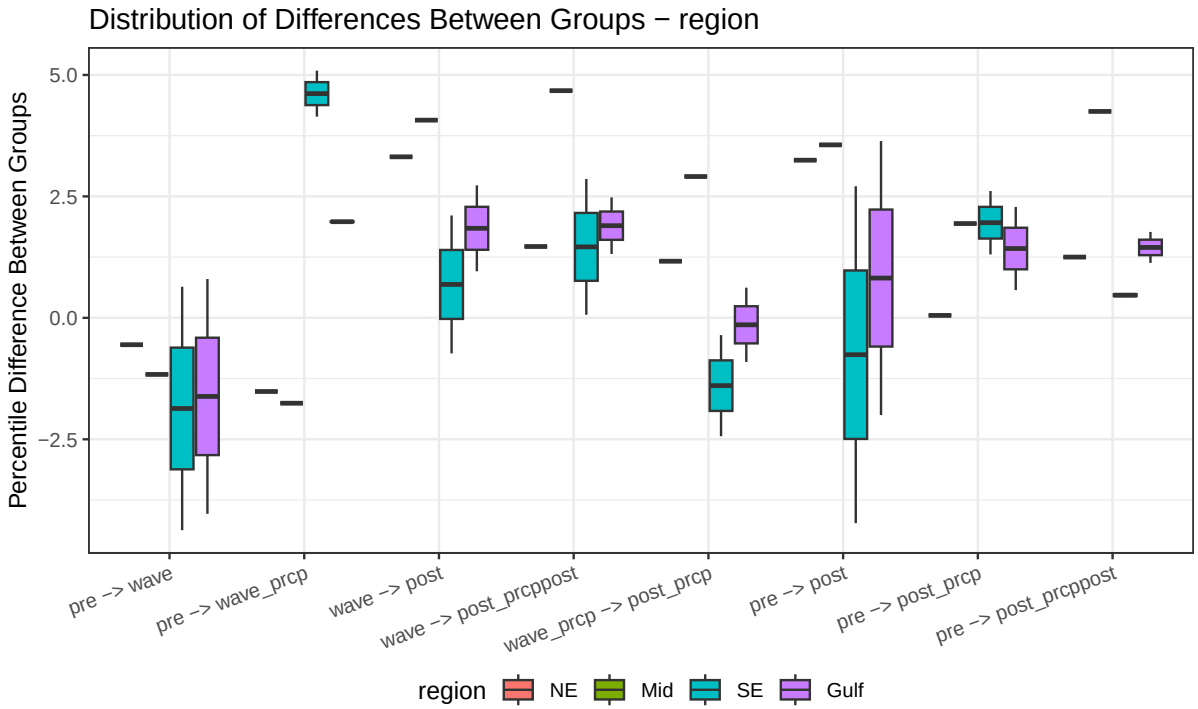

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
5
0
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
salinity
5−15
15−25
>25

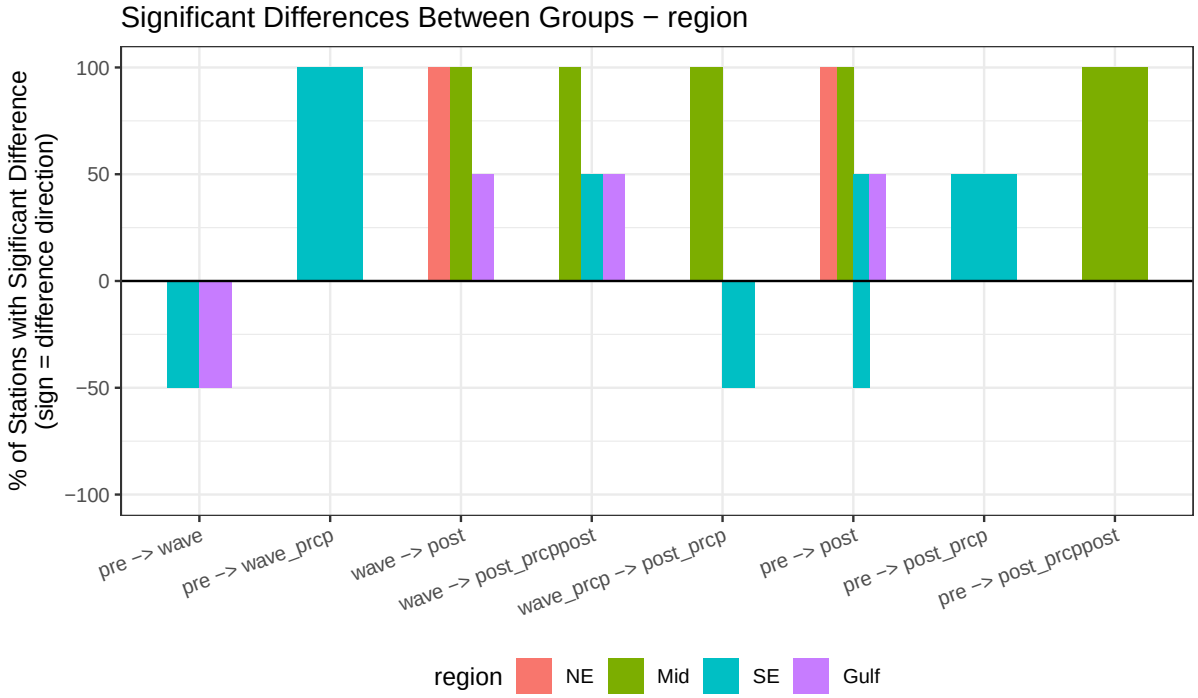

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
pre –> post
pre –> post_prcp
pre –> post_prcppost
salinity
5−15
15−25
>25

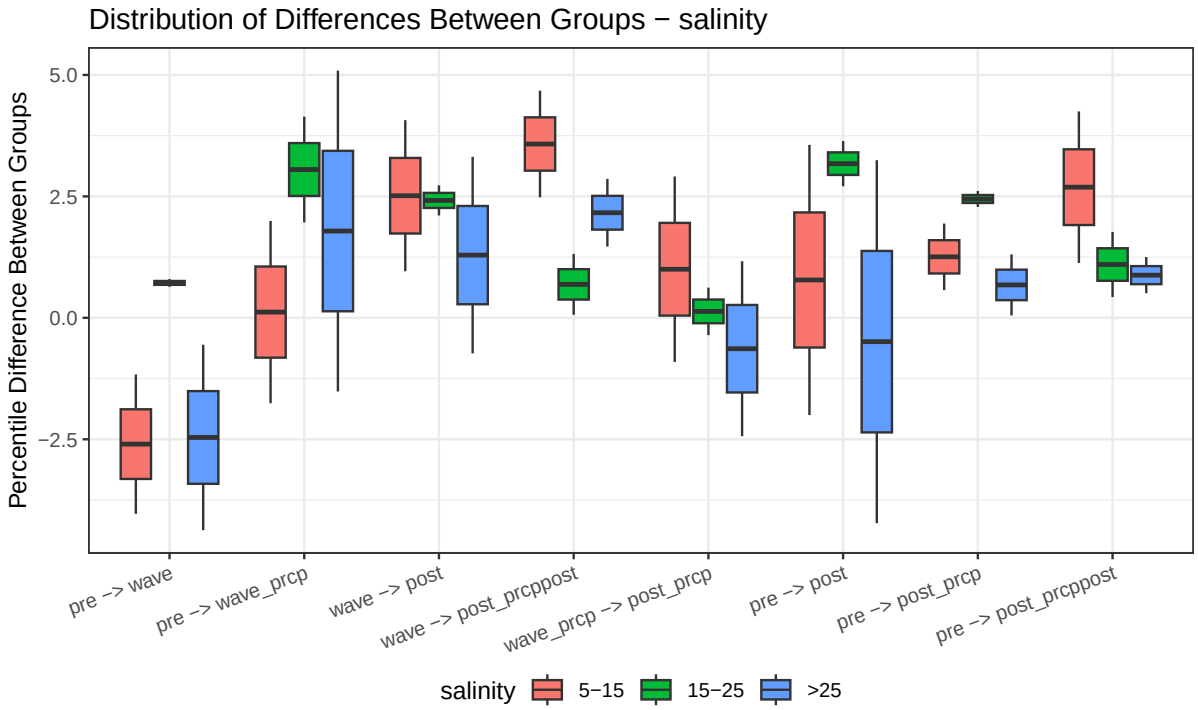
Turbidity

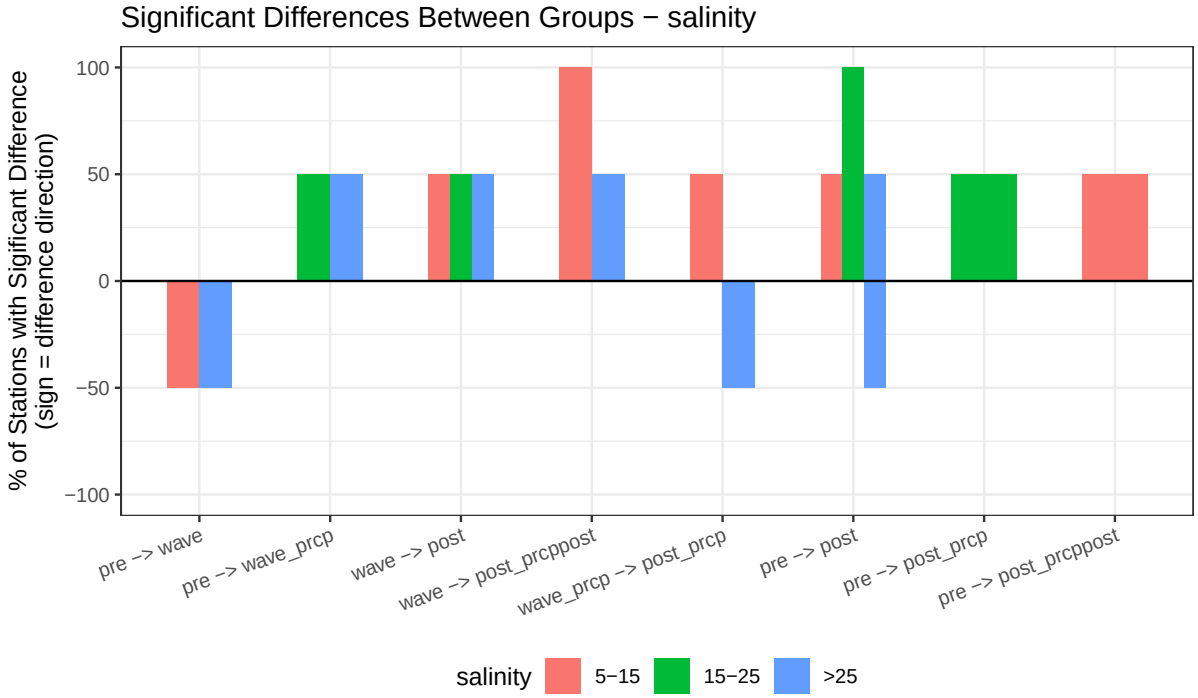
- Turbidity typically increases post heatwave from pre and heatwaves conditions when precip occurs
- Turbidity can drop during heatwaves without large precipitation events in SE and Gulf regions
- Moderate salinity group responds different to other groups; not dropping durings waves without precip and consistently increasing following heatwaves




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
5.0
2.5
0.0
−2.5
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
region
NE
Mid
SE
Gulf


Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
region
NE
Mid
SE
Gulf


Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
5.0
2.5
0.0
−2.5
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
salinity
5−15
15−25
>25

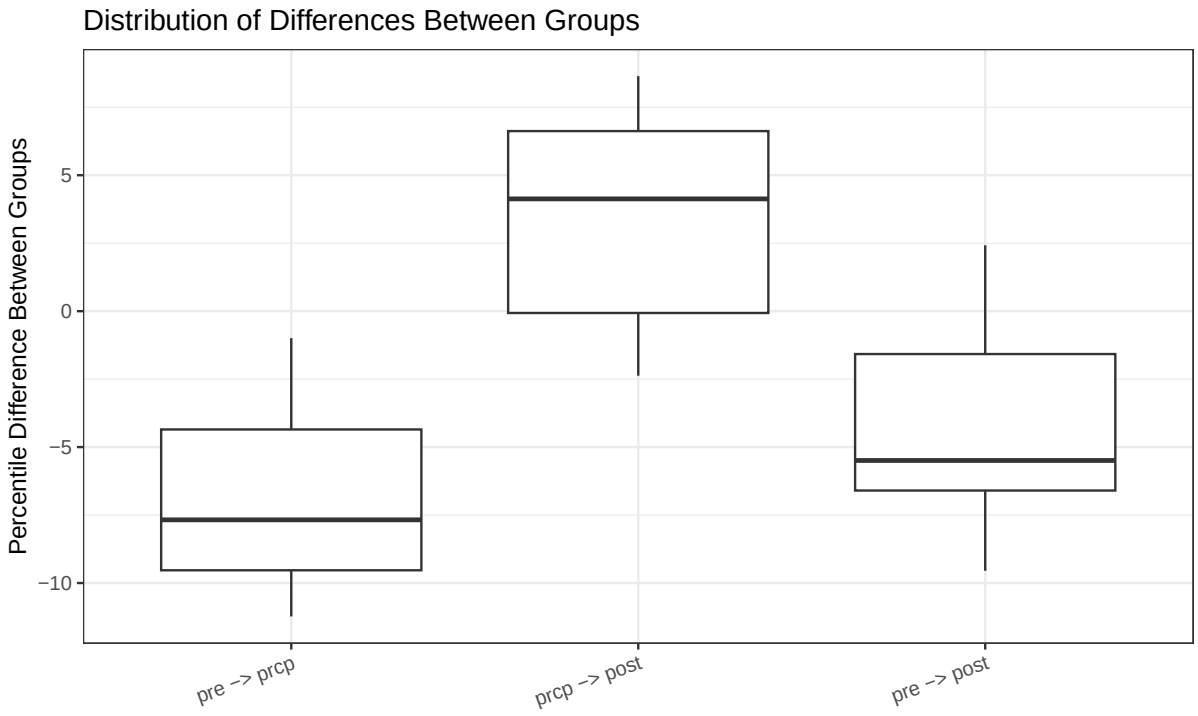

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> wave
pre –> wave_prcp
wave –> post
wave –> post_prcppost
wave_prcp –> post_prcp
pre –> post
pre –> post_prcp
pre –> post_prcppost
salinity
5−15
15−25
>25

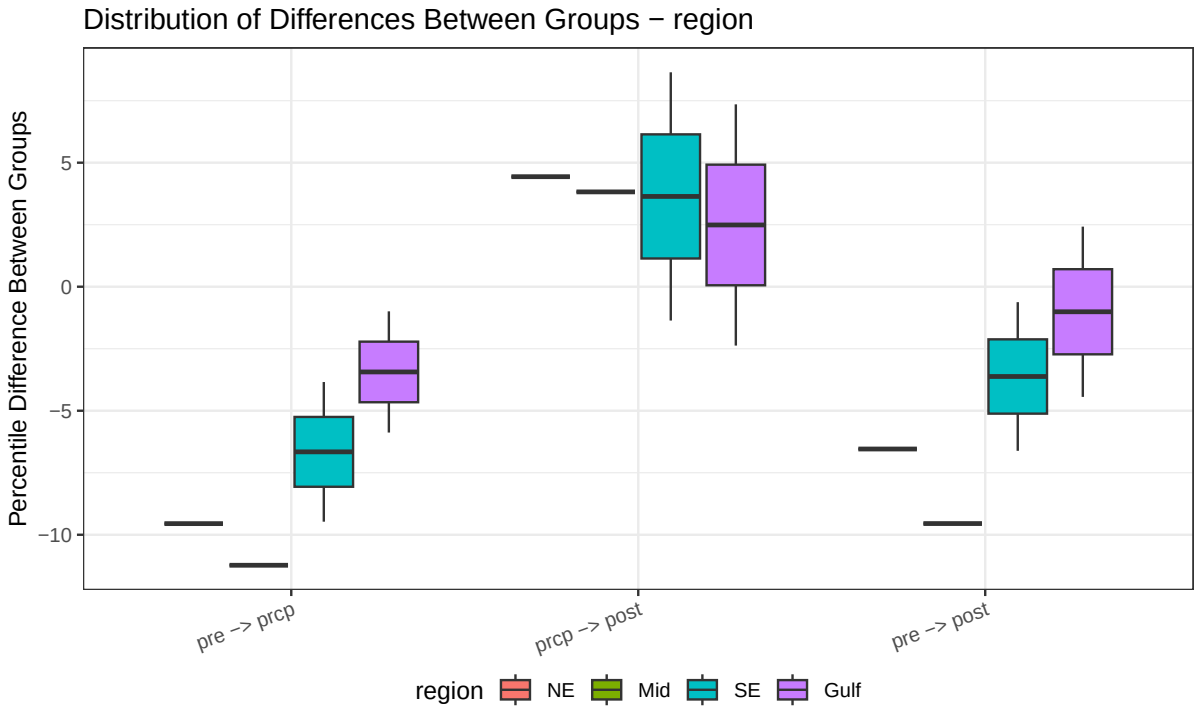
Extreme Precipitation Events without Heatwaves

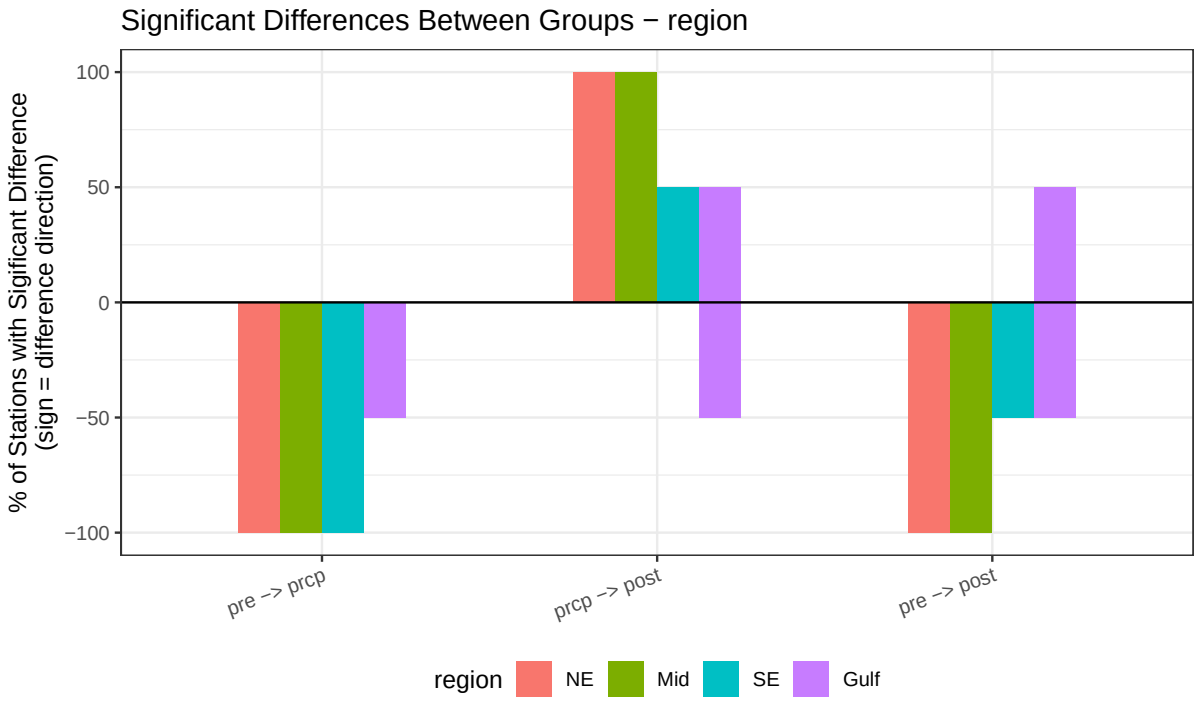
I defined extreme precip events as any day with > 90th percentile total precipitation. I compared these days with 6 days pre-event and 3 days post event.

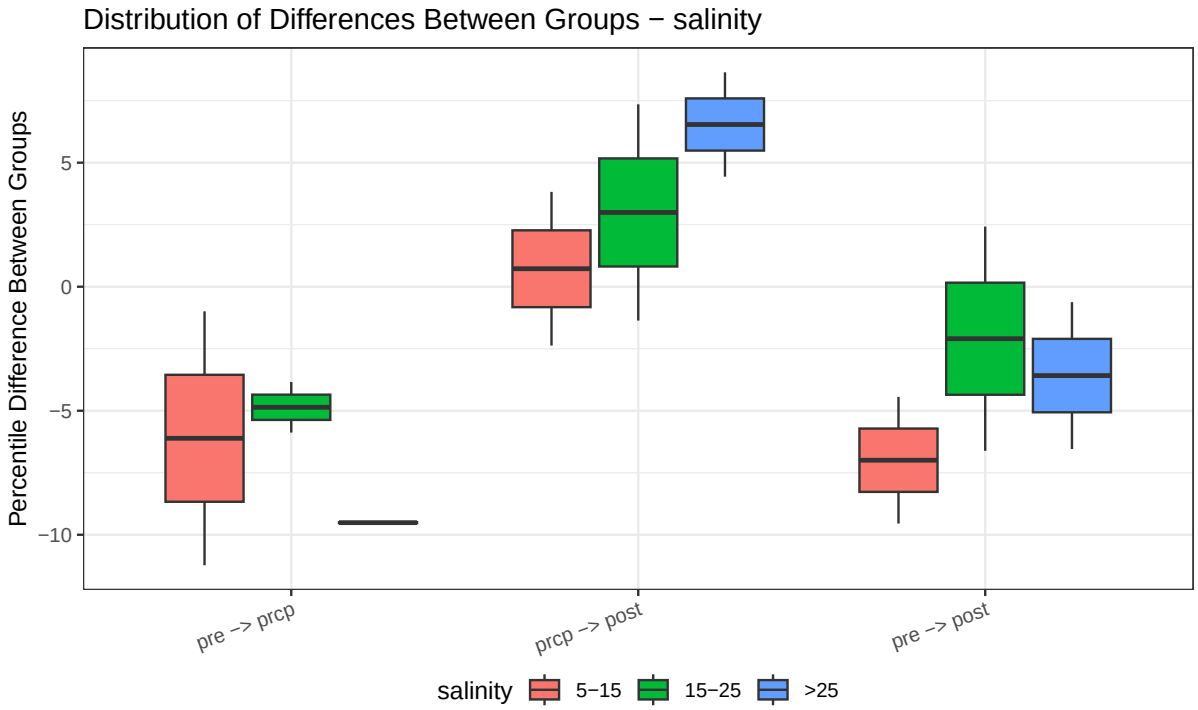
DO

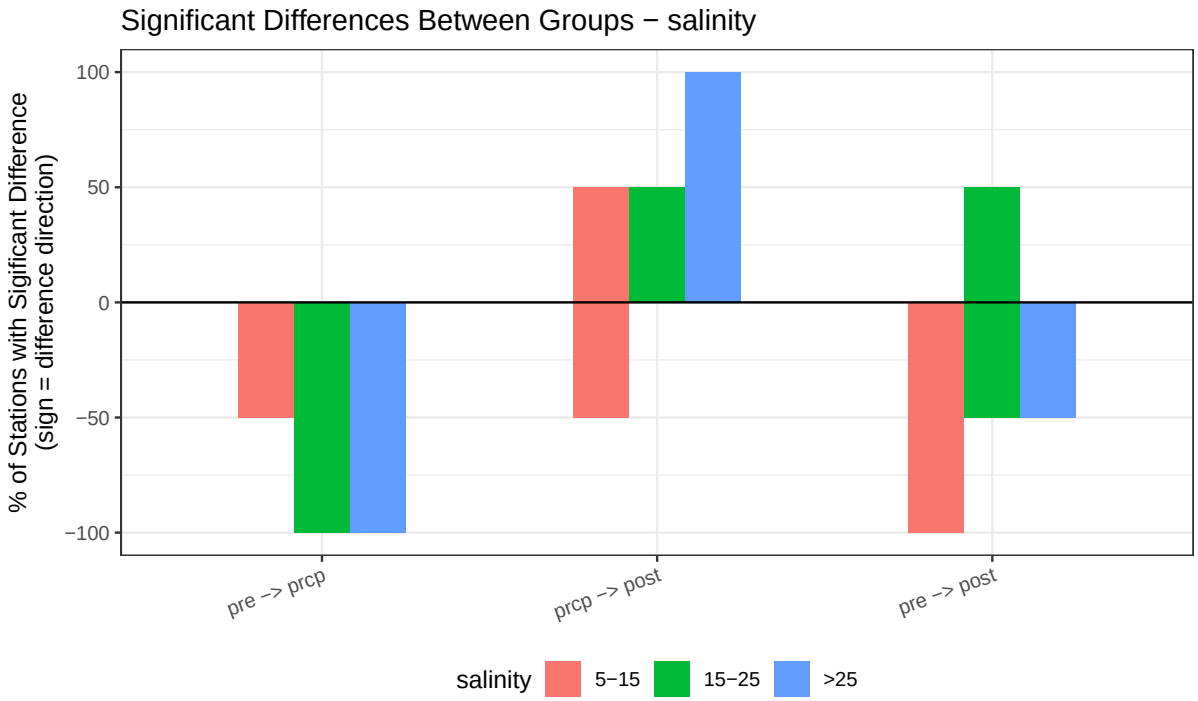
- Drops during precip events
- Rebounds slightly post event
- Pattern is present across regions
- No difference between salinity groups




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
5
0
−5
−10
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf

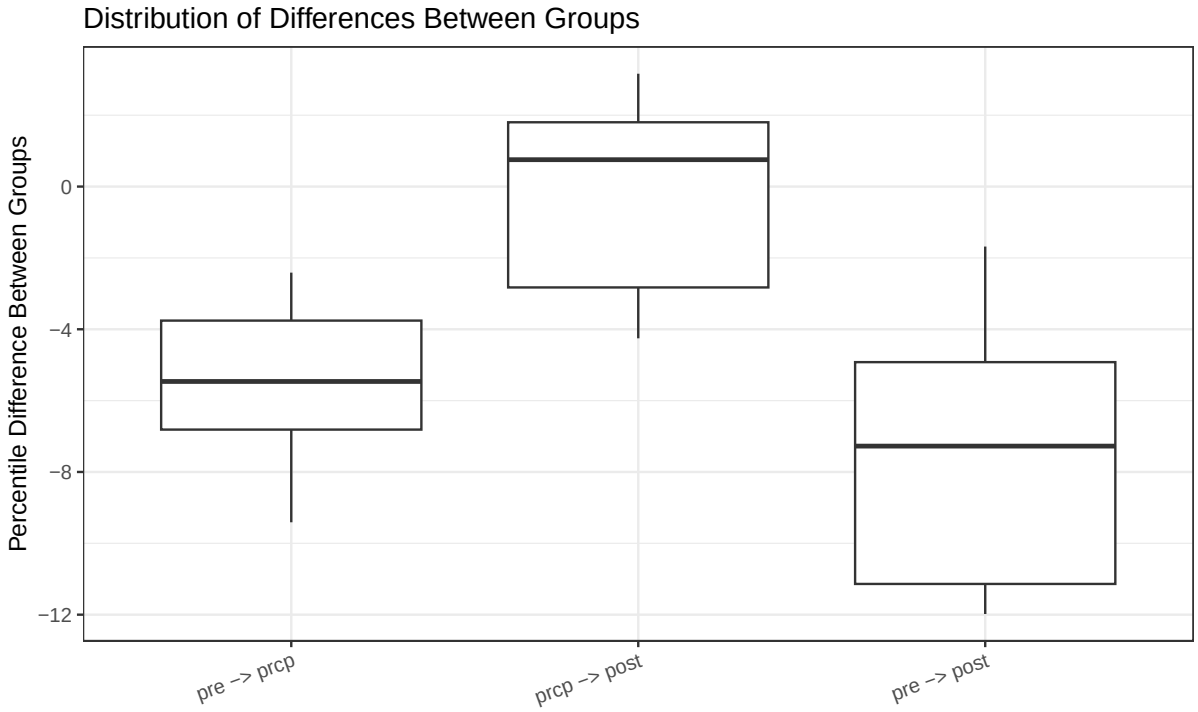

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf

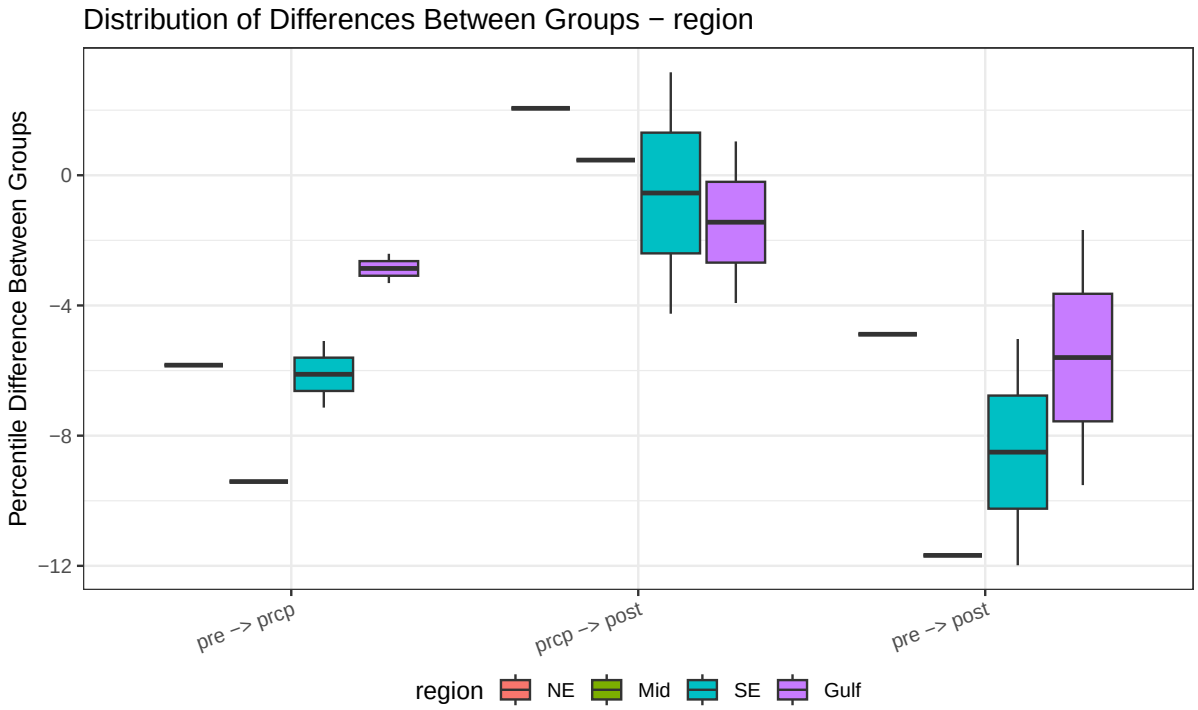

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
5
0
−5
−10
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25

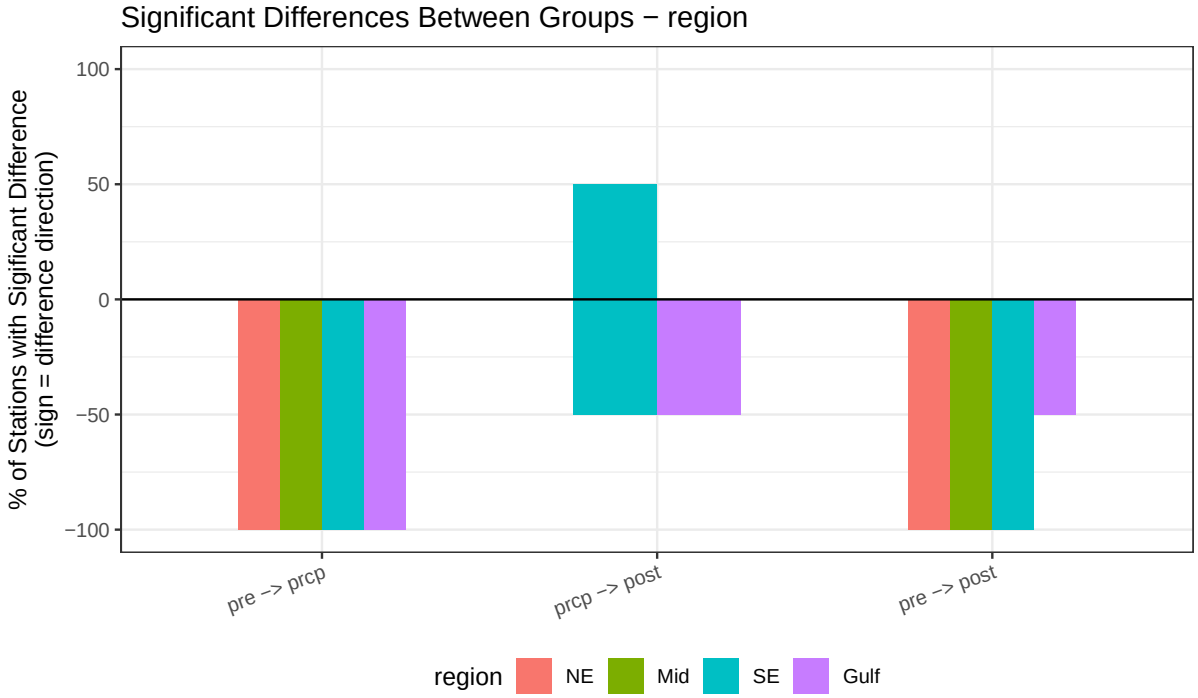

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25

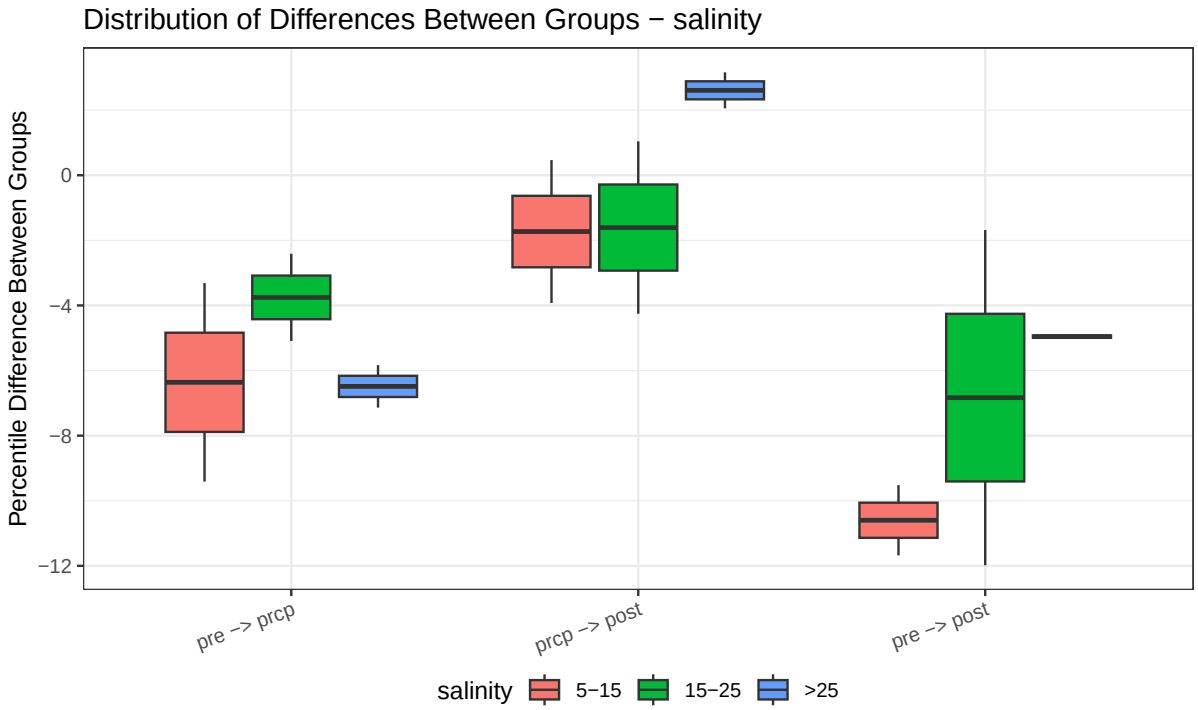
pH

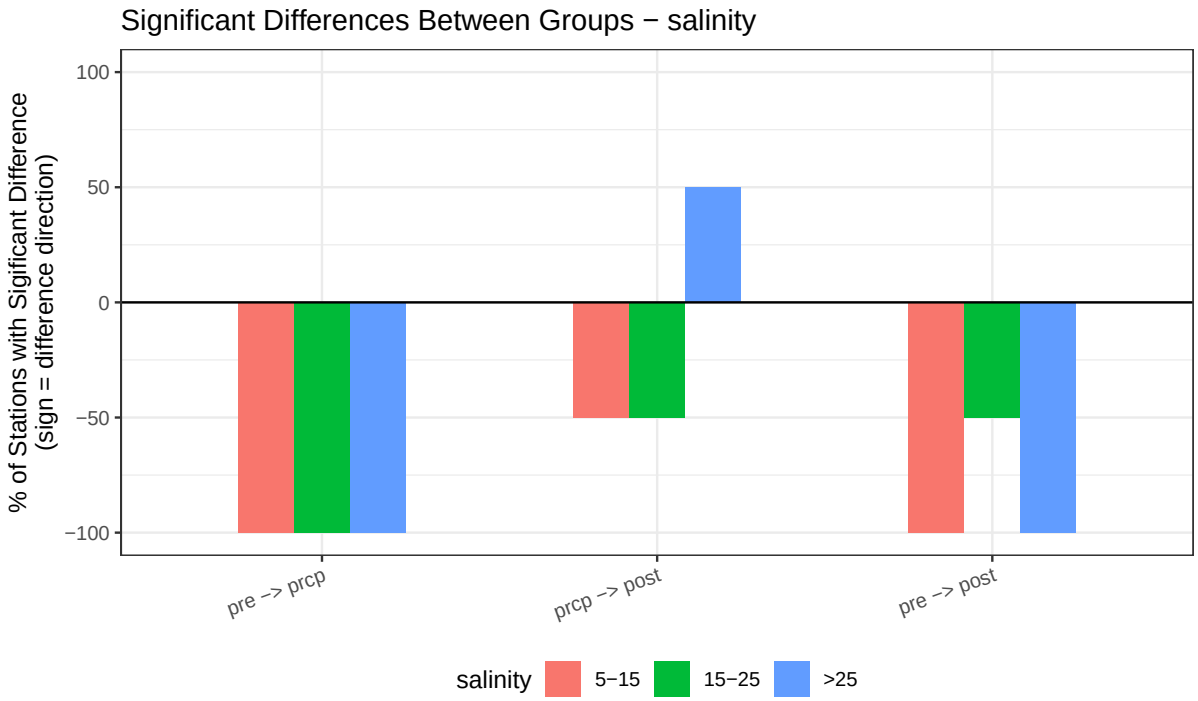
- pH drops during and post precip events.
- Pattern present across regions and salinity groups




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
0
−4
−8
−12
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf

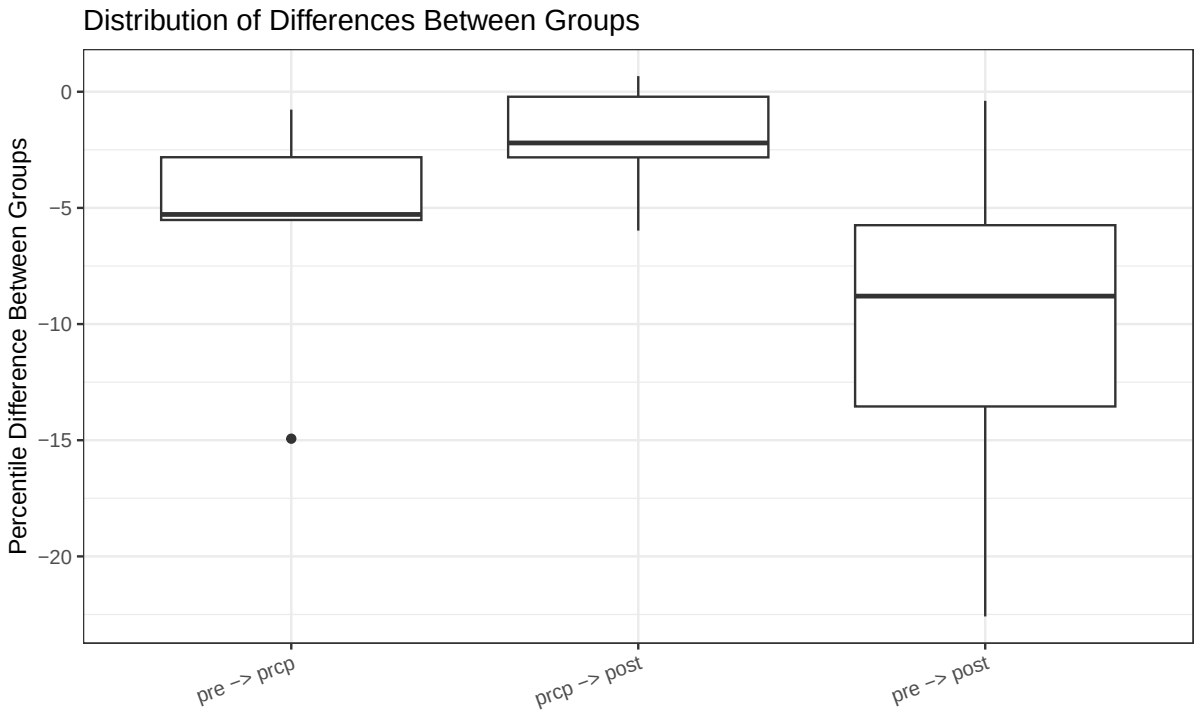

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf

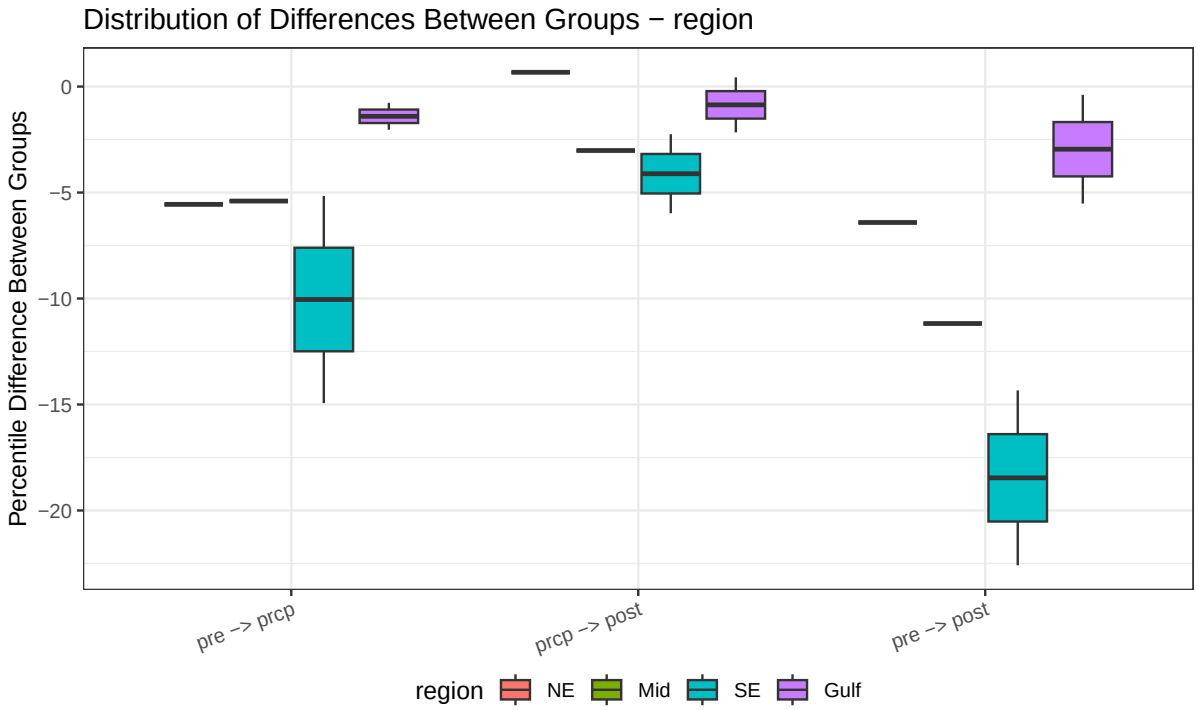

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
0
−4
−8
−12
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25

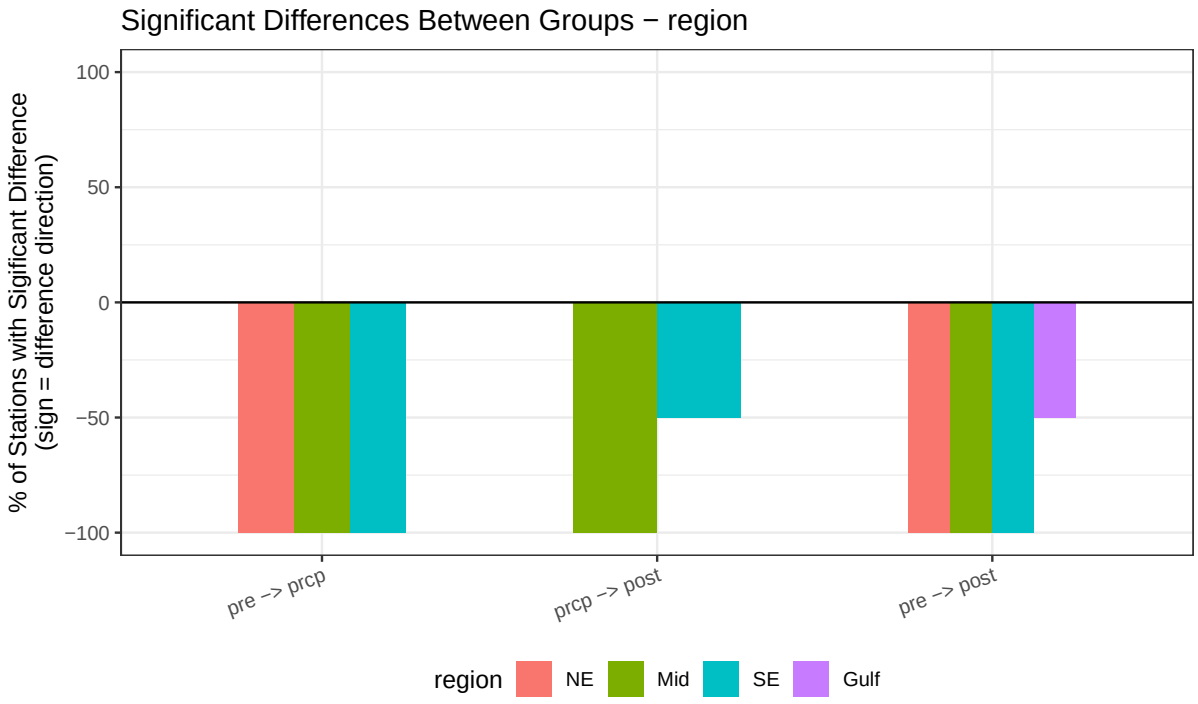

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25

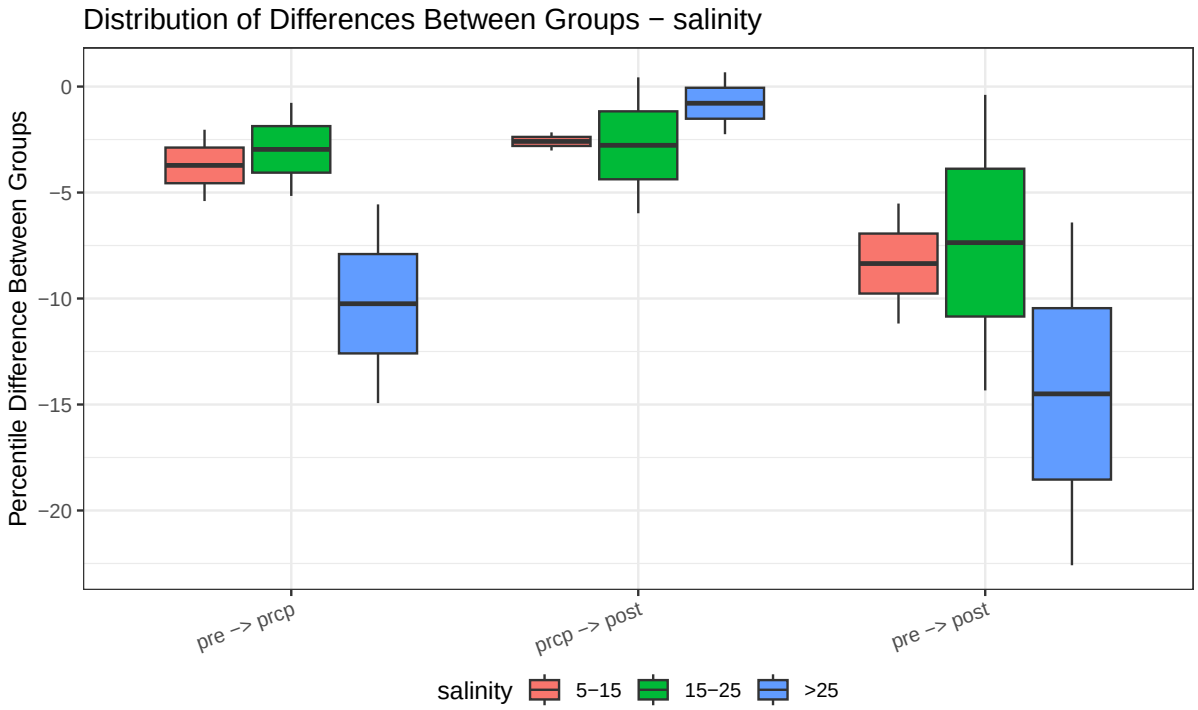
Conductivity

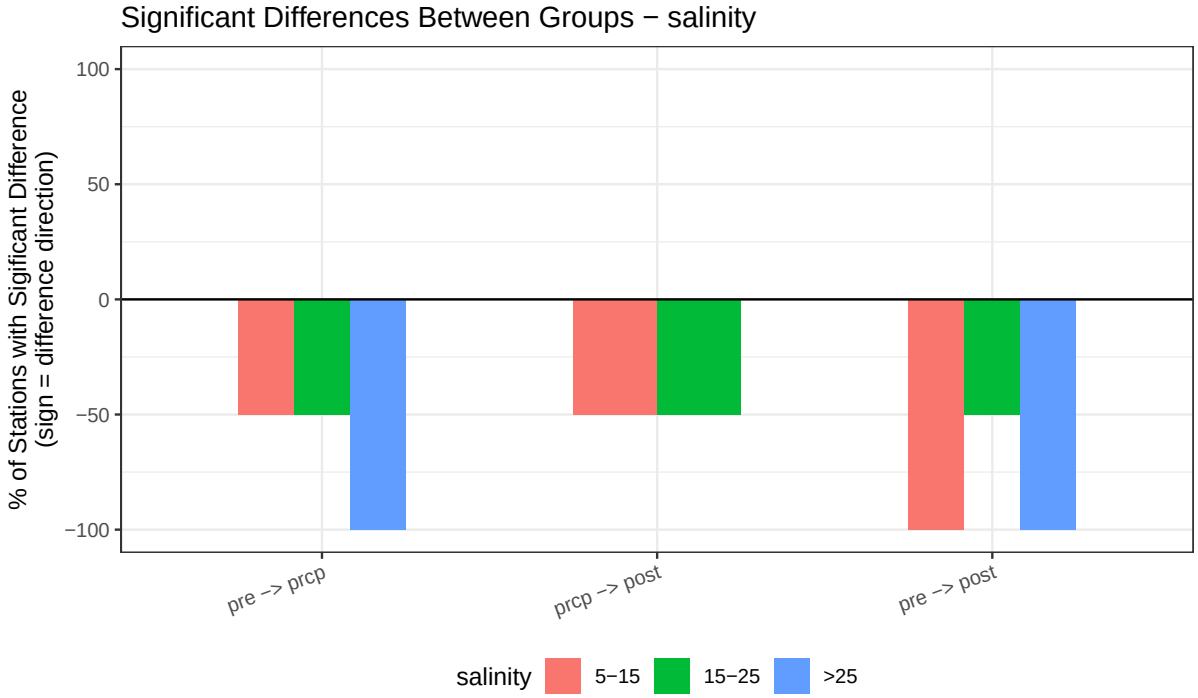
- Slight drop in conductivity during precip events
- No change during precip events in gulf region
- Big drop in conductivity during/post precip event in SE region/high salinity




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
0
−5
−10
−15
−20
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf

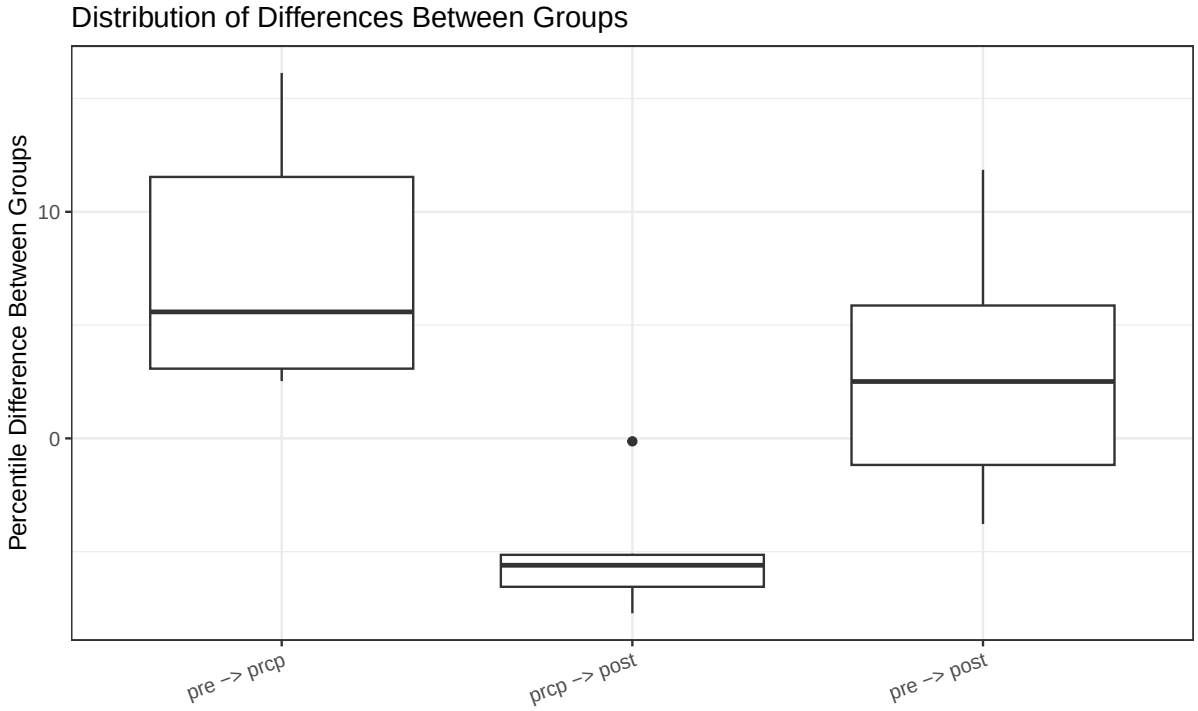

Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf

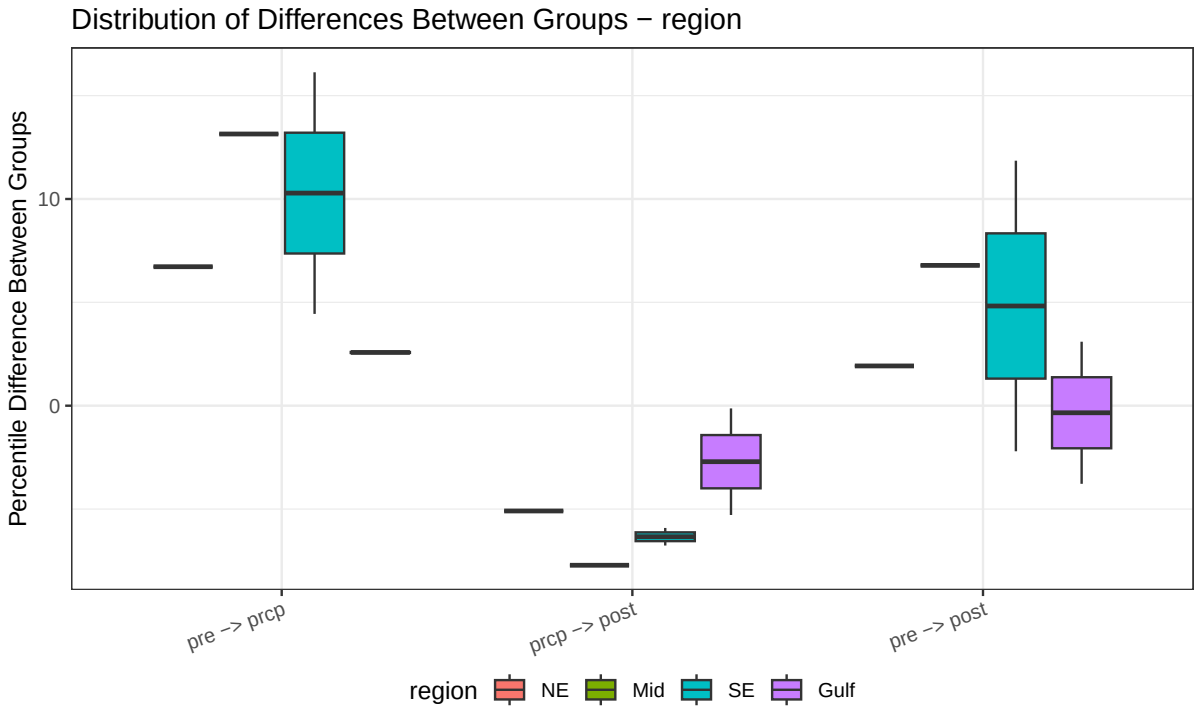

Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
0
−5
−10
−15
−20
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25

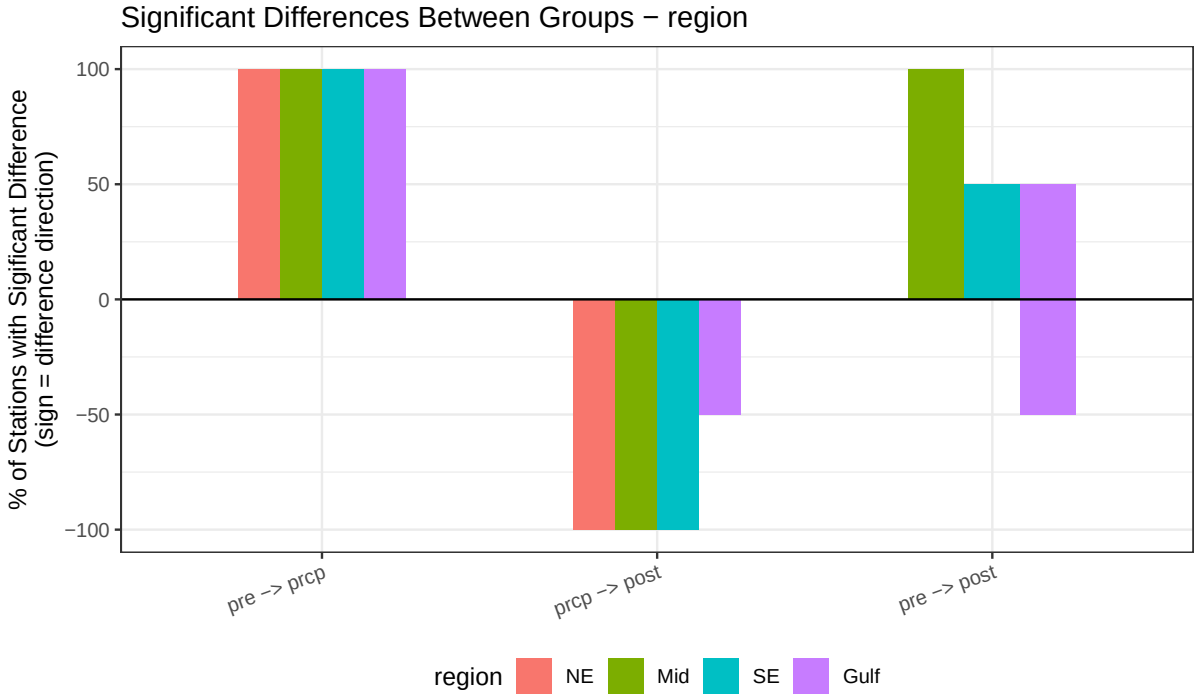

Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25

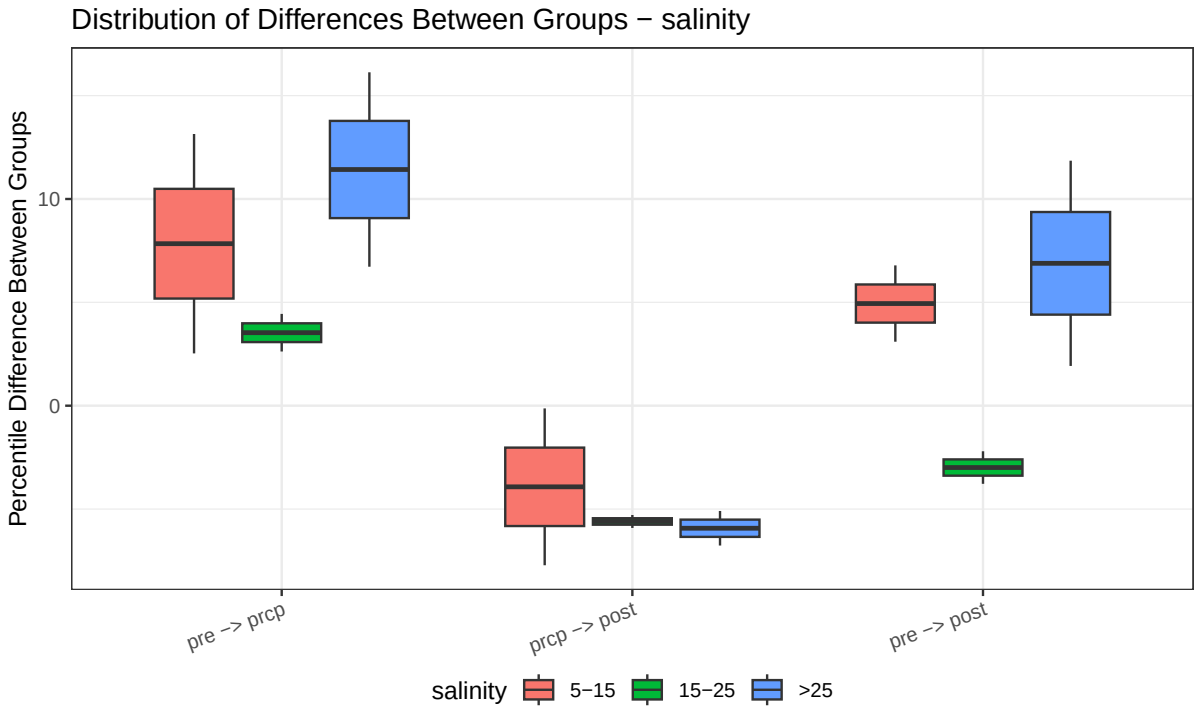
Turbidity

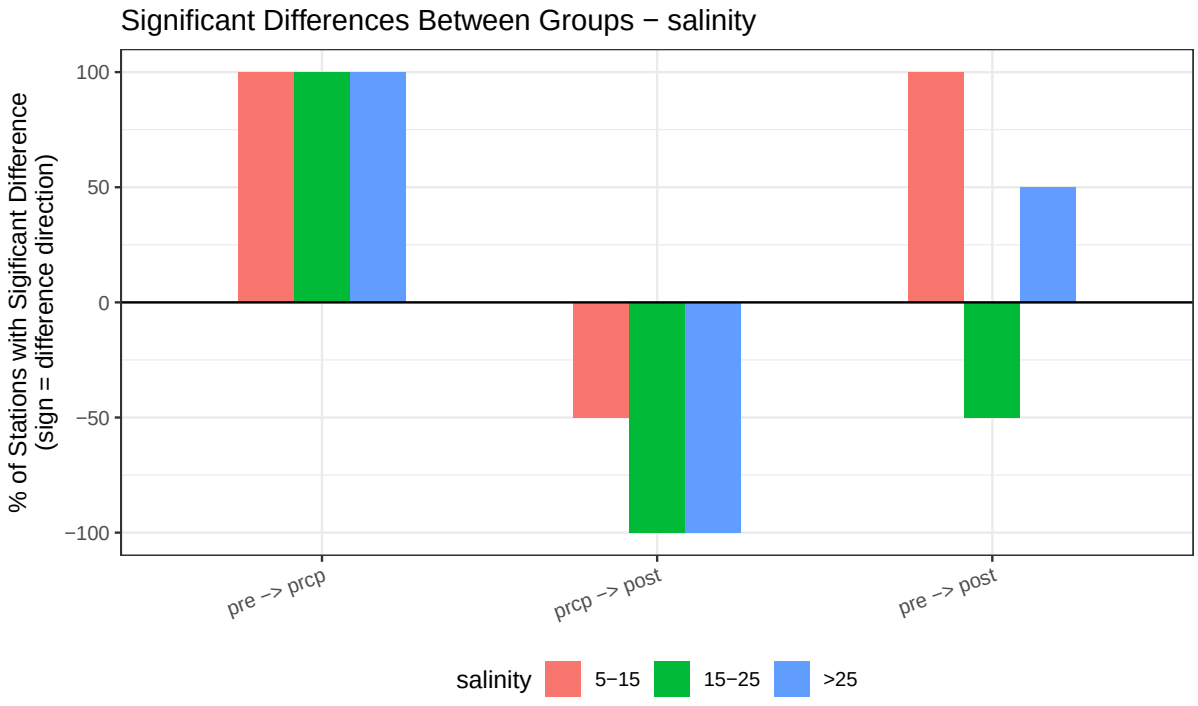
- Turbidity increases during precip events in all regions/salinity groups
- Typically drops from during event to post event, but still remains higher post than pre
- Moderate salinity group is not higher post event that pre, and the high salinity group has the weakest precip impact on turbidity




Distribution of Differences Between Groups − region
Percentile Difference Between Groups
10
0
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf


Significant Differences Between Groups − region
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
region
NE
Mid
SE
Gulf


Distribution of Differences Between Groups − salinity
Percentile Difference Between Groups
10
0
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25


Significant Differences Between Groups − salinity
% of Stations with Sigificant Difference
(sign = difference direction)
100
50
0
−50
−100
pre –> prcp
prcp –> post
pre –> post
salinity
5−15
15−25
>25